Learn Being Forward

COURSES · CAMPUS PARTNERSHIPS · PLACEMENTS

STUDY NOTES · BASIC TO ADVANCED

Logical Reasoning Think Clearly, Solve Fast

Complete notes for the Learn Being Forward Logical Reasoning syllabus: every topic explained with visuals and shortcut tricks, followed by basic, intermediate and advanced questions with fully worked solutions.

27	3	41	675
modules	levels	minimum days	solved questions

How to use these notes

These notes follow the **Learn Being Forward Logical Reasoning syllabus** exactly: the same 27 modules, in the same order, with every topic and subtopic explained step by step. Each subtopic number (for example **6.2.1**) matches the syllabus, so you can always find where you are.

Level	Modules	Minimum	Outcome
1. Beginner	1 to 9	12 days	Solve number and letter series, analogies, classification, coding-decoding, blood relations, directions and ranking with confidence
2. Intermediate	10 to 19	15 days	Master alphabet tests, symbol operations, inequalities, syllogisms, linear and circular seating, puzzles, clocks, calendars and input-output questions
3. Advanced	20 to 27	14 days	Handle statements and assumptions, conclusions and arguments, cause and effect, data sufficiency, critical reasoning, formal logic and advanced puzzles under exam timing
Total	1 to 27	41 days	About 7 weeks at 6 study days per week

1 · Understand
Read the concept and study the diagram until you can explain the rule in your own words.

2 · Learn the trick
Every topic has shortcut tricks in gold boxes. Practise them on the worked examples.

3 · Solve with a timer
Attempt the basic, intermediate and advanced questions on paper before looking at any solution.

4 · Check and log
Compare with the step-by-step solutions and write every mistake in your error log.

Every topic follows the same pattern: the idea in plain words, a visual model (table, tree, number line, path or diagram), the standard method, **shortcut tricks** for exam speed, and worked examples. Each module then gives about 25 practice questions in three levels, **basic**, **intermediate** and **advanced**, in the styles used by campus tests (TCS NQT, Infosys, Wipro, Accenture, Cognizant, Capgemini, AMCAT, eLitmus) and bank and government exams. Every question is fully solved, with a diagram where the reasoning needs one. Answers that can be computed (series, codes, ranks, distances, clock angles, calendar days, arrangements, syllogisms) were checked by a program while these notes were built.

HOW TO READ THE BOXES
Gold-bordered boxes hold **shortcut tricks**, worked examples and notes, red boxes warn about common traps, and navy boxes hold the key idea to remember. Questions are numbered in navy squares; each solution box repeats the question number with the answer in gold, followed by the full working.

Course plan at a glance

Module	Topic	Level	Min days
01	Introduction to logical reasoning	Beginner	1
02	Number series	Beginner	2
03	Letter and alphanumeric series	Beginner	1

Module	Topic	Level	Min days
04	Analogy	Beginner	1
05	Classification (odd one out)	Beginner	1
06	Coding and decoding	Beginner	2
07	Blood relations	Beginner	2
08	Direction sense	Beginner	1
09	Ranking, ordering and position	Beginner	1
10	Alphabet test	Intermediate	1
11	Mathematical operations	Intermediate	1
12	Inequalities	Intermediate	1
13	Syllogism	Intermediate	2
14	Linear seating arrangement	Intermediate	2
15	Circular seating arrangement	Intermediate	2
16	Puzzles	Intermediate	2
17	Clocks	Intermediate	1
18	Calendars	Intermediate	1
19	Input-output	Intermediate	2
20	Statement and assumptions	Advanced	2
21	Conclusions, arguments and courses of action	Advanced	2
22	Cause and effect, decision making	Advanced	2
23	Data sufficiency	Advanced	2
24	Critical reasoning	Advanced	2
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LEVEL 1

Beginner level

Solve number and letter series, analogies, classification, coding-decoding, blood relations, directions and ranking with confidence

9 MODULES · 12 DAYS MINIMUM

Module 01 · Introduction to logical reasoning	1 day
Module 02 · Number series	2 days
Module 03 · Letter and alphanumeric series	1 day
Module 04 · Analogy	1 day
Module 05 · Classification (odd one out)	1 day
Module 06 · Coding and decoding	2 days
Module 07 · Blood relations	2 days
Module 08 · Direction sense	1 day
Module 09 · Ranking, ordering and position	1 day

MODULE 01 · BEGINNER

Introduction to logical reasoning

Minimum duration **1 day** | Level **Beginner**

This module explains what logical reasoning is, why employers and exams test it, where and in what format it appears, and a dependable method for approaching any reasoning question. It ends with a 25-question diagnostic test across all the reasoning areas of this course, fully solved, so you can measure your starting accuracy and speed.

IN THIS MODULE

1.1 What is logical reasoning?
1.2 Why do we need it?
1.3 Where is it tested?
1.4 Question formats
1.5 How to approach reasoning
1.6 Diagnostic test

1.1 What is logical reasoning?

1.1.1 Definition: the ability to analyse information, find patterns and reach valid conclusions

Logical reasoning is the ability to take a set of facts, see the pattern or structure hidden in them, and reach a conclusion that **must** follow, without guessing. Every reasoning question, however it looks, asks you to do three things: **extract** the facts, **organise** them (in a table, tree, line or diagram), and **infer** what follows.



THE REASONING CYCLE USED THROUGHOUT THIS BOOK

1.1.2 Deductive vs inductive reasoning

Deductive reasoning goes from general rules to a certain conclusion: if the statements are true, the conclusion cannot be false (syllogisms, seating arrangements, blood relations). **Inductive** reasoning goes from examples to a probable rule: you observe a pattern and extend it (number series, analogies, odd one out). Recognising which one a question needs tells you how sure your answer can be.

	Deductive	Inductive
Direction	general rule → specific case	specific cases → general rule
Certainty	conclusion is certain if premises are true	conclusion is probable; the simplest consistent rule wins
Example	All coders are logical. Ravi is a coder. So Ravi is logical.	2, 4, 8, 16 … each term doubles, so the next is 32.

	Deductive	Inductive
Topics	syllogism, seating, puzzles, blood relations, inequalities	series, analogy, classification, coding patterns

1.1.3 Analytical reasoning vs verbal reasoning vs non-verbal reasoning

Analytical / logical
Arranging and deducing from facts: series, coding, seating, puzzles, directions, ranking, clocks, calendars. **This book.**

Verbal reasoning
Reasoning expressed in language: statements and assumptions, conclusions, arguments, critical reasoning. Modules 20 to 24 and the Verbal book.

Non-verbal reasoning
Reasoning with figures: figure series, mirror images, paper folding, cubes and dice, embedded figures. The Non-Verbal book.

1.2 Why do we need it?

1.2.1 Tests problem-solving and structured thinking, which employers value

A reasoning score is a quick, objective signal of how a candidate thinks: can they read precisely, organise messy information, and avoid jumping to conclusions? These are the same skills used to understand requirements, plan work and solve unfamiliar problems, which is why almost every campus hiring test has a reasoning section.

1.2.2 Directly related to programming logic and debugging

Reasoning skill	Programming equivalent
Spotting the rule in a number series	finding the loop or formula that generates output
Coding-decoding	encoding, ciphers, string manipulation
Input-output (machine) questions	tracing an algorithm step by step (sorting passes)
Syllogisms and formal logic	boolean conditions, if-else branches, De Morgan's laws
Puzzles with constraints	constraint solving, backtracking, test-case design
Data sufficiency	deciding whether you have enough information before coding

1.2.3 Useful for decisions in work and daily life

Comparing offers, planning a schedule around fixed constraints, checking whether an argument actually supports its claim, or noticing that "some" does not mean "all": everyday decisions improve when you separate what is **known** from what is merely **possible**.

1.3 Where is it tested?

1.3.1 Campus recruitment: TCS NQT, Infosys, Wipro, Accenture, Cognizant, Capgemini, Deloitte

Company test	Typical reasoning content	What to prioritise
TCS NQT	series, coding, seating, blood relations, syllogism, data sufficiency	speed on series and coding; seating sets
Infosys	puzzles, data sufficiency, arrangements, critical reasoning; one of the hardest	puzzles and DS (Modules 16, 23, 26)
Wipro (NLTH/Elite)	series, coding, directions, blood relations, syllogism	beginner and intermediate topics
Accenture	logical and abstract reasoning, flowcharts, input-output	Modules 2 to 9 and 19
Cognizant (GenC)	syllogism, coding, seating, sufficiency, inferences	syllogism and seating
Capgemini	game-based and logical reasoning, inequalities, coding	inequalities, coding, series
Deloitte	analytical puzzles, statements and conclusions	puzzles and verbal logic

1.3.2 AMCAT, eLitmus, CoCubes and other assessment platforms

AMCAT has an adaptive logical ability section; **eLitmus (pH test)** is known for hard puzzles and cryptarithms with negative marking; **CoCubes**, **Mettl** and **HackerEarth** assessments are used by many companies for screening. Their reasoning sections draw from the same topic list as this book.

1.3.3 Bank PO and clerk, SSC, railways and other government exams

IBPS and SBI PO/Clerk exams have large reasoning sections dominated by puzzles, seating, inequalities, syllogism and coding (often 5-question sets); SSC CGL/CHSL and railway exams focus on series, analogy, classification, coding and blood relations, answered very fast.

1.3.4 CAT, XAT, GMAT (critical reasoning) and other management exams

Management exams test **logical reasoning** as sets (arrangements, games, tournaments) and **critical reasoning** as arguments: assumptions, strengthening, weakening and inference (Modules 20 to 24).

1.4 Question formats

1.4.1 Multiple choice, fill in the blank, puzzle sets

Single MCQ

One question, four or five options. Use elimination: two options can usually be rejected quickly.

Fill in the blank (TITA)

Type the answer; no options to check against, so verify your rule on every term.

Puzzle / set

One long description followed by 3 to 5 questions. Solve the arrangement once, then answer all questions quickly.

1.4.2 Adaptive and sectional time limits

In **adaptive** tests the next question gets harder when you answer correctly, so accuracy on early questions matters most. **Sectional time limits** lock each section after its time. With **negative marking** (eLitmus, bank exams) an educated guess is worth it only if you can eliminate at least two options.

Exam style	Typical time per question	Strategy
Campus MCQ section	60 to 75 seconds	skip any question not cracked in 90 s; return later
Bank prelims puzzle set	50 to 60 seconds per question (including set solving)	pick the sets with the most direct clues first
eLitmus / CAT sets	3 to 6 minutes per set	select sets; do not attempt everything

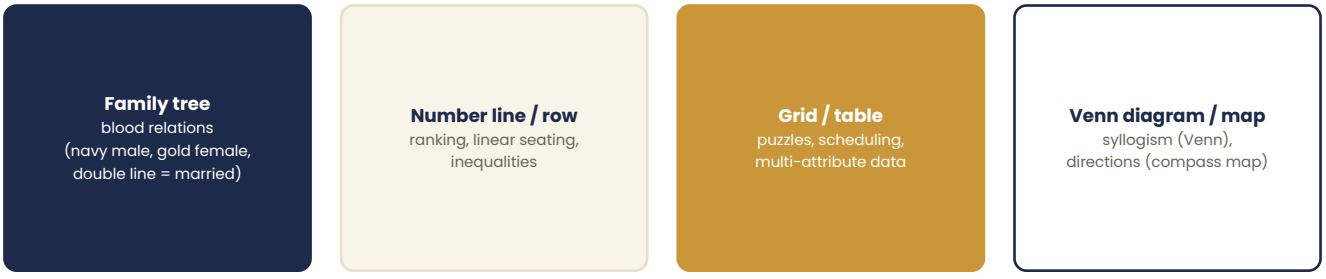
1.5 How to approach reasoning

1.5.1 Read carefully and extract facts

Most wrong answers come from misreading, not from weak logic. Read every word: "only", "not", "immediately", "second to the left", "at least", "facing the centre". Write facts in short form as you read (**A ≠ end**, **B = 3rd from left**), and separate **definite** facts from **conditional** ones ("if … then").

> **WATCH OUT**
> Watch for words that change meaning: "to the left of" (anywhere left) vs "immediate left" (next seat); "older than" vs "at least as old as"; "some" (at least one, possibly all) vs "only some".

1.5.2 Represent information visually (tables, diagrams, family trees)



CHOOSE THE REPRESENTATION THAT MATCHES THE QUESTION TYPE

1.5.3 Eliminate options

Options are information. In a series, check which options fit the parity or size of the next term; in coding, test the first letter only; in seating, an option that contradicts one definite clue is out. Eliminating two options often leaves the answer without full solving.

1.5.4 Build speed through pattern recognition

Speed comes from recognising patterns you have seen before: squares and cubes up to 30 and 20, primes up to 100, alphabet positions (forward and reverse), and standard puzzle structures. Keep an **error log**: for every mistake, note the topic, why it happened (misread, wrong rule, calculation) and the fix.

MEMORY TRICK

Learn the alphabet positions with **EJOTY**: E = 5, J = 10, O = 15, T = 20, Y = 25. Any letter is a step or two from one of these. For reverse positions use **reverse = 27 − position** (M = 13 has reverse 14).



ALPHABET POSITIONS; THE EJOTY ANCHORS ARE HIGHLIGHTED

1.6 Diagnostic test

Take this 25-question diagnostic before starting Module 2. Allow **30 minutes**, then mark yourself with the solutions. Record your accuracy per topic: each question names the module that teaches it.

1.6.1 Diagnostic questions

1.6.1.1 Basic level

Q1 *(Number series, M2)* Find the next term: 3, 7, 15, 31, 63, ?

(a) 95 **(b)** 125 **(c)** 127 **(d)** 129

Q2 *(Letter series, M3)* Find the next term: B, E, H, K, ?

(a) M **(b)** N **(c)** O **(d)** L

Q3 *(Analogy, M4)* Doctor : Hospital :: Teacher : ?

(a) Student **(b)** Book **(c)** School **(d)** Class test

Q4 *(Classification, M5)* Find the odd one out: 21, 35, 49, 63, 74

(a) 21 **(b)** 49 **(c)** 63 **(d)** 74

Q5 *(Coding, M6)* If CAT is coded as DBU, how is DOG coded?

(a) EPH **(b)** EOH **(c)** CNF **(d)** EPG

Q6 *(Blood relations, M7)* Pointing to a man, Riya says, "He is the son of my mother's only son." How is the man related to Riya?

(a) Brother **(b)** Nephew **(c)** Son **(d)** Cousin

Q7 *(Directions, M8)* Arun walks 5 km north, turns right and walks 3 km, then turns right again and walks 5 km. Which direction is he from the start?

(a) North **(b)** East **(c)** West **(d)** South

Q8 *(Ranking, M9)* In a row of students, Meena is 12th from the left and 18th from the right. How many students are in the row?

(a) 28 **(b)** 29 **(c)** 30 **(d)** 31

1.6.1.2 Intermediate level

Q9 *(Alphabet test, M10)* How many letters are there between H and S in the English alphabet?

(a) 9 **(b)** 10 **(c)** 11 **(d)** 12

Q10 *(Symbol operations, M11)* If + means ×, - means ÷, × means - and ÷ means +, find 16 - 4 + 3 × 2 ÷ 5.

(a) 15 **(b)** 13 **(c)** 9 **(d)** 18

Q11 *(Inequalities, M12)* Given P > Q ≥ R = S < T. Which conclusion definitely follows?

(a) Q > S **(b)** P > S **(c)** T > Q **(d)** R > T

Q12 *(Syllogism, M13)* Statements: All pens are books. Some books are bags. Conclusions: I. Some pens are bags. II. Some bags are books.

(a) Only I follows **(b)** Only II follows
(c) Both follow **(d)** Neither follows

Q13 *(Linear seating, M14)* Five friends sit in a row facing north. A is to the immediate left of B. C is at the right end. D is second from the left. E is at the left end. Who is in the middle?

(a) A **(b)** B **(c)** D **(d)** C

Q14 *(Circular seating, M15)* Six people sit around a round table facing the centre. P is opposite S, and Q is to the immediate left of P. Who is opposite Q?

(a) P **(b)** The person to the immediate right of S
(c) The person to the immediate left of S **(d)** S

Q15 *(Clocks, M17)* What is the angle between the hands of a clock at 3:40?

(a) 120° **(b)** 130° **(c)** 140° **(d)** 150°

Q16 *(Calendars, M18)* If 1 January 2026 was a Thursday, what day was 1 January 2027?

(a) Thursday **(b)** Friday **(c)** Saturday **(d)** Wednesday

1.6.1.3 Advanced level

Q17 *(Wrong number, M2)* Find the wrong term: 2, 5, 10, 17, 26, 38, 50

(a) 17 **(b)** 26 **(c)** 38 **(d)** 50

Q18 *(Coding, M6)* In a code, "sky is blue" is "ta na pa", "blue and green" is "ra pa ka", and "green is fresh" is "ka na la". What is the code for "sky"?

(a) ta **(b)** na **(c)** pa **(d)** ka

Q19 *(Blood relations, M7)* A + B means A is the father of B; A - B means A is the wife of B; A × B means A is the brother of B. What does P × Q + R - S mean about P and S?

(a) P is the father-in-law of S **(b)** P is the brother-in-law of S
(c) P is the uncle of S's wife **(d)** P is the son of S

Q20 *(Directions, M8)* From his house Kiran walks 6 km east, then 8 km north. How far is he from his house, and in which direction?

(a) 10 km, north-east **(b)** 14 km, north-east
(c) 10 km, north-west **(d)** 12 km, east

Q21 *(Puzzle, M16)* Three boxes A, B, C are stacked. B is not at the top. C is above A. A is not at the bottom. Which box is at the top?

(a) A **(b)** B
(c) C **(d)** Cannot be determined

Q22 *(Input-output, M19)* A machine sorts words: Input: *cat egg ant dog*. Step 1: *ant cat egg dog*. Step 2: *ant cat dog egg*. What rule is the machine following?

(a) Reverse alphabetical order, one word per step
(b) Alphabetical order, bringing the next word into place each step
(c) Sorting by word length
(d) Moving the last word to the front

Q23 *(Data sufficiency, M23)* What is Rahul's rank in a class of 40? I. Rahul is 5 ranks above Sumit. II. Sumit is 22nd from the bottom.

(a) I alone is sufficient
(b) II alone is sufficient
(c) Both together are needed
(d) Even both are not sufficient

Q24 *(Statement and assumption, M20)* Statement: "Use our new app to book tickets in 10 seconds." Which assumption is implicit?

(a) Everyone owns a smartphone
(b) People prefer booking tickets quickly
(c) Other apps are illegal
(d) Tickets will be free

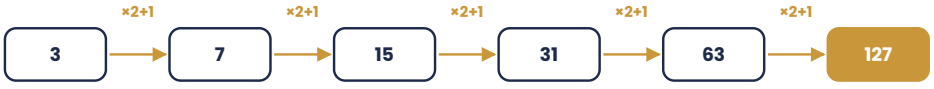
Q25 *(Formal logic, M25)* "If it rains, the match is cancelled." The match was not cancelled. What follows?

(a) It rained
(b) It did not rain
(c) The match was played indoors
(d) Nothing can be concluded

1.6.2 Solutions

Q1 **Answer: (c) 127**

Each term is **× 2 + 1**: 3 × 2 + 1 = 7 … 63 × 2 + 1 = **127**. (Each term is also $2^n - 1$.)



Q2 **Answer: (b) N**

Positions 2, 5, 8, 11 increase by 3, so the next is 14 = **N**.

Q3 **Answer: (c) School**

A professional and the place where they work. A teacher works in a **school**.

Q4 **Answer: (d) 74**

21, 35, 49 and 63 are multiples of 7; **74** is not.

Q5 **Answer: (a) EPH**

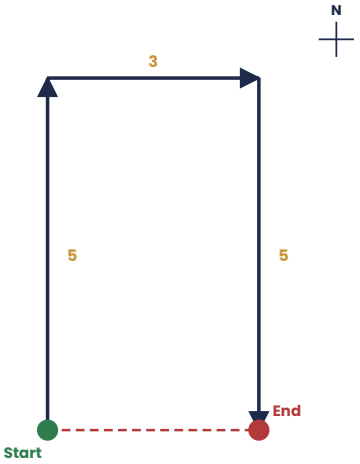
Each letter moves +1: C→D, A→B, T→U. So D→E, O→P, G→H = **EPH**.

Q6 **Answer: (b) Nephew**

Riya's mother's only son is Riya's brother. The man is her brother's son: her **nephew**.

Q7 **Answer: (b) East**

North 5, right = east 3, right = south 5. He is 3 km to the **east** of the start.



Q8 **Answer: (b) 29**

Total = left rank + right rank − 1 = 12 + 18 − 1 = **29**.

Q9 **Answer: (b) 10**

H = 8, S = 19. Letters strictly between = 19 − 8 − 1 = **10**.

Q10 **Answer: (a) 15**

Rewrite: 16 ÷ 4 × 3 − 2 + 5. BODMAS: 16 ÷ 4 = 4; 4 × 3 = 12; 12 − 2 + 5 = **15**.

Q11 **Answer: (b) P > S**

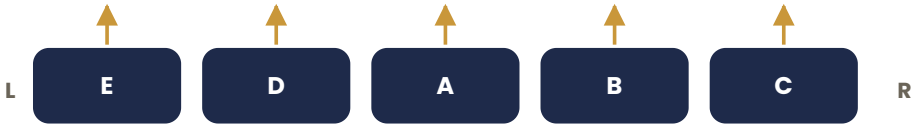
P > Q ≥ R = S gives P > S (one strict sign makes the result strict). Q ≥ S only, and T cannot be compared with Q. Answer: **P > S**.

Q12 **Answer: (b) Only II follows**

"Some books are bags" converts to "Some bags are books", so II follows. Pens may lie in the part of books outside bags, so I is only possible. **Only II**.

Q13 **Answer: (a) A**

E is 1st, D 2nd, C 5th. A and B take 3rd and 4th with A immediately left of B. The middle (3rd) is **A**.



Q14 **Answer: (c) The person to the immediate left of S**

Opposite seats are three places apart. Moving one seat to P's left and then crossing the table is the same as crossing to S and moving one seat to S's left, because "left" is the same rotation for everyone facing the centre: **the person to the immediate left of S**.

Q15 **Answer: (b) 130°**

Angle = |30H − 5.5M| = |90 − 220| = **130°**.

Q16 **Answer: (b) Friday**

2026 is not a leap year: 365 days = 52 weeks + 1 odd day. Thursday + 1 = **Friday**.

Q17 **Answer: (c) 38**

The terms are $n^2 + 1$: 2, 5, 10, 17, 26, 37, 50. So **38** should be 37.

Q18 **Answer: (a) ta**

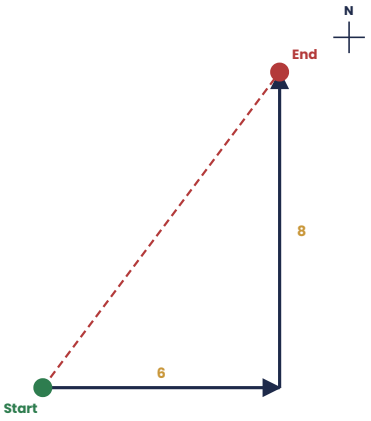
blue (sentences 1 and 2) = pa; is (1 and 3) = na. In sentence 1 the remaining word sky has the remaining code **ta**.

Q19 **Answer: (c) P is the uncle of S's wife**

P × Q: P is Q's brother. Q + R: Q is R's father. R − S: R is S's wife. P is the brother of R's father, i.e. R's uncle: **P is the uncle of S's wife**.

Q20 **Answer: (a) 10 km, north-east**

Distance = $\sqrt{(6^2 + 8^2)} = \sqrt{100}$ = **10 km**, towards the **north-east**.



Q21 **Answer: (c) C**

A is not at the bottom and C is above A, so A is in the middle and C on top; B is at the bottom (consistent with B not at the top). Top: **C**.

Q22 **Answer: (b) Alphabetical order, bringing the next word into place each step**

Step 1 moves the alphabetically first word to the front; step 2 places the next one in order. The machine arranges words in **alphabetical order, one word per step**.

Q23 **Answer: (c) Both together are needed**

II gives Sumit's rank from the top: 40 - 22 + 1 = 19; I then gives Rahul = 14. Neither alone works: **both are needed**.

Q24 **Answer: (b) People prefer booking tickets quickly**

An advertisement stressing speed assumes that speed matters to customers: **people prefer booking quickly**.

Q25 **Answer: (b) It did not rain**

From "if P then Q" and "not Q" we conclude "not P" (contrapositive). So **it did not rain**.

Logical reasoning = extract facts → represent them visually → find the rule → conclude only what must follow → verify. Deductive questions give certain answers; inductive ones need the simplest consistent rule. Learn the alphabet positions (EJOTY), read every qualifier, eliminate options, and keep an error log.

MODULE 02 · BEGINNER

Number series

Minimum duration **2 days** | Level **Beginner**

Number series questions give a sequence of numbers built by a hidden rule and ask for the next term, a missing term, or the one wrong term. This module teaches the patterns that cover almost every exam question (differences, ratios, mixed operations, squares and cubes, primes, Fibonacci-type, alternating series, second- and third-level differences, digit and factorial series), a fixed checking order that finds the rule fast, and the bank-exam wrong-number technique, followed by 25 solved questions.

IN THIS MODULE

2.1 Basics
2.2 Common patterns
2.3 Method
2.4 Wrong-number series
2.5 Practice questions
2.6 Solutions

2.1 Basics

2.1.1 Finding the next term, missing term or wrong term

A **number series** is a list of numbers in which each term is produced from the previous ones (or from its position) by a fixed rule. Three question types use the same skill, finding the rule:

Next term
5, 11, 23, 47, ? → find what comes after the last term.

Missing term
4, 9, ?, 25, 36 → the rule must fit the terms on **both sides** of the gap.

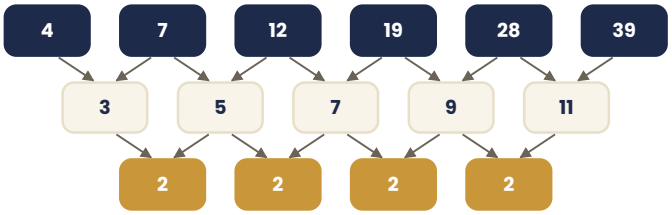
Wrong term
2, 6, 12, 21, 30, 42 → exactly one term breaks the rule; find it (2.4).

Type	Idea	Example
Recursive (term from previous term)	next = f(previous): add, multiply, ×2+1 …	3, 7, 15, 31 (×2 + 1)
Positional (term from its position n)	term = f(n): n^2, n^3, $n^2 + n$, primes …	2, 6, 12, 20, 30 ($n^2 + n$)

2.2 Common patterns

2.2.1 Addition and subtraction series (constant and changing differences)

Write the **differences** between consecutive terms under the series. If they are equal, the series is arithmetic. If they change, look at how they change: increasing by a constant, doubling, odd numbers, primes or squares.



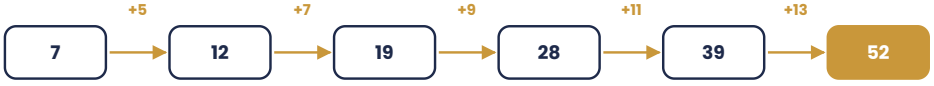
DIFFERENCES 3, 5, 7, 9, 11 AND SECOND DIFFERENCES ALL 2

Differences	Series	Next
constant +6	5, 11, 17, 23, 29	35
increasing by 2: 3, 5, 7, 9	4, 7, 12, 19, 28	39
doubling: 2, 4, 8, 16	3, 5, 9, 17, 33	65
primes: 2, 3, 5, 7, 11	10, 12, 15, 20, 27, 38	51 (+13)
squares: 1, 4, 9, 16	6, 7, 11, 20, 36	61 (+25)
decreasing: -9, -7, -5, -3	100, 91, 84, 79, 76	75 (-1)

WORKED EXAMPLE

Find the next term: 7, 12, 19, 28, 39, ?

Differences 5, 7, 9, 11 increase by 2, so the next difference is 13: 39 + 13 = **52**.



WORKED EXAMPLE: DIFFERENCES GROW BY 2

2.2.2 Multiplication and division series

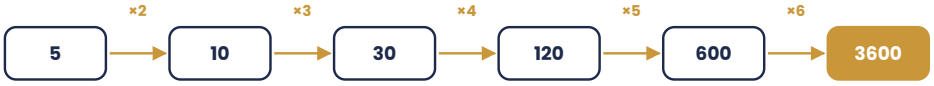
When terms grow or shrink quickly, check **ratios**. The multiplier may be constant (geometric series), increasing (×1, ×2, ×3 …) or alternating (×2, ×3, ×2, ×3).

Rule	Series	Next
× 3	2, 6, 18, 54, 162	486
× 1, × 2, × 3, × 4 …	3, 3, 6, 18, 72	360 (× 5)
× 0.5, × 1, × 1.5, × 2 …	64, 32, 32, 48, 96	240 (× 2.5)
÷ 2	1600, 800, 400, 200	100
× 2, × 3 alternately	1, 2, 6, 12, 36, 72	216

WORKED EXAMPLE

Find the next term: 5, 10, 30, 120, 600, ?

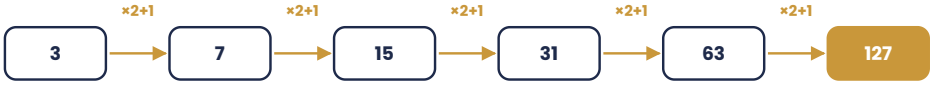
Ratios 2, 3, 4, 5 increase by one, so multiply by 6: 600 × 6 = **3600**.



WORKED EXAMPLE: INCREASING MULTIPLIERS

2.2.3 Mixed operations (× 2 + 1, × 3 - 2)

The most common campus pattern combines a multiplication and an addition: **next = previous × a + b**. Spot it when the ratio is close to a whole number but not exact (7 → 15 is "just over × 2"). The added part may also change: × 1 + 1, × 2 + 2, × 3 + 3 ….



× 2 + 1 AT EVERY STEP

Rule	Series	Next
× 2 + 1	3, 7, 15, 31, 63	127
× 3 - 2	2, 4, 10, 28, 82	244
× 1 + 1, × 2 + 2, × 3 + 3 …	1, 2, 6, 21, 88	445 (× 5 + 5)
× 2 - 1, × 2 - 2, × 2 - 3 …	5, 9, 16, 29, 54	103 (× 2 - 5)

SHORTCUT TRICK

If the ratio is nearly constant, compute **term × ratio** and look at what is left over: 7, 15, 31 → 15 - 7×2 = 1, 31 - 15×2 = 1, so the rule is × 2 + 1. The leftover pattern is the second half of the rule.

WORKED EXAMPLE

Find the next term: 4, 10, 22, 46, 94, ?

Ratio about 2: 4 × 2 = 8 (+2), 10 × 2 = 20 (+2), 22 × 2 = 44 (+2). Rule × 2 + 2: 94 × 2 + 2 = **190**.

2.2.4 Squares and cubes series (n², n³, n² ± 1)

Learn squares to 30 and cubes to 20 by heart; then terms like 120, 168, 288 or 3374 jump out as "a square minus one" or "a cube minus a cube". Typical forms: n², n² ± 1, n² + n = n(n + 1), n³, n³ ± 1, n³ ± n.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n²	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225
n³	1	8	27	64	125	216	343	512	729	1000	1331	1728	2197	2744	3375

Rule	Series	Next
n² - 1	0, 3, 8, 15, 24, 35	48
n² + n	2, 6, 12, 20, 30	42
odd squares	1, 9, 25, 49, 81	121

Rule	Series	Next
$n^3 + 1$	2, 9, 28, 65, 126	217
$n^3 - n$	0, 6, 24, 60, 120	210
n^2 and n^3 alternately	1, 8, 9, 64, 25, 216	49

WORKED EXAMPLE

Find the next term: 3, 8, 15, 24, 35, ?

$3 = 2^2 - 1$, $8 = 3^2 - 1$, $15 = 4^2 - 1$ … so the next is $7^2 - 1 =$ **48**. (Also: differences 5, 7, 9, 11, 13.)

2.2.5 Prime number series

Primes to 100: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97 (25 primes). Series use consecutive primes, alternate primes, primes squared, or prime differences.

Rule	Series	Next
consecutive primes	17, 19, 23, 29, 31	37
alternate primes	2, 5, 11, 17, 23	31
squares of primes	4, 9, 25, 49, 121	169
prime × next prime	6, 15, 35, 77, 143	221 (13 × 17)

2.2.6 Fibonacci-type series

Each term is the **sum of the previous two** (1, 1, 2, 3, 5, 8, 13 …), of the previous three, or a variation such as sum of the previous two plus 1, or their product. Spot it when a term equals two earlier terms added together.

Rule	Series	Next
a + b	2, 3, 5, 8, 13, 21	34
a + b + c	1, 1, 2, 4, 7, 13, 24	44
a + b - 1	3, 4, 6, 9, 14, 22	35
a × b	1, 2, 2, 4, 8, 32	256

WORKED EXAMPLE

Find the next term: 2, 4, 7, 12, 20, 33, ?

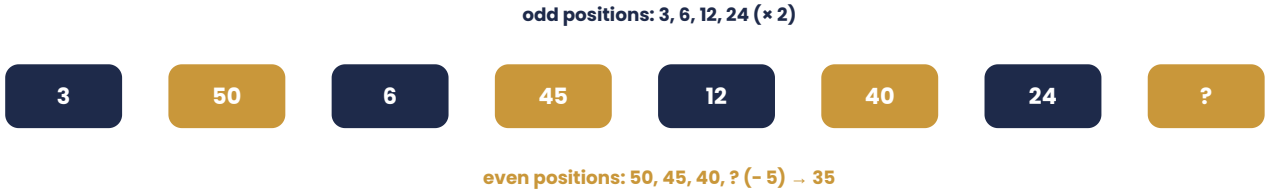
Differences 2, 3, 5, 8, 13 form a Fibonacci series; the next difference is 21: 33 + 21 = **54**.



WORKED EXAMPLE: FIBONACCI DIFFERENCES

2.2.7 Alternate (two interleaved) series

Long series or terms that go up and down are often **two series woven together**: terms 1, 3, 5, 7 follow one rule and terms 2, 4, 6 another. Split them and solve each separately; the answer belongs to whichever series is next.



TWO SERIES INTERLEAVED: THE 8TH TERM BELONGS TO THE EVEN-POSITION SERIES

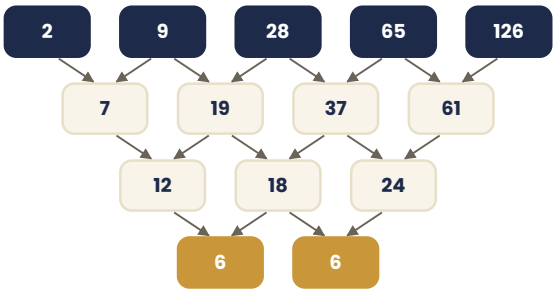
WORKED EXAMPLE

Find the next term: 1, 20, 3, 17, 5, 14, 7, ?

Odd positions 1, 3, 5, 7 (+2); even positions 20, 17, 14 (−3). The 8th term continues the even series: 14 − 3 = **11**.

2.2.8 Second- and third-level difference series

If the first differences show no pattern, take differences of the differences. Constant second differences come from a quadratic rule; constant third differences from a cubic rule (for example n^3). Draw the difference triangle and extend it from the bottom row upwards.



CUBES PLUS ONE: THIRD DIFFERENCES ARE CONSTANT (6)

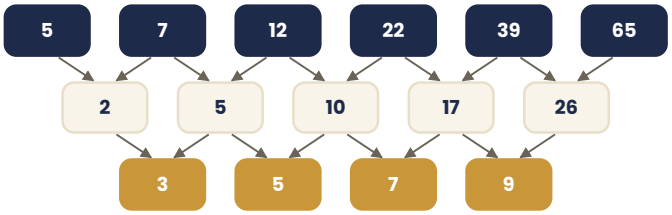
SHORTCUT TRICK

Constant second difference d means the n^2 coefficient is d/2. For 4, 7, 12, 19, 28 the second difference is 2, so the rule starts with n^2: 4 − 1 = 3, 7 − 4 = 3 … the rule is $n^2 + 3$, and the next term is $6^2 + 3 = 39$.

WORKED EXAMPLE

Find the next term: 5, 7, 12, 22, 39, ?

First differences 2, 5, 10, 17 have second differences 3, 5, 7. Next second difference 9, next first difference 26: 39 + 26 = **65**.



WORKED EXAMPLE: SECOND DIFFERENCES 3, 5, 7, 9

2.2.9 Digit-based and factorial series

Some rules use the digits of a term (add the digit sum: 12 → 12 + 3 = 15) or factorials (1, 2, 6, 24, 120, 720), or multiply by the position: 1, 2, 6, 24, 120 is n! (× 2, × 3, × 4, × 5).

Rule	Series	Next
add digit sum	12, 15, 21, 24, 30	33 (30 + 3)
n!	1, 2, 6, 24, 120	720
n! + 1	2, 3, 7, 25, 121	721
alternately +1 and +10	12, 22, 23, 33, 34	44

WORKED EXAMPLE

Find the next term: 1, 2, 4, 7, 28, 33, ?

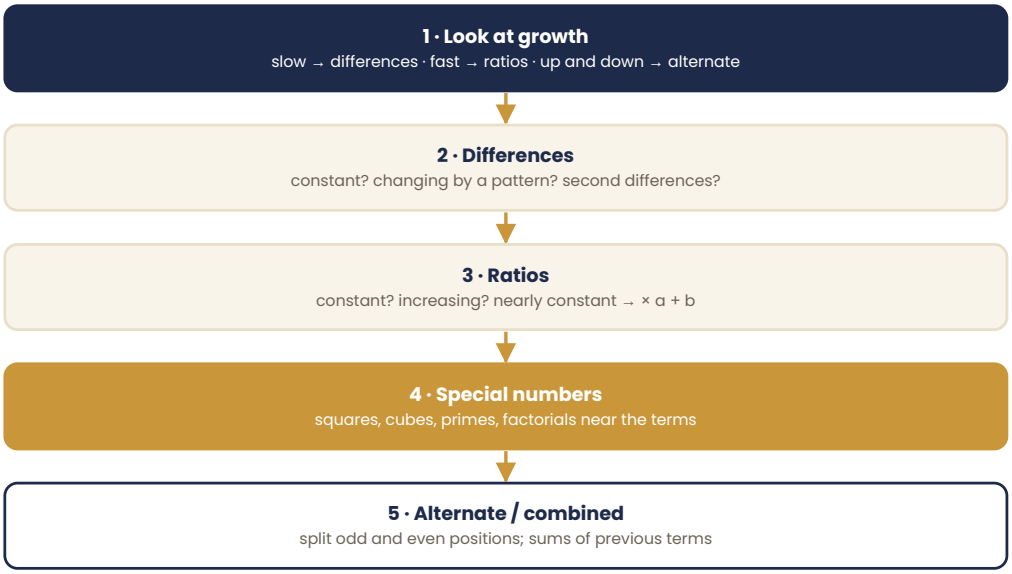
Alternating operations: +1, ×2, +3, ×4, +5, so next is × 6: 33 × 6 = **198**.



ALTERNATING ADD AND MULTIPLY WITH INCREASING NUMBERS

2.3 Method

2.3.1 Check differences, then ratios, then special numbers



A FIXED CHECKING ORDER THAT CRACKS MOST SERIES WITHIN 30 SECONDS

WORKED EXAMPLE

5, 11, 23, 47, 95, ?

Fast growth, ratio about 2: 5 × 2 + 1 = 11, 11 × 2 + 1 = 23 … the next term is 95 × 2 + 1 = **191**.

WORKED EXAMPLE

2, 12, 36, 80, 150, ?

Fast but not geometric. Near cubes? 1 + 1, 8 + 4, 27 + 9, 64 + 16, 125 + 25: $n^3 + n^2$. Next: 216 + 36 = **252**.

SHORTCUT TRICK

Check the **last digits** and parity of the options: if the rule produces only even numbers, most options fall away instantly. For "missing term" questions, the answer must fit the term **after** the gap too.

2.4 Wrong-number series (bank exam style)

Exactly one term is wrong. Find the rule from the **majority** of terms, then regenerate the series and compare term by term. A wrong term spoils **two** differences (before and after it), so look for a single break in an otherwise clean pattern.



expected differences +4, +6, +8, +10, +12 → 21 should be 20 ($n^2 + n$)

ONE WRONG TERM BREAKS TWO NEIGHBOURING DIFFERENCES

SHORTCUT TRICK
Test the rule on the **first three and last two** terms: the wrong term is rarely the first one, and confirming the rule at both ends pins it down.

2.5 Practice questions

2.5.1 Basic level

Q1 7, 12, 17, 22, 27, ?
(a) 30 (b) 31 (c) 32 (d) 33

Q2 4, 8, 16, 32, 64, ?
(a) 96 (b) 128 (c) 112 (d) 256

Q3 2, 5, 10, 17, 26, ?
(a) 35 (b) 36 (c) 37 (d) 38

Q4 1, 8, 27, 64, ?, 216
(a) 100 (b) 120 (c) 125 (d) 128

Q5 3, 5, 7, 11, 13, ?
(a) 15 (b) 17 (c) 19 (d) 16

Q6 1, 1, 2, 3, 5, 8, ?
(a) 11 (b) 12 (c) 13 (d) 14

Q7 120, 110, 101, 93, 86, ?
(a) 78 (b) 79 (c) 80 (d) 81

Q8 2, 6, 18, 54, ?
(a) 108 (b) 162 (c) 148 (d) 216

2.5.2 Intermediate level

Q9 5, 11, 23, 47, 95, ?
(a) 190 (b) 191 (c) 192 (d) 189

Q10 3, 4, 10, 33, 136, ?
(a) 681 (b) 685 (c) 545 (d) 680

Q11 0, 6, 24, 60, 120, ?
(a) 180 (b) 200 (c) 210 (d) 216

Q12 2, 3, 8, 27, 112, ?
(a) 560 (b) 565 (c) 555 (d) 448

Q13 4, 6, 12, 14, 28, 30, ?
(a) 32 (b) 60 (c) 58 (d) 62

Q14 6, 11, 21, 36, 56, ?

(a) 76 (b) 81 (c) 86 (d) 91

Q15 2, 12, 36, 80, 150, ?

(a) 216 (b) 242 (c) 252 (d) 258

Q16 9, 50, 16, 45, 23, 40, ?

(a) 30 (b) 35 (c) 28 (d) 33

2.5.3 Advanced level

Q17 1, 2, 6, 21, 88, ?

(a) 440 (b) 445 (c) 450 (d) 444

Q18 4, 6, 9, 13.5, 20.25, ?

(a) 30.375 (b) 27 (c) 30 (d) 31.5

Q19 3, 10, 29, 66, 127, ?

(a) 216 (b) 218 (c) 217 (d) 220

Q20 7, 9, 13, 21, 37, ?

(a) 53 (b) 69 (c) 67 (d) 73

Q21 Find the wrong term: 3, 7, 15, 31, 63, 128, 255

(a) 15 (b) 31 (c) 128 (d) 255

Q22 Find the wrong term: 1, 4, 9, 16, 23, 36, 49

(a) 16 (b) 23 (c) 36 (d) 49

Q23 Find the wrong term: 5, 6, 14, 41, 105, 232

(a) 6 (b) 14 (c) 41 (d) 232

Q24 1, 3, 4, 7, 11, 18, 29, ?

(a) 40 (b) 45 (c) 47 (d) 48

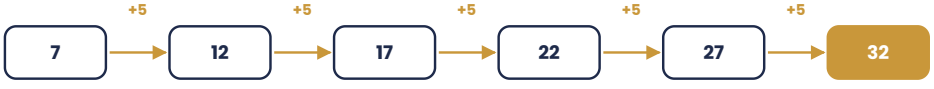
Q25 240, 120, 40, 10, 2, ?

(a) 1 (b) 0.5 (c) 1/3 (d) 0

2.6 Solutions

Q1 **Answer: (c) 32**

Constant difference +5: 27 + 5 = **32**.

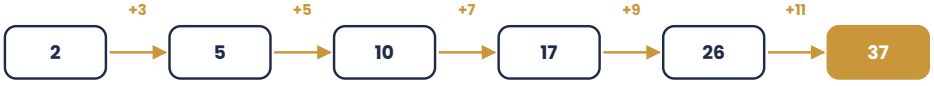


Q2 **Answer: (b) 128**

Each term × 2: 64 × 2 = **128**.

Q3 **Answer: (c) 37**

Differences 3, 5, 7, 9 (odd numbers); next 11: 26 + 11 = **37** (the terms are $n^2 + 1$).



Q4 **Answer: (c) 125**

Cubes 1^3 … 6^3: the missing term is 5^3 = **125**.

Q5 **Answer: (b) 17**

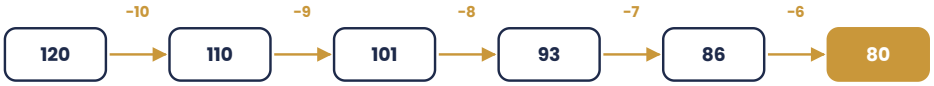
Consecutive primes: 3, 5, 7, 11, 13, **17**.

Q6 **Answer: (c) 13**

Fibonacci: 5 + 8 = **13**.

Q7 **Answer: (c) 80**

Differences -10, -9, -8, -7; next -6: 86 - 6 = **80**.

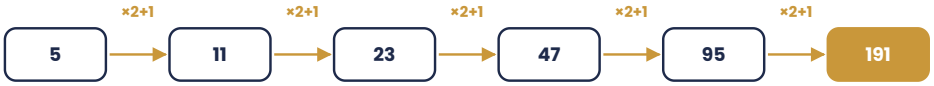


Q8 **Answer: (b) 162**

Each term × 3: 54 × 3 = **162**.

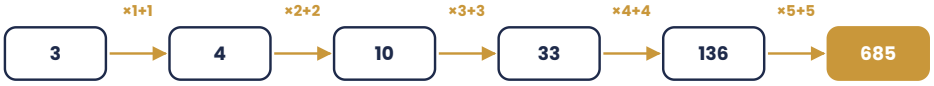
Q9 **Answer: (b) 191**

Rule × 2 + 1: 95 × 2 + 1 = **191**.



Q10 **Answer: (b) 685**

Rule × 1 + 1, × 2 + 2, × 3 + 3, × 4 + 4, so next = 136 × 5 + 5 = **685**.

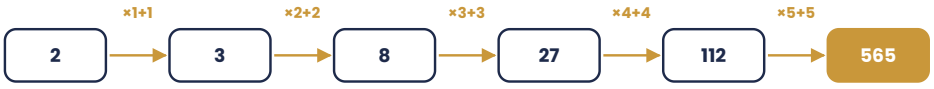


Q11 **Answer: (c) 210**

Terms are $n^3 - n$; next 216 − 6 = **210**.

Q12 **Answer: (b) 565**

Rule × 1 + 1, × 2 + 2, × 3 + 3, × 4 + 4; next 112 × 5 + 5 = **565**.



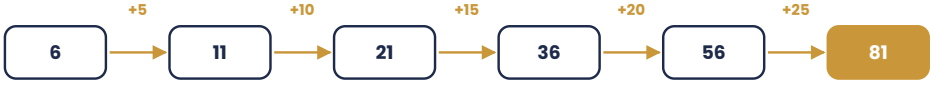
Q13 **Answer: (b) 60**

Alternating + 2 and × 2: 30 × 2 = **60**.



Q14 **Answer: (b) 81**

Differences 5, 10, 15, 20; next 25: 56 + 25 = **81**.



Q15 **Answer: (c) 252**

Terms are $n^3 + n^2$; next 216 + 36 = **252**.

Q16 **Answer: (a) 30**

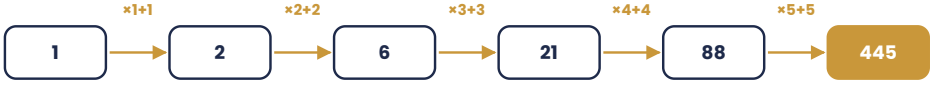
Odd positions 9, 16, 23 (+7) → **30**; even positions 50, 45, 40 (-5). The 7th term is in the odd series.



gold terms: +7 each; white terms: -5 each

Q17 **Answer: (b) 445**

Rule × 1 + 1, × 2 + 2, × 3 + 3, × 4 + 4; next 88 × 5 + 5 = **445**.



Q18 **Answer: (a) 30.375**

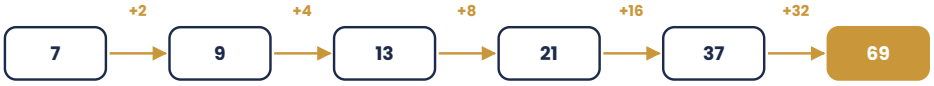
Each term × 1.5: 20.25 × 1.5 = **30.375**.

Q19 **Answer: (b) 218**

Terms are $n^3 + 2$; next 216 + 2 = **218**.

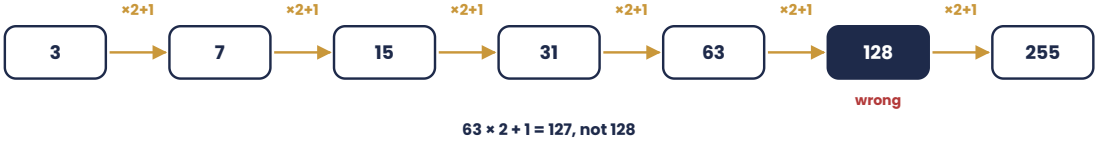
Q20 **Answer: (b) 69**

Differences 2, 4, 8, 16; next 32: 37 + 32 = **69**.



Q21 **Answer: (c) 128**

Rule × 2 + 1: 63 × 2 + 1 = 127, and 127 × 2 + 1 = 255. So **128** is wrong.

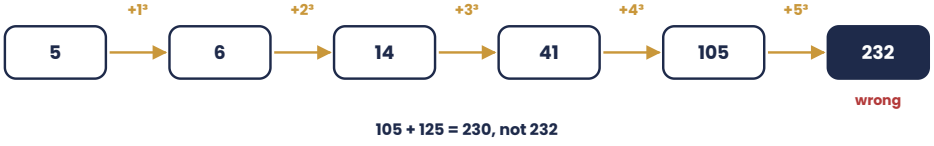


Q22 **Answer: (b) 23**

Squares 1^2 to 7^2: the fifth term should be 25, so **23** is wrong.

Q23 **Answer: (d) 232**

Differences should be cubes: 5 + 1 = 6, 6 + 8 = 14, 14 + 27 = 41, 41 + 64 = 105, 105 + 125 = 230. So **232** is wrong.

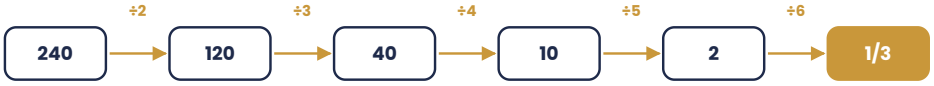


Q24 **Answer: (c) 47**

Each term is the sum of the previous two: 18 + 29 = **47**.

Q25 **Answer: (c) 1/3**

Divide by 2, 3, 4, 5, then 6: 2 ÷ 6 = **1/3**.



Find the rule in a fixed order: growth → differences (and second differences) → ratios (and × a + b) → squares, cubes, primes, factorials → alternate positions and sums of earlier terms. Memorise squares to 30, cubes to 20 and primes to 100. For a missing term the rule must fit both neighbours; for a wrong term, fix the rule from the majority and regenerate the series.

MODULE 03 · BEGINNER

Letter and alphanumeric series

Minimum duration **1 day** | Level **Beginner**

Letter series work exactly like number series once every letter is turned into its position number. This module builds the alphabet tools you need (forward and reverse positions, EJOTY, opposite pairs, wrap-around), the common letter patterns (skips, clusters, alternating and repeating blanks), the mixed letter-number-symbol series, and the arrangement questions that ask for positions and counts in a long string, followed by 25 solved questions.

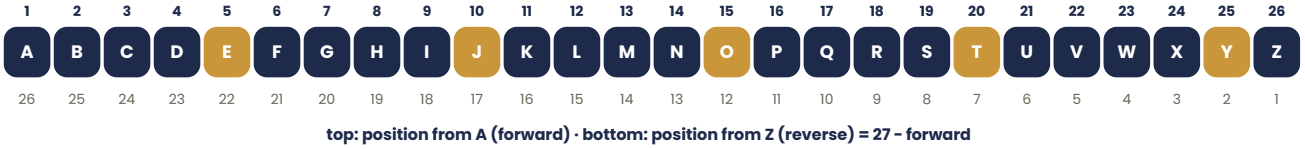
IN THIS MODULE

3.1 Alphabet tools
3.2 Letter series patterns
3.3 Alphanumeric arrangements
3.4 Shortcut tricks
3.5 Practice questions
3.6 Solutions

3.1 Alphabet tools

3.1.1 Forward and reverse positions

Every letter question becomes a number question once you know the **position** of each letter. Forward position counts from A = 1 to Z = 26. Reverse position counts from Z = 1, and the two always add to **27**.



THE ALPHABET WITH FORWARD AND REVERSE POSITIONS (EJOTY IN GOLD)

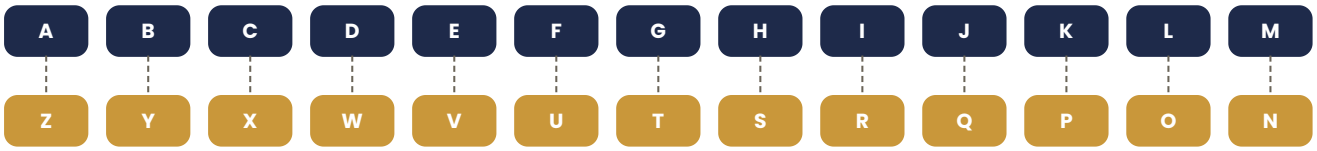
SHORTCUT TRICK

Remember **EJOTY** = 5, 10, 15, 20, 25. Any letter is at most two steps from one of them: R = T - 2 = 18, M = O - 2 = 13, W = Y - 2 = 23. Reverse position = 27 - forward: the reverse position of G (7) is 20.

Need	Rule	Example
k-th letter from the left	letter at position k	14th from left = N
k-th letter from the right	letter at position 27 - k	10th from right = Q (17)
m-th to the right of the k-th from left	position k + m	7th right of 14th = U (21)
m-th to the left of the k-th from right	position (27 - k) - m	4th left of 8th from right = O (19 - 4 = 15)
letter midway between X and Y	(P(X) + P(Y)) / 2	between G and Q → (7 + 17)/2 = 12 = L

3.1.2 Opposite letters and wrap-around

Two letters are **opposites** when their positions add to 27: A–Z, B–Y, C–X … M–N. Coding questions use this constantly. When a series runs past Z it **wraps** back to A: position 27 is A, 28 is B (subtract 26).



THE 13 OPPOSITE PAIRS: EACH COLUMN ADDS TO 27

SHORTCUT TRICK
Mnemonics for opposites: **AZ**, **BY** (by), **CX**, **DW** (Dr. Watson), **EV** (electric vehicle), **FU**, **GT**, **HS** (high school), **IR** (Indian Railways), **JQ**, **KP**, **LO** (love), **MN**.

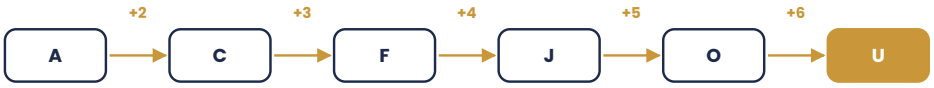
WATCH OUT
When you count "k-th to the right of" a letter, add k; do not add k - 1. The 3rd letter to the right of D is G (4 + 3), not F.

3.2 Letter series patterns

3.2.1 Skip series (constant and changing gaps)

Convert letters to positions and treat the series as a number series: constant gaps, increasing gaps, decreasing gaps, or alternate series.

WORKED EXAMPLE
A, C, F, J, O, ?
Positions 1, 3, 6, 10, 15: gaps +2, +3, +4, +5, so the next gap is +6 → 21 = **U**.



INCREASING GAPS IN A LETTER SERIES

WORKED EXAMPLE
Q, S, V, Z, E, ?
Gaps +2, +3, +4, +5 (Z = 26, 26 + 5 = 31 → 31 - 26 = 5 = E), then +6 → **K**. The series wraps past Z.

3.2.2 Letter clusters (groups of two or three letters)

In cluster series each position inside the group follows its own rule. Write the clusters one under another and read **down each column**.

Cluster	1st letter	2nd letter	3rd letter
BDF	B	D	F

Cluster	1st letter	2nd letter	3rd letter
HJL	H (+6)	J (+6)	L (+6)
NPR	N (+6)	P (+6)	R (+6)
TVX	**T**	**V**	**X**

WORKED EXAMPLE

AZ, BY, CX, DW, ?

First letters go forward (A, B, C, D → E); second letters go backward (Z, Y, X, W → V); each cluster is an opposite pair. Answer **EV**.

3.2.3 Alternating and mixed series

If the gaps do not settle, split the series into odd and even positions. A, Z, C, X, E, V, ? is A, C, E, **G** (forward by 2) interleaved with Z, X, V (backward by 2).



gold: A, C, E, G (+2) white: Z, X, V (−2)

TWO LETTER SERIES INTERLEAVED

3.2.4 Repeating patterns with blanks

Questions like **aab_aa_ba_bb** hide a short repeating unit. Find the unit length (try 2, 3, 4, 5), write the string in blocks of that length, and the blanks become obvious.

Given	Unit	Blocks	Blanks
aab_aa_ba_bb	aabb	aab_ | aa_b | a_bb	b, b, a
ab_ab_ab_	abc	ab_ | ab_ | ab_	c, c, c
_bc_ca_ab	abc	_bc | _ca | _ab (rotating)	a, b, c

TIP

Count the given letters: if the string has 12 places and the unit is 4 letters, there are exactly 3 blocks. The number of a's and b's in the full string must split evenly across blocks.

3.3 Alphanumeric arrangements

3.3.1 Mixed letter-number series

Series such as **KM5, IP8, GS11, EV14, ?** have one rule per character position: K, I, G, E (−2) → C; M, P, S, V (+3) → Y; 5, 8, 11, 14 (+3) → 17. Answer **CY17**.

Term	Letter 1 (−2)	Letter 2 (+3)	Number (+3)
KM5	K	M	5

Term	Letter 1 (-2)	Letter 2 (+3)	Number (+3)
IP8	I	P	8
GS11	G	S	11
EV14	E	V	14
CY17	**C**	**Y**	**17**

3.3.2 Position and counting questions in a long string

Bank exams give a long arrangement of letters, digits and symbols and ask about positions and patterns. We use this arrangement for the examples and for questions 18 to 22:



navy = letters · gold = digits · white = symbols · small numbers = position from the left

THE PRACTICE ARRANGEMENT (27 ELEMENTS)

Question type	How to answer
k-th from the left / right	From the right: position = (total + 1) - k. Here total = 27, so 10th from the right = 18th from the left.
m-th to the right of the k-th from left	position k + m (to the left: k - m)
"How many X are immediately preceded by Y and followed by Z?"	Scan once, looking only at each X; check its left and right neighbour. Remember that the first element has no left neighbour and the last has no right neighbour.
"If all symbols are dropped …"	Rewrite the string without them first, then count.

WATCH OUT

"Immediately preceded by" means the element just to the **left**; "immediately followed by" means the element just to the **right**. Mixing them up is the most common error in these questions.

3.4 Shortcut tricks

- Write the alphabet with positions in the margin before you start; it costs 15 seconds and saves many mistakes.
- Use EJOTY anchors (5, 10, 15, 20, 25) and opposites (sum 27) instead of counting on fingers.
- For positions beyond 26, subtract 26 (wrap-around); for positions below 1, add 26.
- Letters at positions 3, 6, 9 … are **CFILORUX**; at positions 4, 8, 12 … are **DHLPTX**; vowels are at 1, 5, 9, 15, 21.
- For clusters, read each column separately; for long strings, mark the type of each element (letter, digit, symbol) above it once.

3.5 Practice questions

3.5.1 Basic level

Q1 A, C, F, J, O, ?

(a) T (b) U (c) V (d) W

Q2 Z, X, V, T, R, ?

(a) Q (b) O (c) P (d) N

Q3 B, E, H, K, N, ?

(a) P (b) Q (c) R (d) O

Q4 AZ, BY, CX, DW, ?

(a) EU (b) FV (c) EV (d) VE

Q5 Which letter is 7th to the right of the 14th letter from the left in the English alphabet?

(a) T (b) U (c) V (d) S

Q6 Which letter is 10th from the right end of the English alphabet?

(a) P (b) Q (c) R (d) J

Q7 AB, DE, GH, JK, ?

(a) LM (b) MN (c) NO (d) KL

Q8 If every letter of FAN is replaced by its opposite letter (A ↔ Z, B ↔ Y ...), what is the result?

(a) UZM (b) UZN (c) VZM (d) TZM

Q9 ACE, BDF, CEG, ?

(a) DFG (b) DEH (c) DFH (d) EGI

3.5.2 Intermediate level

Q10 C, G, L, R, ?

(a) X (b) Y (c) Z (d) W

Q11 KM5, IP8, GS11, EV14, ?

(a) CY17 (b) BY17 (c) CX17 (d) CY18

Q12 A, Z, C, X, E, V, ?

(a) F (b) G (c) H (d) U

Q13 Fill the blanks: aab_aa_ba_bb

(a) bba (b) bab (c) abb (d) bbb

Q14 BDF, HJL, NPR, ?

(a) TVW (b) SUW (c) TVX (d) UWY

Q15 2Z5, 7Y7, 14X9, 23W11, ?

(a) 34V13 (b) 32V13 (c) 34U13 (d) 34V12

Q16 Which letter is exactly midway between G and Q in the English alphabet?

(a) K **(b)** L **(c)** M **(d)** J

Q17 If the first half of the alphabet (A to M) is written in reverse order, which letter will be 5th to the left of the 16th letter from the left?

(a) K **(b)** C **(c)** D **(d)** B

3.5.3 Advanced level

Q18 In the arrangement R4E%K7#BH3@MQ9P$1TA5&U8WD2*, which element is 8th to the right of the 15th element from the left?

(a) 1 **(b)** 8 **(c)** U **(d)** 5

Q19 In the same arrangement, how many digits are immediately preceded by a consonant and immediately followed by a symbol?

(a) 2 **(b)** 3 **(c)** 4 **(d)** 1

Q20 In the same arrangement, how many symbols are immediately preceded by a digit?

(a) 3 **(b)** 4 **(c)** 5 **(d)** 2

Q21 If all the symbols are removed from the arrangement, which element will be 10th from the right?

(a) 1 **(b)** T **(c)** P **(d)** 9

Q22 In the arrangement, which element is exactly midway between the 6th element from the left and the 10th element from the right?

(a) Q **(b)** M **(c)** @ **(d)** 9

Q23 Q, S, V, Z, E, ?

(a) J **(b)** K **(c)** L **(d)** I

Q24 AYBZC, DWEXF, GUHVI, ?

(a) JSKTL **(b)** JTKSL **(c)** JSKUL **(d)** KSLTM

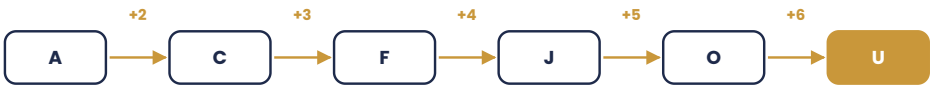
Q25 Find the wrong term: C, F, J, O, T, B

(a) F **(b)** J **(c)** T **(d)** B

3.6 Solutions

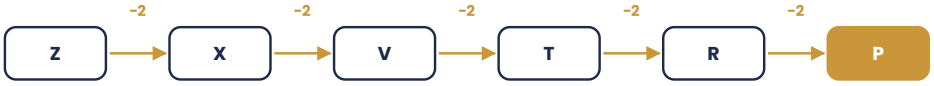
Q1 **Answer: (b) U**

Gaps +2, +3, +4, +5; next +6: O (15) + 6 = 21 = **U**.



Q2 **Answer: (c) P**

Each step −2: R (18) − 2 = 16 = **P**.



Q3 **Answer: (b) Q**

Each step +3: N (14) + 3 = 17 = **Q**.

Q4 **Answer: (c) EV**

First letters A, B, C, D, E; second letters Z, Y, X, W, V. Answer **EV** (each pair is an opposite pair).

Q5 **Answer: (b) U**

14 + 7 = 21, so the letter is **U**.

Q6 **Answer: (b) Q**

10th from the right = (27 - 10) = 17th from the left = **Q**.

Q7 **Answer: (b) MN**

Each pair starts 3 letters after the previous start (A, D, G, J, M): **MN**.

Q8 **Answer: (a) UZM**

F (6) → 21 = U; A → Z; N (14) → 13 = M. Answer **UZM**.

Q9 **Answer: (c) DFH**

Each letter of the cluster moves forward by 1: **DFH**.

Q10 **Answer: (b) Y**

Gaps +4, +5, +6, next +7: R (18) + 7 = 25 = **Y**.



Q11 **Answer: (a) CY17**

First letters -2 (K, I, G, E → C), second letters +3 (M, P, S, V → Y), numbers +3 (→ 17). Answer **CY17**.

Q12 **Answer: (b) G**

Odd positions A, C, E (+2) → **G**; even positions Z, X, V (-2).



Q13 **Answer: (a) bba**

The unit is "aabb": aab**b** | aa**b**b | a**a**bb. Blanks in order: **b, b, a**.

Q14 **Answer: (c) TVX**

Each letter of the cluster moves +6 from one term to the next: N → T, P → V, R → X. Answer **TVX**.

Q15 **Answer: (a) 34V13**

First numbers 2, 7, 14, 23 (+5, +7, +9, next +11) → 34; letters Z, Y, X, W → V; last numbers 5, 7, 9, 11 → 13. Answer **34V13**.

Q16 **Answer: (b) L**

(7 + 17) ÷ 2 = 12 = **L**.

Q17 **Answer: (b) C**

Positions 1 to 13 now hold M, L, K … A; positions 14 to 26 are unchanged. 5th to the left of the 16th is position 11. In the reversed half, position k holds the letter at 14 - k: 14 - 11 = 3 = **C**.

Q18 **Answer: (b) 8**

15 + 8 = 23rd from the left, which is **8**.

Q19 **Answer: (b) 3**

Check every digit: K**7#**, H**3@**, D**2*** qualify (4 has R before but E after; 9 has P after; 1 has $ before; 5 has A, a vowel, before; 8 has W after). Count = **3**.

Q20 **Answer: (b) 4**

7**#**, 3**@**, 5**&** and 2***** (the last element counts too). % follows E and $ follows P. Count = **4**.

Q21 **Answer: (c) P**

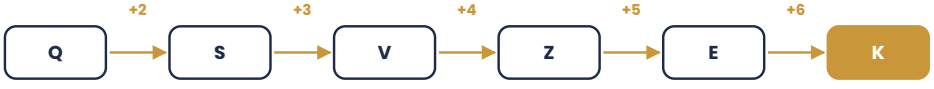
Without symbols: R4EK7BH3MQ9P1TA5U8WD2 (21 elements). 10th from the right = 12th from the left = **P**.

Q22 **Answer: (b) M**

6th from the left is position 6 (7); 10th from the right is position 28 - 10 = 18 (T). Midway = (6 + 18) ÷ 2 = 12th = **M**.

Q23 **Answer: (b) K**

Gaps +2, +3, +4, +5 (Z → E wraps: 26 + 5 = 31 - 26 = 5), next +6: E (5) + 6 = 11 = **K**.

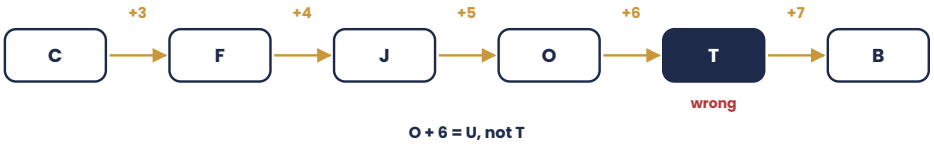


Q24 **Answer: (a) JSKTL**

Letters 1, 3, 5 run A B C, D E F, G H I, J K L. Letters 2 and 4 run Y, W, U, S and Z, X, V, T. Answer **JSKTL**.

Q25 **Answer: (c) T**

Gaps should be +3, +4, +5, +6, +7: C, F, J, O, **U**, B (21 + 7 = 28 → 2 = B). So **T** is wrong.



Turn letters into positions (A = 1 … Z = 26, reverse = 27 - forward) and every letter series becomes a number series. Use EJOTY and opposite pairs, wrap past Z by subtracting 26, read clusters column by column, and in long arrangements mark each element's type once before counting.

MODULE 04 · BEGINNER

Analogy

Minimum duration **1 day** | Level **Beginner**

An analogy question gives one related pair and asks you to complete a second pair with the same relationship (A : B :: C : ?). This module lists the relationship types used in exams (worker and tool, product and raw material, country and currency, animal and young or home, part and whole, synonyms and antonyms, instrument and measure), then number analogies (squares, cubes, $n^2 \pm 1$, $n(n + 1)$, digit sums) and letter analogies (shifts, opposites and reversal), followed by 25 solved questions.

IN THIS MODULE

4.1	Basics	4.2	Word analogies
4.3	Number analogies	4.4	Letter analogies
4.5	Practice questions	4.6	Solutions

4.1 Basics

4.1.1 Reading A : B :: C : ?

Read the notation as "A is to B as C is to ?". The method has two steps: **(1) state the relationship between A and B in one exact sentence**, then **(2) apply the same sentence to C**. The more exact the sentence, the fewer options survive.



A PAINTER USES A BRUSH; A WRITER USES A PEN

WORKED EXAMPLE

Doctor : Hospital :: Teacher : ?

Vague sentence: "a doctor is related to a hospital" fits almost anything. Exact sentence: "a doctor **works in** a hospital". Apply to teacher: a teacher works in a **school**.

WATCH OUT

The relationship must also keep its **direction**. Calf : Cow is "young : adult"; if the second pair is written Dog : ?, the answer is the adult-to-young direction only if the first pair was adult : young. Always check which side is which.

4.1.2 Question formats

Format	Example	What to do
Complete the pair	Japan : Yen :: India : ?	Find B for C

Format	Example	What to do
Choose the similar pair	Mason : Wall → Author : Book	Test the sentence on each option
Find the odd pair	Pick the pair that does not share the relation	Test all, one fails
Double analogy	? : Brush :: Writer : ?	Two blanks; test both together

4.2 Word analogies

4.2.1 Common relationship types

Relationship	Example pair	Another pair
Worker : Tool	Carpenter : Saw	Surgeon : Scalpel
Worker : Product	Mason : Wall	Author : Book
Worker : Workplace	Teacher : School	Chef : Kitchen
Raw material : Product	Cotton : Cloth	Clay : Brick
Country : Currency	Japan : Yen	Bangladesh : Taka
Country : Capital	France : Paris	Kenya : Nairobi
Animal : Young	Cow : Calf	Dog : Puppy
Animal : Home	Bee : Hive	Lion : Den
Animal : Sound	Lion : Roar	Horse : Neigh
Part : Whole	Petal : Flower	Page : Book
Instrument : Measures	Thermometer : Temperature	Barometer : Pressure
Unit : Quantity	Kilogram : Mass	Litre : Volume
Synonym	Brave : Courageous	Happy : Glad
Antonym	Hot : Cold	Early : Late
Cause : Effect	Fire : Smoke	Rain : Flood
Degree (less : more)	Warm : Hot	Drizzle : Downpour

> **TIP**
> When two options both fit, make the sentence sharper. “Bird : Nest :: Bee : ?” → “a bird **builds and lives in** a nest” picks **hive** over **flower**.

4.2.2 Animal, currency and instrument lists to know

Animal	Young	Home	Sound
Cat	Kitten	—	Mew

Animal	Young	Home	Sound
Dog	Puppy	Kennel	Bark
Lion	Cub	Den	Roar
Horse	Foal	Stable	Neigh
Bee	Larva	Hive	Buzz
Bird	Chick	Nest	Chirp
Spider	Spiderling	Web	—
Rabbit	Kit	Burrow	Squeak

Country	Currency	Country	Currency
India	Rupee	Japan	Yen
Bangladesh	Taka	China	Yuan
UK	Pound sterling	Russia	Rouble
Kuwait	Dinar	South Africa	Rand
Thailand	Baht	Saudi Arabia	Riyal

4.3 Number analogies

4.3.1 Standard number relations

For numbers, test these relations in order until one fits the first pair exactly: multiple, square, cube, $n^2 \pm k$, $n^3 \pm k$, $n(n+1)$, digit sum or product, reversed digits.

Relation	First pair	Apply to
$n \to n^2$	3 : 9	7 : 49
$n \to n^3$	4 : 64	5 : 125
$n \to n^2 + 1$	6 : 37	8 : 65
$n \to n(n+1)$	7 : 56	9 : 90
$n^2 \to n^3$	25 : 125	36 : 216
$n^3 \to (n+1)^3 + 1$	8 : 28	27 : 65
digit sum	12 : 3	45 : 9



NUMBER ANALOGY: SQUARE AND ADD 1

SHORTCUT TRICK

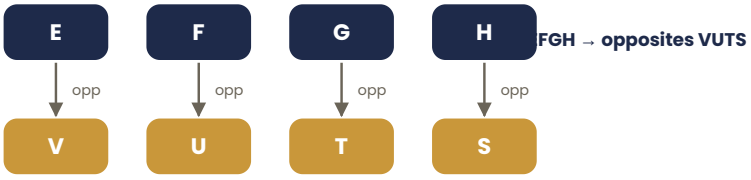
Check that the rule works **only one way**. 3 : 9 could be ×3 or n^2, which gives 7 : 21 or 7 : 49. If both appear in the options, look for a second clue (other given pairs) or prefer the rule that uses the number itself (n^2) over a fixed multiplier.

4.4 Letter analogies

4.4.1 Shifts, opposites and reversal

Write the positions of the letters of A and B, find what happened to each letter (+k, -k, opposite, reversed order), then do the same to C.

Pair	What happened	Apply
CAT : DBU	each letter +1	DOG → EPH
ABC : ZYX	each letter → its opposite	DEF → WVU
ABCD : WXYZ	opposites, then order reversed	EFGH → STUV
ACEG : DFHJ	each letter +3	QSUW → TVXZ
MOTHER : NPUIFS	each letter +1	FATHER → GBUIFS



ABCD : WXYZ APPLIED TO EFGH: OPPOSITES, THEN REVERSE THE ORDER

4.5 Practice questions

4.5.1 Basic level

Q1 Painter : Brush :: Writer : ?

(a) Book (b) Pen (c) Paper (d) Ink

Q2 Japan : Yen :: India : ?

(a) Dollar (b) Taka (c) Rupee (d) Dinar

Q3 Cow : Calf :: Dog : ?

(a) Kitten (b) Puppy (c) Cub (d) Foal

Q4 Bird : Nest :: Bee : ?

(a) Hive (b) Den (c) Burrow (d) Stable

Q5 3 : 9 :: 7 : ?

(a) 14 (b) 21 (c) 49 (d) 63

Q6 4 : 64 :: 5 : ?

(a) 25 (b) 100 (c) 125 (d) 150

Q7 ABC : ZYX :: DEF : ?

(a) UVW (b) WVU (c) VWX (d) WXV

Q8 Hot : Cold :: Early : ?

(a) Soon (b) Quick (c) Late (d) Fast

Q9 Cotton : Cloth :: Clay : ?

(a) Brick (b) Sand (c) Earth (d) Mud

4.5.2 Intermediate level

Q10 6 : 37 :: 8 : ?

(a) 63 (b) 64 (c) 65 (d) 72

Q11 7 : 56 :: 9 : ?

(a) 72 (b) 81 (c) 90 (d) 99

Q12 CAT : DBU :: DOG : ?

(a) EPH (b) EPG (c) FPH (d) DPH

Q13 ABCD : WXYZ :: EFGH : ?

(a) VUTS (b) STUV (c) TUVW (d) RSTU

Q14 12 : 3 :: 45 : ?

(a) 9 (b) 11 (c) 5 (d) 20

Q15 Thermometer : Temperature :: Barometer : ?

(a) Rain (b) Pressure (c) Height (d) Humidity

Q16 Kilogram : Mass :: Litre : ?

(a) Length (b) Weight (c) Volume (d) Density

Q17 25 : 125 :: 36 : ?

(a) 196 (b) 216 (c) 343 (d) 180

4.5.3 Advanced level

Q18 8 : 28 :: 27 : ?

(a) 64 (b) 65 (c) 82 (d) 126

Q19 11 : 122 :: 13 : ?

(a) 169 (b) 170 (c) 171 (d) 196

Q20 ACEG : DFHJ :: QSUW : ?

(a) TVXZ (b) RTVX (c) SUWY (d) TVYZ

Q21 MOTHER : NPUIFS :: FATHER : ?

(a) GBUIFS (b) GBUJFS (c) GAUIFS (d) EBUIFS

Q22 Choose the pair that has the same relationship as Mason : Wall

(a) Chef : Kitchen (b) Author : Book
(c) Doctor : Hospital (d) Farmer : Field

Q23 2 : 9 :: 3 : ?

(a) 27 (b) 28 (c) 26 (d) 30

Q24 ZXV : ACE :: YWU : ?

(a) BDF (b) BCE (c) CDF (d) ACF

Q25 Lion : Roar :: Horse : ?

(a) Bray (b) Neigh (c) Bleat (d) Grunt

4.6 Solutions

Q1 **Answer: (b) Pen**

A painter **uses** a brush as the tool of his work; a writer uses a **pen**. Book and paper are products and materials, not tools.



Q2 **Answer: (c) Rupee**

Country : currency. The currency of India is the **rupee**.

Q3 **Answer: (b) Puppy**

Adult : young. The young of a dog is a **puppy**.

Q4 **Answer: (a) Hive**

Animal : home it builds. A bee lives in a **hive**.

Q5 **Answer: (c) 49**

$9 = 3^2$; so $7^2 =$ **49**. (×3 would give 21, but squaring is the rule that uses the number itself.)

Q6 **Answer: (c) 125**

$64 = 4^3$; $5^3 =$ **125**.

Q7 **Answer: (b) WVU**

Each letter is replaced by its opposite (A–Z, B–Y, C–X). D–W, E–V, F–U → **WVU**.

Q8 **Answer: (c) Late**

Antonyms. The opposite of early is **late**.

Q9 **Answer: (a) Brick**

Raw material : product made from it. Clay is made into **brick**.

Q10 **Answer: (c) 65**

$37 = 6^2 + 1$, so $8^2 + 1 = \mathbf{65}$.



Q11 **Answer: (c) 90**

$56 = 7 \times 8 = n(n + 1)$, so $9 \times 10 = \mathbf{90}$.

Q12 **Answer: (a) EPH**

Each letter moves +1: C → D, A → B, T → U. So D → E, O → P, G → H: **EPH**.

Q13 **Answer: (b) STUV**

Opposites of A, B, C, D are Z, Y, X, W; written in reverse order they give WXYZ. Opposites of E, F, G, H are V, U, T, S; reversed: **STUV**.

Q14 **Answer: (a) 9**

3 = 1 + 2 (digit sum). 4 + 5 = **9**.

Q15 **Answer: (b) Pressure**

Instrument : what it measures. A barometer measures atmospheric **pressure**.

Q16 **Answer: (c) Volume**

Unit : quantity measured. The litre is a unit of **volume**.

Q17 **Answer: (b) 216**

$25 = 5^2$, $125 = 5^3$. $36 = 6^2$, so $6^3 = \mathbf{216}$.

Q18 **Answer: (b) 65**

$8 = 2^3$ and $28 = 3^3 + 1$ (next cube plus 1). $27 = 3^3$, so $4^3 + 1 = \mathbf{65}$.

Q19 **Answer: (b) 170**

$122 = 11^2 + 1$, so $13^2 + 1 = \mathbf{170}$.

Q20 **Answer: (a) TVXZ**

Each letter +3: A→D, C→F, E→H, G→J. So Q→T, S→V, U→X, W→Z: **TVXZ**.

Q21 **Answer: (a) GBUIFS**

Each letter of MOTHER moves +1. FATHER → **GBUIFS**.

Q22 **Answer: (b) Author : Book**

A mason **builds** a wall (worker : product). An author writes a book (**Author : Book**). The others are worker : workplace.

Q23 **Answer: (b) 28**

$9 = 2^3 + 1$, so $3^3 + 1 =$ **28**.

Q24 **Answer: (a) BDF**

Each letter of ZXV is replaced by its opposite: Z–A, X–C, V–E. Y–B, W–D, U–F → **BDF**.

Q25 **Answer: (b) Neigh**

Animal : sound. A horse **neighs** (bray is a donkey, bleat a goat or sheep, grunt a pig).

State the A : B relationship as one exact sentence that keeps its direction, then apply it to C. For words, learn the standard relation types; for numbers, test squares, cubes, $n^2 \pm 1$, $n(n + 1)$ and digit sums; for letters, write positions and look for shifts, opposites and reversal.

MODULE 05 · BEGINNER

Classification (odd one out)

Minimum duration **1 day** | Level **Beginner**

Classification questions give four or five items of which all but one share a property; you pick the item that does not belong. The skill is finding the property shared by the majority, and checking that exactly one item fails it. This module covers word groups, number properties (primes, squares, cubes, multiples, digit patterns, pair rules) and letter groups (gaps, opposites, vowels), followed by 25 solved questions.

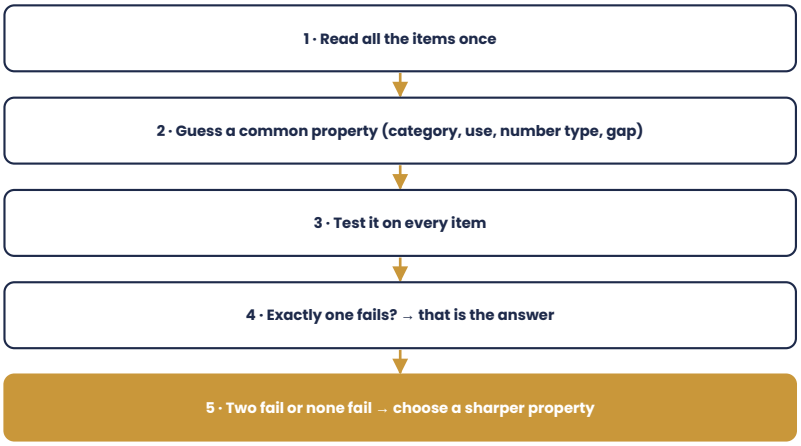
IN THIS MODULE

5.1 Basics
5.2 Word classification
5.3 Number classification
5.4 Letter classification
5.5 Practice questions
5.6 Solutions

5.1 Basics

5.1.1 The method

Do not look for the odd item first. Look for the **property that most items share**, then confirm that exactly one item fails it. If two items fail, your property is wrong; if none fail, it is too weak.



THE FIVE-STEP METHOD FOR ODD ONE OUT



fruits that grow above ground: Potato is a vegetable (a tuber)

ONE ITEM FAILS THE SHARED PROPERTY

WATCH OUT
Prefer the **most specific** common property. In "Rose, Lotus, Lily, Marigold, Neem", all are plants (too weak); four are flowers and Neem is a tree, so Neem is the answer.

5.2 Word classification

5.2.1 Common grouping bases

Basis	Group	Odd one
Category	Apple, Mango, Banana, Grapes	Potato (vegetable)
Sub-category	Guitar, Violin, Sitar, Veena (string)	Flute (wind)
Function	Pen, Pencil, Marker, Crayon (write)	Eraser
Planets and satellites	Mercury, Venus, Mars, Jupiter	Moon (a satellite)
Units	Metre, Kilometre, Centimetre (length)	Litre (volume)
Shapes	Square, Rectangle, Rhombus (4 sides)	Triangle

TIP
For words, ask in order: what kind of thing is it? what is it used for? where is it found? what is it made of? Usually the second or third question separates the odd one.

5.3 Number classification

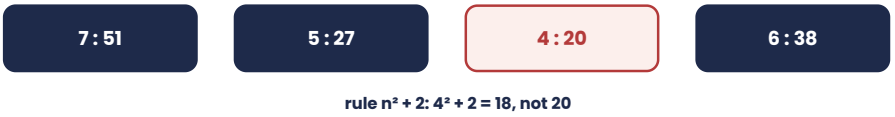
5.3.1 Properties to test

Property	Group	Odd one
Prime	3, 5, 7, 11, 13	9
Perfect square	16, 25, 36, 64	48
Perfect cube	27, 64, 125, 216	144
Multiple of 7	21, 35, 49, 63	72
$n(n + 1)$	132, 156, 182, 210	250
$n^2 + 1$	5, 10, 17, 26	35
Digit rule (middle = first + 1)	341, 562, 783	935

SHORTCUT TRICK
Memorise primes up to 100. The numbers that **look** prime but are not are the usual traps: 51 = 3 × 17, 57 = 3 × 19, 87 = 3 × 29, 91 = 7 × 13.

5.3.2 Number pairs

Pairs like **7 : 51, 5 : 27, 4 : 20, 6 : 38** use one rule from the first number to the second. Find the rule from three pairs and test the fourth: $7^2 + 2 = 51$, $5^2 + 2 = 27$, $6^2 + 2 = 38$, but $4^2 + 2 = 18 \neq 20$, so **4 : 20** is odd.



NUMBER-PAIR CLASSIFICATION

5.4 Letter classification

5.4.1 Gap patterns, opposites and vowels

Write the positions of the letters in each group and compare the **gaps**. Other common bases are opposite pairs (sum 27), vowels versus consonants, and consecutive letters.

Group	Positions	Gaps
BDG	2, 4, 7	+2, +3
FHK	6, 8, 11	+2, +3
MOR	13, 15, 18	+2, +3
HJN	8, 10, 14	**+2, +4** (odd)

> **TIP**
> If the groups are pairs, check the opposite rule first (AZ, BY, CX ...: positions add to 27). If they are three letters, check gaps; if four letters, check whether they are consecutive.

5.5 Practice questions

5.5.1 Basic level

Q1 Find the odd one out: Apple, Mango, Potato, Banana

(a) Apple (b) Mango (c) Potato (d) Banana

Q2 Find the odd one out: Rose, Lotus, Lily, Neem

(a) Rose (b) Lotus (c) Lily (d) Neem

Q3 Find the odd one out: 3, 5, 7, 9, 11

(a) 3 (b) 7 (c) 9 (d) 11

Q4 Find the odd one out: 16, 25, 36, 48, 64

(a) 25 (b) 36 (c) 48 (d) 64

Q5 Find the odd one out: Guitar, Violin, Sitar, Flute

(a) Guitar (b) Violin (c) Sitar (d) Flute

Q6 Find the odd one out: 27, 64, 125, 144, 216

(a) 64 (b) 125 (c) 144 (d) 216

Q7 Find the odd one out: ACE, BDF, GIK, MNP

(a) ACE (b) BDF (c) GIK (d) MNP

Q8 Find the odd one out: Mercury, Venus, Moon, Mars

(a) Mercury (b) Venus (c) Moon (d) Mars

Q9 Find the odd one out: 21, 35, 49, 63, 72

(a) 35 (b) 49 (c) 63 (d) 72

5.5.2 Intermediate level

Q10 Find the odd one out: 341, 562, 783, 935

(a) 341 (b) 562 (c) 783 (d) 935

Q11 Find the odd pair: 7 : 51, 5 : 27, 4 : 20, 6 : 38

(a) 7 : 51 (b) 5 : 27 (c) 4 : 20 (d) 6 : 38

Q12 Find the odd one out: 132, 156, 182, 210, 250

(a) 132 (b) 182 (c) 210 (d) 250

Q13 Find the odd one out: BDG, FHK, MOR, HJN

(a) BDG (b) FHK (c) MOR (d) HJN

Q14 Find the odd one out: AZ, BY, CX, DV

(a) AZ (b) BY (c) CX (d) DV

Q15 Find the odd one out: 17, 19, 23, 27, 29

(a) 17 (b) 19 (c) 27 (d) 29

Q16 Find the odd one out: Kilometre, Metre, Litre, Centimetre

(a) Kilometre (b) Metre (c) Litre (d) Centimetre

Q17 Find the odd pair: 3 – 27, 4 – 64, 5 – 125, 6 – 206

(a) 3 – 27 (b) 4 – 64 (c) 5 – 125 (d) 6 – 206

5.5.3 Advanced level

Q18 Find the odd one out: 5, 10, 17, 26, 35

(a) 10 (b) 17 (c) 26 (d) 35

Q19 Find the odd one out: 1236, 2349, 3418, 4527

(a) 1236 (b) 2349 (c) 3418 (d) 4527

Q20 Find the odd one out: CEG, KMO, SUW, PRU

(a) CEG (b) KMO (c) SUW (d) PRU

Q21 Find the odd one out: EFGH, JKLM, OPQR, TVWX

(a) EFGH (b) JKLM (c) OPQR (d) TVWX

Q22 Find the odd one out: 29, 37, 41, 47, 51

(a) 29 (b) 41 (c) 47 (d) 51

Q23 Find the odd pair: 48 : 6, 63 : 7, 80 : 8, 99 : 10

(a) 48 : 6 (b) 63 : 7 (c) 80 : 8 (d) 99 : 10

Q24 Find the odd one out: BCD, FGH, JKL, NOP

(a) BCD (b) FGH (c) JKL (d) NOP

Q25 Find the odd one out: Square, Rectangle, Rhombus, Triangle

(a) Square (b) Rectangle (c) Rhombus (d) Triangle

5.6 Solutions

Q1 **Answer: (c) Potato**

Apple, mango and banana are fruits; **potato** is a vegetable.

Q2 **Answer: (d) Neem**

Rose, lotus and lily are flowers; **neem** is a tree.

Q3 **Answer: (c) 9**

All are primes except **9** = 3 × 3.

Q4 **Answer: (c) 48**

16, 25, 36, 64 are perfect squares; **48** is not.

Q5 **Answer: (d) Flute**

Guitar, violin and sitar are string instruments; the **flute** is a wind instrument.

Q6 **Answer: (c) 144**

27, 64, 125, 216 are cubes (3^3, 4^3, 5^3, 6^3); **144** = 12^2 is not a cube.

Q7 **Answer: (d) MNP**

ACE, BDF and GIK have gaps +2, +2. **MNP** has gaps +1, +2.

Q8 **Answer: (c) Moon**

Mercury, Venus and Mars are planets; the **Moon** is a satellite of the Earth.

Q9 **Answer: (d) 72**

21, 35, 49, 63 are multiples of 7; **72** is not.

Q10 **Answer: (d) 935**

In 341, 562 and 783 the middle digit is the first digit + 1 (3→4, 5→6, 7→8). In **935** the middle digit is 3, not 10.

Q11 **Answer: (c) 4 : 20**

Rule $n^2 + 2$: 49 + 2 = 51, 25 + 2 = 27, 36 + 2 = 38, but 16 + 2 = 18. So **4 : 20** is odd.

7 : 51	5 : 27	4 : 20	6 : 38

$n^2 + 2$ fails only for 4 : 20

Q12 **Answer: (d) 250**

132 = 11 × 12, 156 = 12 × 13, 182 = 13 × 14, 210 = 14 × 15. **250** is not of the form n(n + 1).

Q13 **Answer: (d) HJN**

BDG, FHK, MOR have gaps +2, +3. **HJN** has gaps +2, +4.

Q14 **Answer: (d) DV**

AZ, BY, CX are opposite pairs (positions add to 27). D + V = 4 + 22 = 26, so **DV** is odd (DW would fit).

Q15 **Answer: (c) 27**

All are prime except **27** = 3^3.

Q16 **Answer: (c) Litre**

Kilometre, metre and centimetre measure length; the **litre** measures volume.

Q17 **Answer: (d) 6 – 206**

Rule n → n^3: 6^3 = 216, not 206. So **6 – 206** is odd.

Q18 **Answer: (d) 35**

5, 10, 17, 26 are $n^2 + 1$ for n = 2, 3, 4, 5. The next would be 37; **35** is odd.

Q19 **Answer: (d) 4527**

In each number the last digit equals the sum of the first three: 1 + 2 + 3 = 6, 2 + 3 + 4 = 9, 3 + 4 + 1 = 8. For **4527**, 4 + 5 + 2 = 11 ≠ 7.

Q20 **Answer: (d) PRU**

CEG, KMO, SUW have gaps +2, +2. **PRU** has gaps +2, +3.

Q21 **Answer: (d) TVWX**

The first three are four consecutive letters. **TVWX** skips U.

Q22 **Answer: (d) 51**

All are prime except **51** = 3 × 17.

Q23 **Answer: (d) 99 : 10**

Rule: first = n(n + 2): 6 × 8 = 48, 7 × 9 = 63, 8 × 10 = 80. For n = 10, 10 × 12 = 120, so **99 : 10** is odd (99 belongs with 9).

Q24 **Answer: (d) NOP**

All are three consecutive letters, but BCD, FGH and JKL contain only consonants. **NOP** contains the vowel O.

Q25 **Answer: (d) Triangle**

Square, rectangle and rhombus have four sides; the **triangle** has three.

Find the property shared by the majority, then confirm exactly one item fails it. If two fail, sharpen the property. For numbers test prime, square, cube, multiple, n(n + 1), $n^2 \pm k$ and digit rules; for letters write positions and compare gaps, opposites and vowels.

MODULE 06 · BEGINNER

Coding and decoding

Minimum duration **2 days** | Level **Beginner**

In coding-decoding a word, number or sentence is converted by a hidden rule, and you must find the rule and apply it to another word, or work backwards. This module covers letter-shift codes, opposite and reversal codes, letter-to-number codes, substitution codes, sentence (word-matching) codes and the conditional symbol codes used in bank exams, with a fixed decoding checklist and 25 solved questions.

IN THIS MODULE

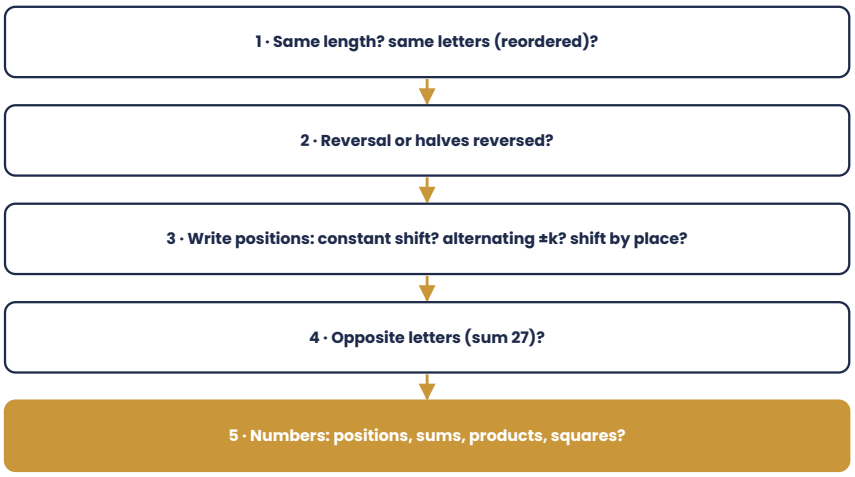
6.1	Basics	6.2	Letter codes
6.3	Number codes	6.4	Substitution and sentence codes
6.5	Conditional symbol coding	6.6	Practice questions
6.7	Solutions		

6.1 Basics

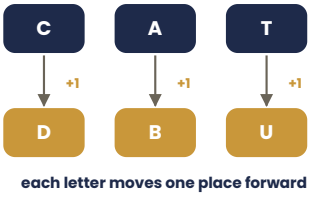
6.1.1 Types of coding and the decoding checklist

Every coding question gives at least one example (word → code). Compare them **letter by letter**, write the positions, and test the standard rules in a fixed order.

Type	Example	Rule
Letter shift	CAT → DBU	each letter +1
Opposite letters	FAN → UZM	A ↔ Z, B ↔ Y …
Reversal	PENCIL → LICNEP	write backwards
Halves reversed	COMPUTER → PMOCRETU	reverse each half
Letter → number	CAT → 24	sum of positions
Substitution	sky → sea, sea → water	names are swapped
Sentence code	“pit na sa” = “you are good”	match common words
Conditional symbol	EMIT → 9@4$	table + rules



DECODING CHECKLIST: TEST THE RULES IN THIS ORDER

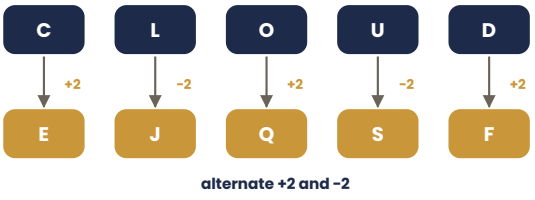


THE SIMPLEST CODE: A CONSTANT SHIFT

6.2 Letter codes

6.2.1 Constant, alternating and position-based shifts

Rule	Example	Check
+1 for every letter	MONDAY → NPOEBZ	M→N, O→P, N→O …
-1 for every letter	TEA → SDZ	T→S, E→D, A→Z
+2, -2 alternately	CLOUD → EJQSF	C+2, L−2, O+2, U−2, D+2
+1, +2, +3 … by position	CAT → DCW	C+1, A+2, T+3



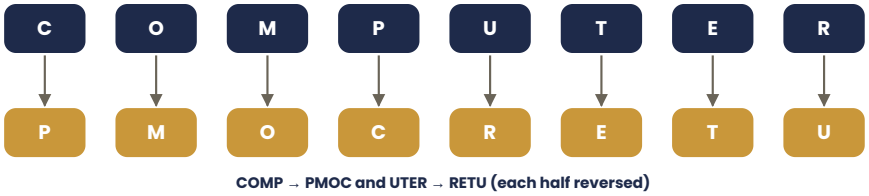
ALTERNATING SHIFT

SHORTCUT TRICK

If the code letters are close to the word letters, it is a shift. If they are far apart and the word is short, try **opposites** first (A–Z, B–Y …): FAN → UZM is a pure opposite code.

6.2.2 Reversal, halves and opposite-then-reverse

Some codes keep the same letters in a new order: whole word reversed (PENCIL → LICNEP) or each half reversed (COMPUTER → COMP|UTER → PMOC|RETU). Others combine two steps, such as opposites and then reversal: GOD → T, L, W → reversed **WLT**.



HALVES REVERSED

6.3 Number codes

6.3.1 Letter positions, sums and products

Rule	Example	Apply
positions written in order	BAG → 2-1-7	FED → 6-5-4
sum of positions	CAT → 3 + 1 + 20 = 24	DOG → 4 + 15 + 7 = 26
product of positions	BAD → 2 × 1 × 4 = 8	BED → 2 × 5 × 4 = 40
reverse positions	A → 26, B → 25	CAB → 24-26-25

6.3.2 Letter-to-digit matching (two words given)

When two words and their number codes are given, every letter has a fixed digit. Line up each word with its code and build a key.

WORKED EXAMPLE

ROSE = 6821, CHAIR = 73456. Find SEARCH.

From ROSE: R6, O8, S2, E1. From CHAIR: C7, H3, A4, I5, R6 (R = 6 again, which confirms the key). SEARCH = S2 E1 A4 R6 C7 H3 = **214673**.

Letter	R	O	S	E	C	H	A	I
Digit	6	8	2	1	7	3	4	5

6.4 Substitution and sentence codes

6.4.1 Substitution codes

“If sky is called sea, sea is called water, water is called air …, where do fish live?” Answer with the **new name** of the real answer: fish live in water, and water is called **air**.

WATCH OUT
Do not answer with the real word. The question asks for the word used in the code language.

6.4.2 Sentence codes (word matching)

Each sentence is coded word for word, but the order of code words is jumbled. A word that appears in two sentences has the code that appears in both codes.

Sentence	Code
you are good	pit na sa
good and bad	na ka la
they are honest	sa ja pa

"good" is in sentences 1 and 2 → common code **na**. "are" is in sentences 1 and 3 → common code **sa**. So in sentence 1, "you" = the remaining code **pit**.



COMMON WORDS SHARE COMMON CODES

6.5 Conditional symbol coding

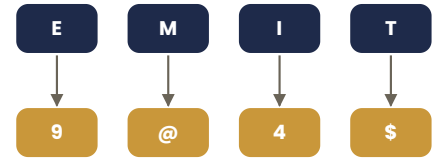
6.5.1 Table plus conditions

Bank exams give a table of letter → code and some conditions that change the codes of the first and last letters. Apply the **first condition that matches**; if none matches, code each letter from the table.

Letter	P	E	R	A	T	M	I	K
Code	7	$	2	%	9	@	4	#

1. If the first letter is a vowel and the last letter is a consonant, their codes are interchanged.
2. If both the first and the last letters are consonants, both are coded as &.
3. If the first letter is a consonant and the last letter is a vowel, the last letter is coded as *.

WORKED EXAMPLE
Code EMIT
E is a vowel and T a consonant → condition (i): swap the codes of E ($) and T (9). Table codes $ @ 4 9 become **9@4$**.



condition (i): codes of the first and last letters are swapped

APPLYING A CONDITION

TIP

Classify only the first and last letters (vowel or consonant) before looking at the conditions. The middle letters always come straight from the table.

6.6 Practice questions

6.6.1 Basic level

Q1 If CAT is coded as DBU, how is DOG coded?

(a) EPH (b) EPG (c) DPH (d) FQI

Q2 If FAN is coded as UZM, how is BAT coded?

(a) YZG (b) YZH (c) ZYG (d) YAG

Q3 If CAT = 24 (sum of letter positions), what is DOG?

(a) 24 (b) 26 (c) 28 (d) 25

Q4 If PENCIL is written as LICNEP, how is MARKER written?

(a) REKRAM (b) REKARM (c) MREKRA (d) RAKREM

Q5 If ROSE = 6821 and CHAIR = 73456, what is SEARCH?

(a) 214673 (b) 214763 (c) 241673 (d) 216473

Q6 If TEA is coded as SDZ, how is MILK coded?

(a) LHKJ (b) NJML (c) LHJK (d) LIKJ

Q7 If BAG is written as 2-1-7, how is FED written?

(a) 6-5-3 (b) 6-5-4 (c) 5-6-4 (d) 6-4-5

Q8 If sky is called sea, sea is called water, water is called air and air is called cloud, where do fish live?

(a) air (b) water (c) sea (d) cloud

Q9 If MONDAY is coded as NPOEBZ, how is FRIDAY coded?

(a) GSJEBZ (b) GSJEBY (c) GSIEBZ (d) GTJEBZ

6.6.2 Intermediate level

Q10 If COMPUTER is coded as PMOCRETU, how is KEYBOARD coded?

(a) BYEKDRAO (b) BYEKDROA (c) KEYBDRAO (d) BEYKDRAO

Q11 If DELHI = 73541 and CALCUTTA = 82589662, what is CALICUT?

(a) 8251896 **(b)** 8254896 **(c)** 8521896 **(d)** 8251869

Q12 If SCHOOL is coded as T6K (next letter of the first letter, number of letters, previous letter of the last letter), how is GARDEN coded?

(a) H6M **(b)** H6O **(c)** F6M **(d)** H5M

Q13 In a code, "pit na sa" means "you are good", "na ka la" means "good and bad" and "sa ja pa" means "they are honest". What is the code for "you"?

(a) pit **(b)** na **(c)** sa **(d)** ka

Q14 If CLOUD is coded as EJQSF, how is RAIN coded?

(a) TYKL **(b)** TCKL **(c)** TYKP **(d)** SYKL

Q15 If ACE = 9 and BDF = 12, what is HJL?

(a) 27 **(b)** 30 **(c)** 33 **(d)** 36

Q16 If GOD is coded as WLT, how is CAT coded?

(a) XZG **(b)** GZX **(c)** GXZ **(d)** ZXG

Q17 If BAD = 8 and CAB = 6, what is BED?

(a) 11 **(b)** 20 **(c)** 40 **(d)** 30

6.6.3 Advanced level

Q18 Use the table (P 7, E $, R 2, A %, T 9, M @, I 4, K #) and the three conditions in 6.5.1. How is PRAM coded?

(a) 72%@ **(b)** &2%& **(c)** &2%@ **(d)** 72%&

Q19 Using the same table and conditions, how is EMIT coded?

(a) $@49 **(b)** 9@4$ **(c)** &@4& **(d)** $@4*

Q20 Using the same table and conditions, how is TRIP coded?

(a) 924& **(b)** &247 **(c)** &24& **(d)** 9247

Q21 Using the same table and conditions, how is MARE coded?

(a) @%2$ **(b)** @%2* **(c)** &%2& **(d)** $%2@

Q22 Using the same table and conditions, how is IKRA coded?

(a) %#24 **(b)** 4#2% **(c)** [illegible]& **(d)** 4#2*

Q23 If TEMPLE is coded as VGORNG, how is CHURCH coded?

(a) EJWTEJ **(b)** EIWTEJ **(c)** DJWTEJ **(d)** EJVTEJ

Q24 In a code, "ko pa ta" means "eat sweet fruits", "ta ri ko" means "sweet fruits ripe" and "ri su na" means "ripe red apples". What is the code for "ripe"?

(a) ri **(b)** ko **(c)** su **(d)** ta

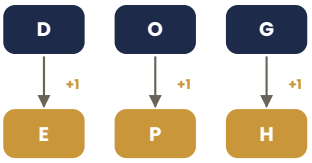
Q25 If CAT is coded as DCW (each letter moved forward by its position number), how is BAG coded?

(a) CCJ **(b)** CBJ **(c)** CCI **(d)** DCJ

6.7 Solutions

Q1 **Answer: (a) EPH**

Each letter +1 (C → D, A → B, T → U). DOG → **EPH**.



Q2 **Answer: (a) YZG**

Each letter is replaced by its opposite (F–U, A–Z, N–M). B–Y, A–Z, T–G → **YZG**.

Q3 **Answer: (b) 26**

D + O + G = 4 + 15 + 7 = **26**.

Q4 **Answer: (a) REKRAM**

The word is written backwards. MARKER → **REKRAM**.

Q5 **Answer: (a) 214673**

Key: R6, O8, S2, E1, C7, H3, A4, I5. SEARCH = 2 1 4 6 7 3 = **214673**.

Q6 **Answer: (a) LHKJ**

Each letter -1 (T → S, E → D, A → Z). M → L, I → H, L → K, K → J → **LHKJ**.

Q7 **Answer: (b) 6-5-4**

Letter positions in order: F6, E5, D4 → **6-5-4**.

Q8 **Answer: (a) air**

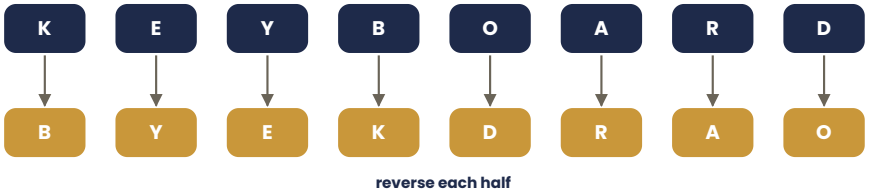
Fish live in water, and water is called **air**.

Q9 **Answer: (a) GSJEBZ**

Each letter +1. F → G, R → S, I → J, D → E, A → B, Y → Z → **GSJEBZ**.

Q10 **Answer: (a) BYEKDRAO**

Each half is reversed: KEYB → BYEK, OARD → DRAO. Answer **BYEKDRAO**.



Q11 **Answer: (a) 8251896**

Key: D7, E3, L5, H4, I1, C8, A2, U9, T6. CALICUT = C8 A2 L5 I1 C8 U9 T6 = **8251896**.

Q12 **Answer: (a) H6M**

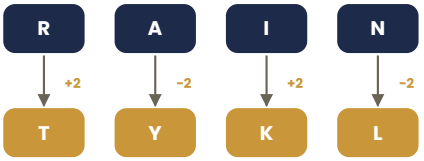
G → H, 6 letters, N → M. Answer **H6M**.

Q13 **Answer: (a) pit**

"good" is common to sentences 1 and 2 → na. "are" is common to sentences 1 and 3 → sa. The remaining code in sentence 1 is **pit** = you.

Q14 **Answer: (a) TYKL**

Alternately +2 and −2: R+2 = T, A−2 = Y (wraps), I+2 = K, N−2 = L → **TYKL**.



Q15 **Answer: (b) 30**

The code is the sum of positions: 1 + 3 + 5 = 9, 2 + 4 + 6 = 12. HJL = 8 + 10 + 12 = **30**.

Q16 **Answer: (b) GZX**

Opposites of G, O, D are T, L, W; written in reverse order: WLT. For CAT: opposites X, Z, G, reversed → **GZX**.

Q17 **Answer: (c) 40**

The code is the product of positions: 2 × 1 × 4 = 8, 3 × 1 × 2 = 6. BED = 2 × 5 × 4 = **40**.

Q18 **Answer: (b) &2%&**

P and M are both consonants → condition (ii): both become &. Middle letters from the table: R2, A%. Answer **&2%&**.

Q19 **Answer: (b) 9@4$**

E (vowel) first, T (consonant) last → condition (i): swap codes $ and 9. Answer **9@4$**.

Q20 **Answer: (c) &24&**

T and P are both consonants → both coded &. Middle: R2, I4. Answer **&24&**.

Q21 **Answer: (b) @%2***

M (consonant) first, E (vowel) last → condition (iii): last letter coded *. Answer **@%2***.

Q22 **Answer: (b) 4#2%**

I and A are both vowels: no condition applies, so every letter comes from the table: **4#2%**.

Q23 **Answer: (a) EJWTEJ**

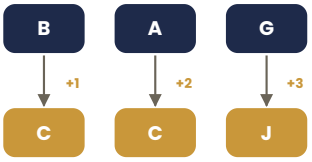
Each letter +2. C→E, H→J, U→W, R→T, C→E, H→J → **EJWTEJ**.

Q24 **Answer: (a) ri**

"ripe" appears in sentences 2 and 3; the only code common to "ta ri ko" and "ri su na" is **ri**.

Q25 **Answer: (a) CCJ**

B + 1 = C, A + 2 = C, G + 3 = J → **CCJ**.



Compare word and code letter by letter. Test, in order: same letters reordered (reversal, halves), constant or alternating shifts, opposites, then number rules (positions, sums, products). For sentence codes, the code common to two sentences belongs to the word common to them. For conditional codes, classify the first and last letters and apply the first matching rule.

MODULE 07 · BEGINNER

Blood relations

Minimum duration **2 days** | Level **Beginner**

Blood-relation questions describe people through chains of relations ("my father's only son's daughter") and ask how two of them are related. The reliable method is to draw a small family tree with fixed symbols for male, female, marriage and children. This module covers the relation vocabulary, the tree notation, pointing and introducing questions, coded relations (+, −, ×, ÷ and symbol codes) and family puzzles, followed by 25 solved questions, each with its tree.

IN THIS MODULE

7.1 Basics
7.2 Pointing and introducing questions
7.3 Coded relations
7.4 Family puzzles
7.5 Practice questions
7.6 Solutions

7.1 Basics

7.1.1 Relation vocabulary

Relation	Meaning	Relation	Meaning
Grandfather (paternal)	father's father	Uncle	father's or mother's brother; aunt's husband
Grandfather (maternal)	mother's father	Aunt	father's or mother's sister; uncle's wife
Grandson / granddaughter	son's or daughter's son / daughter	Nephew / niece	brother's or sister's son / daughter
Father-in-law	husband's or wife's father	Cousin	uncle's or aunt's child
Brother-in-law	spouse's brother or sister's husband	Sister-in-law	spouse's sister or brother's wife
Daughter-in-law	son's wife	Son-in-law	daughter's husband

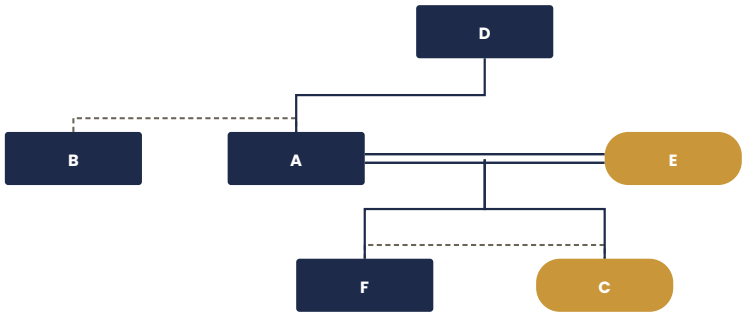
SHORTCUT TRICK

Compress chains from the speaker outward: "my father's only son" = **me** (if I am male) or **my brother** (if I am female). "My mother's husband" = **my father**. "My grandfather's only son" = **my father**. Replace each pair of words with one relation before moving on.

7.1.2 Family-tree notation

Use the same symbols every time: a **navy rectangle** for a male, a **gold rounded box** for a female, a **white box** when the gender is unknown, a **double line** for marriage, a **vertical line** from parents to children, and

a **dashed bracket** for siblings. Each generation sits on its own row.



males navy · females gold · = marriage · dashed = siblings

A THREE-GENERATION FAMILY TREE

WATCH OUT

Never assume a gender that the question does not give. "A is the child of B" leaves A as son **or** daughter; a good option list often includes "cannot be determined".

7.2 Pointing and introducing questions

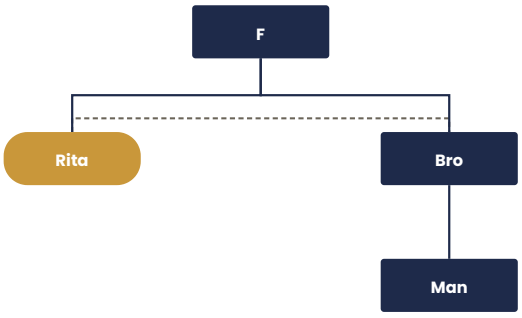
7.2.1 Work from the speaker

In "Pointing to X, Y said …", the chain of relations starts from the **speaker Y**. Resolve it from the innermost part outwards, then state X's relation to Y.

WORKED EXAMPLE

Pointing to a man, Rita said, "He is the son of my father's only son."

Rita is female, so her father's only son is her **brother**. The man is the son of her brother: her **nephew**.



father's only son = Rita's brother → the man is her nephew

POINTING QUESTION

TIP

Mark the speaker's box with a star or underline. Most errors come from starting the chain from the person pointed at instead of the speaker.

7.3 Coded relations

7.3.1 Symbol codes

Relations are replaced by symbols. Translate each symbol into words, two people at a time, and draw the tree as you go. We use this code throughout the module:

Code	Meaning
A + B	A is the father of B
A − B	A is the wife of B
A × B	A is the brother of B
A ÷ B	A is the daughter of B

WORKED EXAMPLE

What does M × A + N mean?

M × A: M is the brother of A. A + N: A is the father of N. So M is the brother of N's father: M is the **uncle** (paternal) of N.



DECODING A CODED RELATION

SHORTCUT TRICK

Every symbol also tells you a **gender**: + and × make the first person male, − and ÷ make the first person female. Write m or f under each letter as you read the expression.

7.4 Family puzzles

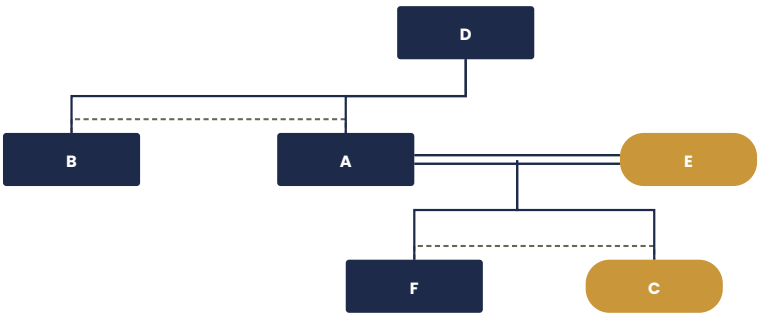
7.4.1 Several people, several clues

Puzzles give 5 to 8 people and clues about relations, couples and counts ("two fathers, three brothers and a mother"). Draw the definite links first (parent of, married to), then place the rest, then use the counts to check genders.

WORKED EXAMPLE

Six persons A, B, C, D, E, F. C is the sister of F. B is the brother of E's husband. D is the father of A and grandfather of F. There are two fathers, three brothers and a mother.

D is the father of A and the grandfather of F, so A is F's father. B is the brother of E's husband; the only married man who fits is A, so E is A's wife and B is A's brother. C and F are A and E's children. Fathers: D, A. Brothers: A, B, F. Mother: E.



two fathers (D, A) · three brothers (A, B, F) · one mother (E)

THE SOLVED FAMILY PUZZLE

7.5 Practice questions

7.5.1 Basic level

Q1 A is the father of B. B is the sister of C. How is C related to A?

(a) Son (b) Daughter
(c) Son or daughter (d) Nephew

Q2 Pointing to a man, Rita said, "He is the son of my father's only son." How is the man related to Rita?

(a) Son (b) Brother (c) Nephew (d) Cousin

Q3 X is the brother of Y. Z is the father of X. W is the mother of Z. How is Y's father related to W?

(a) Son (b) Brother (c) Husband (d) Grandson

Q4 Pointing to a photograph, Arun said, "She is the daughter of my grandfather's only son." How is she related to Arun?

(a) Cousin (b) Sister (c) Niece (d) Aunt

Q5 How is my mother's brother's son related to me?

(a) Brother (b) Nephew (c) Cousin (d) Uncle

Q6 A's father is B's son. How is B related to A?

(a) Father (b) Grandfather or grandmother
(c) Uncle (d) Brother

Q7 Introducing a woman, Ravi said, "Her mother is the only daughter of my mother." How is the woman related to Ravi?

(a) Sister (b) Niece (c) Daughter (d) Cousin

Q8 P is the wife of Q. R is the father of Q. How is P related to R?

(a) Daughter (b) Sister-in-law
(c) Daughter-in-law (d) Niece

7.5.2 Intermediate level

Q9 A + B: A is the father of B; A − B: A is the wife of B; A × B: A is the brother of B; A ÷ B: A is the daughter of B. If P ÷ R + Q, how is P related to Q?

(a) Daughter **(b)** Sister **(c)** Mother **(d)** Aunt

Q10 Using the same code, if P × Q − R, how is P related to R?

(a) Brother **(b)** Brother-in-law **(c)** Father-in-law **(d)** Uncle

Q11 Using the same code, which expression means that M is the uncle of N?

(a) M × A + N **(b)** M + A × N **(c)** M ÷ A + N **(d)** M − A + N

Q12 Using the same code, if A − B + C ÷ D, how is A related to C?

(a) Sister **(b)** Aunt **(c)** Mother **(d)** Daughter

Q13 Six persons A, B, C, D, E, F: C is the sister of F; B is the brother of E's husband; D is the father of A and the grandfather of F; there are two fathers, three brothers and a mother. Who is the mother?

(a) A **(b)** C **(c)** D **(d)** E

Q14 In the same family, how is B related to F?

(a) Father **(b)** Uncle **(c)** Brother **(d)** Grandfather

Q15 In the same family, how many male members are there?

(a) 3 **(b)** 4 **(c)** 5 **(d)** 2

Q16 Pointing to a man, a woman said, “His mother is the only daughter of my mother.” How is the woman related to the man?

(a) Sister **(b)** Mother **(c)** Aunt **(d)** Grandmother

7.5.3 Advanced level

Q17 A is the son of B. C, B's sister, has a son D and a daughter E. F is the maternal uncle of D. How is A related to D?

(a) Brother **(b)** Nephew **(c)** Cousin **(d)** Uncle

Q18 In the same family, how is E related to F?

(a) Daughter **(b)** Sister **(c)** Niece **(d)** Cousin

Q19 A $ B: A is the mother of B; A # B: A is the brother of B; A @ B: A is the husband of B; A % B: A is the son of B. In P # Q $ R @ S, how is P related to R?

(a) Father **(b)** Maternal uncle **(c)** Brother **(d)** Father-in-law

Q20 Using the same code, in M % N @ O $ P, how is M related to P?

(a) Father **(b)** Son **(c)** Brother **(d)** Uncle

Q21 Introducing a boy, a girl said, “He is the son of the daughter of the father of my uncle.” How is the boy related to the girl?

(a) Brother **(b)** Cousin
(c) Brother or cousin **(d)** Nephew

Q22 In a family of six, P, Q, R, S, T, U, there are two married couples. Q is the father of T. U is the mother of Q and the grandmother of R. S is the wife of Q and the mother of R. P is U's husband. T is a boy and R is a girl. How is P related to T?

(a) Father **(b)** Uncle **(c)** Grandfather **(d)** Brother

Q23 In the same family, how is S related to U?

(a) Daughter **(b)** Daughter-in-law
(c) Sister **(d)** Niece

Q24 Using the + − × ÷ code of question 9, which expression shows that Q is the grandson of T?

(a) Q × R ÷ S ÷ T **(b)** Q ÷ R ÷ T **(c)** T × R + Q **(d)** Q × R − T

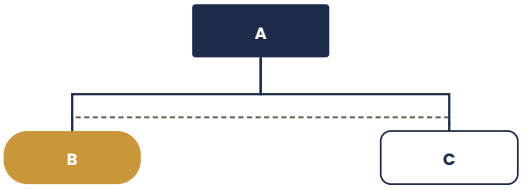
Q25 A is B's sister. C is B's mother. D is C's father. E is D's mother. How is A related to D?

(a) Granddaughter **(b)** Daughter
(c) Great-granddaughter **(d)** Niece

7.6 Solutions

Q1 **Answer: (c) Son or daughter**

B and C are A's children. C's gender is not given, so C is A's **son or daughter**.



C is A's child; C's gender is not given

Q2 **Answer: (c) Nephew**

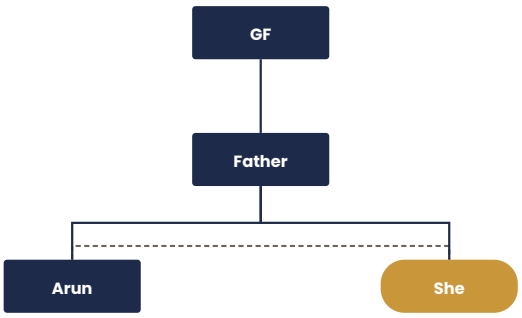
Rita's father's only son is her brother (Rita is female). The man is her brother's son: her **nephew**.

Q3 **Answer: (a) Son**

X and Y are siblings, so Y's father is Z. W is Z's mother, so Z is W's **son**.

Q4 **Answer: (b) Sister**

Arun's grandfather's only son is Arun's father. The daughter of Arun's father is his **sister**.



grandfather's only son = Arun's father

Q5 **Answer: (c) Cousin**

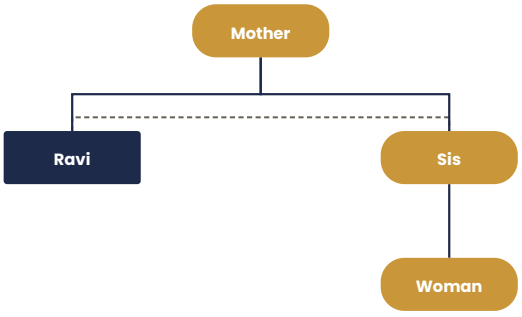
Mother's brother = maternal uncle. His son is my **cousin**.

Q6 **Answer: (b) Grandfather or grandmother**

B is the parent of A's father, so B is A's grandparent. B's gender is not given: **grandfather or grandmother**.

Q7 **Answer: (b) Niece**

The only daughter of Ravi's mother is Ravi's sister (Ravi is male). The woman is his sister's daughter: his **niece**.



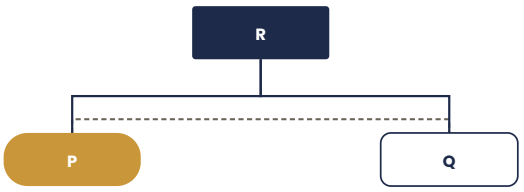
the woman is Ravi's sister's daughter: his niece

Q8 **Answer: (c) Daughter-in-law**

P is the wife of R's son Q, so P is R's **daughter-in-law**.

Q9 **Answer: (b) Sister**

P ÷ R: P is R's daughter. R + Q: R is Q's father. P and Q have the same father, and P is female: P is Q's **sister**.



P ÷ R + Q → P is Q's sister

Q10 **Answer: (b) Brother-in-law**

P × Q: P is Q's brother. Q − R: Q is R's wife. P is the brother of R's wife: R's **brother-in-law**.



P is the brother of R's wife: brother-in-law

Q11 **Answer: (a) M × A + N**

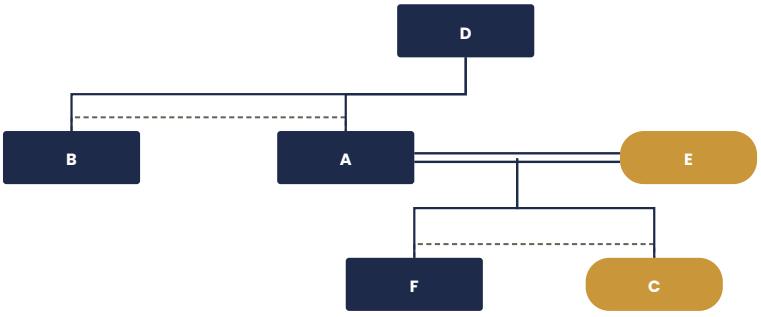
M × A + N: M is A's brother and A is N's father, so M is N's **uncle**. The others give father (M + A × N), sister (M ÷ A + N) and mother (M − A + N).

Q12 **Answer: (c) Mother**

A − B: A is B's wife. B + C: B is C's father. So A is the wife of C's father: C's **mother** (C ÷ D just says C is D's daughter; D must be B or A).

Q13 **Answer: (d) E**

As solved in 7.4.1, A is F's father and E is A's wife, so **E** is the mother of C and F.



Q14 **Answer: (b) Uncle**

B is the brother of A, F's father, so B is F's **uncle**.

Q15 **Answer: (b) 4**

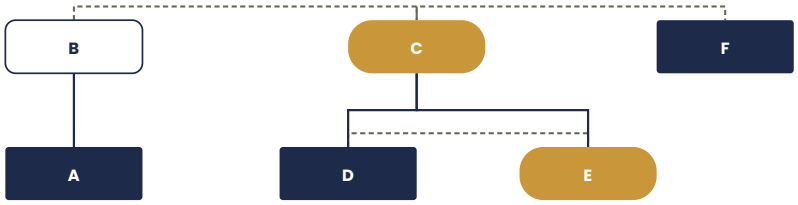
Males: D, A, B (brothers and fathers) and F (C's brother). C and E are female. Count = **4**.

Q16 **Answer: (b) Mother**

The only daughter of the woman's mother is the woman herself. So the man's mother is the woman: she is his **mother**.

Q17 **Answer: (c) Cousin**

B and C are siblings. A is B's son and D is C's son, so A and D are **cousins**.



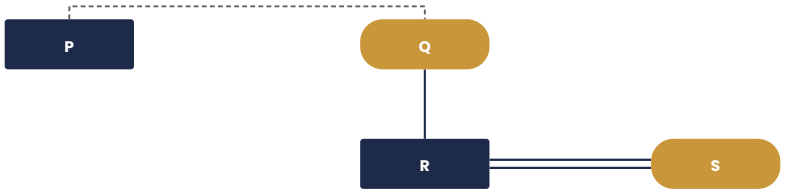
B, C, F are siblings

Q18 **Answer: (c) Niece**

F is D's maternal uncle, so F is C's brother. E is C's daughter, so E is F's **niece**.

Q19 **Answer: (b) Maternal uncle**

P # Q: P is Q's brother. Q $ R: Q is R's mother. So P is the brother of R's mother: R's **maternal uncle**. (R @ S just says R is S's husband.)



P # Q $ R → P is R's maternal uncle

Q20 **Answer: (c) Brother**

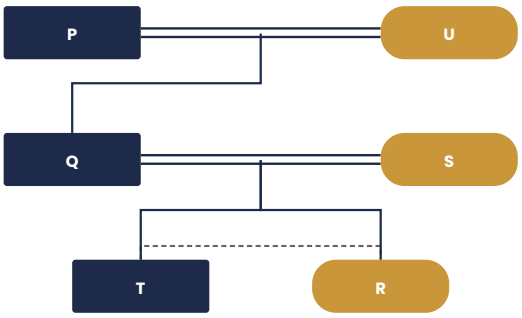
M % N: M is N's son. N @ O: N is O's husband. O $ P: O is P's mother. So M and P are children of the couple N and O, and M is male: M is P's **brother**.

Q21 **Answer: (c) Brother or cousin**

The father of her uncle is her grandfather. His daughter is either the girl's mother or her aunt. Her son is then the girl's brother or her cousin: **brother or cousin**.

Q22 **Answer: (c) Grandfather**

P is the husband of U, who is Q's mother, so P is Q's father. Q is T's father, so P is T's **grandfather**.



Q23 **Answer: (b) Daughter-in-law**

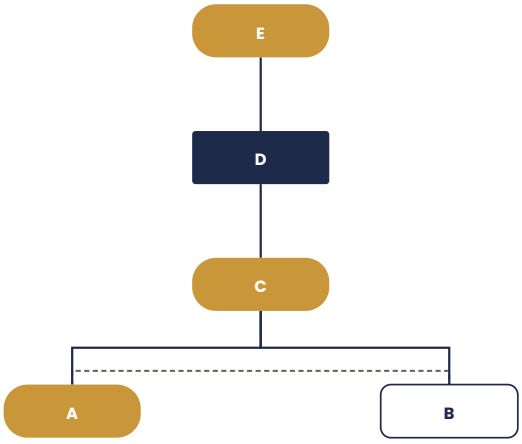
S is the wife of U's son Q: U's **daughter-in-law**.

Q24 **Answer: (a) Q × R ÷ S ÷ T**

Q × R: Q (male) is R's brother. R ÷ S: R is S's daughter. S ÷ T: S is T's daughter. So Q is a child of T's daughter and is male: T's **grandson**. (Q ÷ R ÷ T gives a granddaughter.)

Q25 **Answer: (a) Granddaughter**

A and B are C's children; D is C's father, so A is D's grandchild, and A is female: **granddaughter**.



A is D's granddaughter

Draw the tree every time: navy for male, gold for female, white for unknown, double line for marriage, one row per generation. Start pointing questions from the speaker, compress chains two words at a time, read genders from the code symbols, and never assume a gender the question does not give.

MODULE 08 · BEGINNER

Direction sense

Minimum duration **1 day** | Level **Beginner**

Direction questions describe a walk (distances and turns) and ask where the person ends up, how far from the start, or which way they face. This module covers the eight directions, left and right turns and angle turns, distance by Pythagoras, final direction, renamed-direction questions and shadow questions, with a path diagram for every worked example and 25 solved questions.

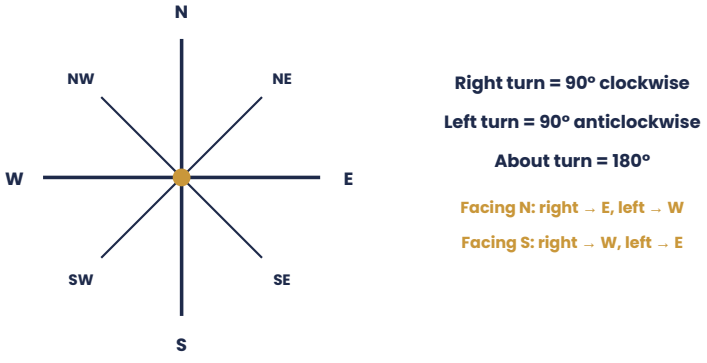
IN THIS MODULE

8.1	Basics	8.2	Turns and facing
8.3	Distance and final position	8.4	Shadows
8.5	Practice questions	8.6	Solutions

8.1 Basics

8.1.1 The eight directions

North is at the top of the page, East on the right. The four main directions are 90° apart; the four corner directions (NE, SE, SW, NW) lie halfway between them, 45° from each neighbour. Measure angles clockwise from North: N 0°, NE 45°, E 90°, SE 135°, S 180°, SW 225°, W 270°, NW 315°.



THE COMPASS AND THE MEANING OF TURNS

SHORTCUT TRICK

A right turn always adds 90° to the angle and a left turn subtracts 90°. Facing West (270°), a right turn gives 360° = North, a left turn gives 180° = South. No need to imagine yourself turning.

8.2 Turns and facing

8.2.1 Left, right and angle turns

Start facing	Right	Left	About turn
North	East	West	South
East	South	North	West
South	West	East	North
West	North	South	East

WORKED EXAMPLE

Facing West, a man turns 45° clockwise, then 180° clockwise, then 270° anticlockwise. Which way does he face?

270° + 45° = 315° (NW); + 180° = 135° (SE); - 270° = -135° = 225°: **South-west**.

8.2.2 Renamed directions

"If South-east is called North, what is West called?" The renaming is a rotation: SE (135°) becomes N (0°), so every direction is rotated by -135°. West (270°) - 135° = 135° = **South-east**.

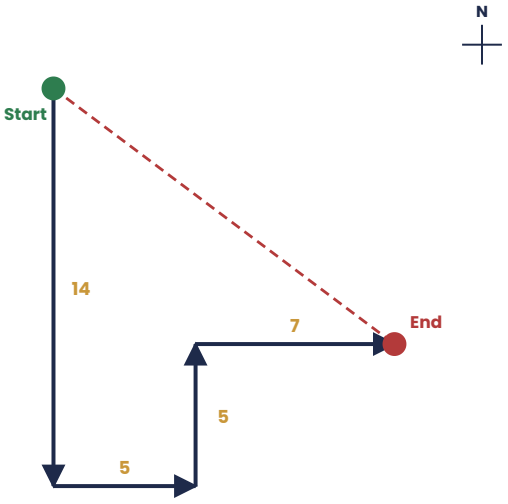
TIP

Compute the rotation once (new angle - old angle) and apply it to the asked direction. Draw two compasses only if you are unsure.

8.3 Distance and final position

8.3.1 Net east-west and north-south

Add up all east (+) and west (-) distances into one number x, and all north (+) and south (-) distances into y. The shortest distance from the start is $\sqrt{(x^2 + y^2)}$; the direction comes from the signs of x and y.



x = 5 + 7 = 12 east, y = −14 + 5 = −9 → 15 m South-east

NET DISPLACEMENT IN A WALK

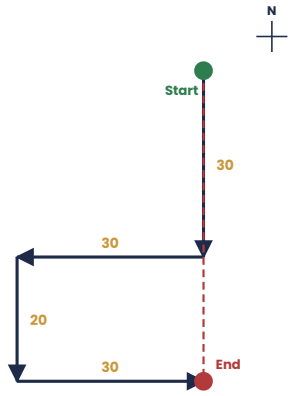
x (east +)	y (north +)	Direction from start
+	+	North-east
+	−	South-east
−	+	North-west
−	−	South-west
0	+ or −	North or South
+ or −	0	East or West

SHORTCUT TRICK
Learn the Pythagorean triples: 3-4-5, 6-8-10, 5-12-13, 8-15-17, 9-12-15, 12-16-20, 7-24-25. Almost every exam distance comes from one of them.

8.3.2 Walks given as turns

When the question says "turns right / left", first convert every leg to an absolute direction (keep track of the current facing), then add up x and y.

WORKED EXAMPLE
A man walks 30 m South, turns right and walks 30 m, turns left and walks 20 m, turns left again and walks 30 m. Where is he?
Facing S, right → W; facing W, left → S; facing S, left → E. Legs: S30, W30, S20, E30. x = 0, y = −50: **50 m South** of the start.



S30, W30, S20, E30 → 50 m South

CONVERTING TURNS TO DIRECTIONS

8.4 Shadows

8.4.1 Sun position and shadows

The sun rises in the **east** and sets in the **west**. A shadow points **away from the sun**: in the morning shadows fall to the west, in the evening to the east. At noon there is (almost) no shadow, so noon questions are not asked.

Time	Sun	Shadow falls towards
Morning (6–11 a.m.)	East	West
Evening (2–6 p.m.)	West	East

WORKED EXAMPLE

One evening, Ravi and Sunil were talking face to face. Sunil's shadow fell exactly to his right. Which way was Ravi facing?

Evening shadow points east, and it is on Sunil's right, so Sunil faces North. Ravi faces Sunil, so Ravi faces **South**.

WATCH OUT

In shadow questions, first fix the shadow direction from the time, then use the person's left/right to find where they face.

8.5 Practice questions

8.5.1 Basic level

Q1 A man walks 5 km North, turns right and walks 3 km. In which direction is he from the start?

(a) North **(b)** North-east **(c)** East **(d)** North-west

Q2 Ravi walks 3 m North and then 4 m East. How far is he from the start?

(a) 7 m **(b)** 5 m **(c)** 1 m **(d)** 6 m

Q3 Facing North, a girl turns right, then right again, then left. Which way does she face now?

(a) North **(b)** East **(c)** South **(d)** West

Q4 Facing South, a boy turns 90° clockwise and then 180°. Which way does he face?

(a) North **(b)** East **(c)** West **(d)** South

Q5 A man walks 20 m South, turns left and walks 15 m, then turns left and walks 20 m. How far is he from the start?

(a) 15 m **(b)** 20 m **(c)** 25 m **(d)** 35 m

Q6 In the morning, A stands facing the sun. In which direction does his shadow fall?

(a) East **(b)** West **(c)** North **(d)** South

Q7 If South-east is called North, what will West be called?

(a) North-east **(b)** South-east **(c)** South-west **(d)** North-west

Q8 A man walks 6 km West, turns left and walks 8 km. How far is he from the start?

(a) 14 km **(b)** 10 km **(c)** 12 km **(d)** 2 km

8.5.2 Intermediate level

Q9 A man walks 14 m South, 5 m East, 5 m North and 7 m East. How far is he from the start?

(a) 12 m **(b)** 15 m **(c)** 17 m **(d)** 13 m

Q10 In the same walk, in which direction is he from the start?

(a) North-east **(b)** South-east **(c)** South-west **(d)** East

Q11 A man walks 1 km East, 2 km North, 3 km West, 4 km South and 5 km East. How far is he from the start?

(a) $\sqrt{13}$ km **(b)** 5 km **(c)** 3 km **(d)** $\sqrt{5}$ km

Q12 Facing West, a man turns 45° clockwise, then 180° clockwise, then 270° anticlockwise. Which way does he face?

(a) North-west **(b)** South-east **(c)** South-west **(d)** North-east

Q13 One evening, Ravi and Sunil were talking face to face. Sunil's shadow fell exactly to his right. Which way was Ravi facing?

(a) North **(b)** South **(c)** East **(d)** West

Q14 A is 40 m South-west of B. C is 40 m South-east of B. In which direction is C from A?

(a) North **(b)** East **(c)** South **(d)** North-east

Q15 P is 6 m North of Q. R is 8 m East of Q. What is the distance between P and R?

(a) 14 m **(b)** 10 m **(c)** 12 m **(d)** 2 m

Q16 Starting from his house, Amit walks 10 m North, turns right and walks 5 m, turns right and walks 10 m, then turns left and walks 5 m. Where is he now?

(a) 5 m East **(b)** 10 m East **(c)** 10 m North **(d)** 15 m East

8.5.3 Advanced level

Q17 A man walks 30 m South. Turning right he walks 30 m, then turning left he walks 20 m, and turning left again he walks 30 m. How far is he from the start?

(a) 30 m (b) 50 m (c) 80 m (d) 20 m

Q18 Facing East, a woman walks 15 m, turns left and walks 10 m, turns left and walks 25 m, then turns left and walks 10 m. Where is she now?

(a) 10 m West (b) 10 m East (c) 25 m West (d) 15 m West

Q19 If North-east is called West, what will South be called?

(a) North (b) North-east (c) South-east (d) North-west

Q20 At 7 a.m. a man walks 10 m towards his shadow, turns right and walks 10 m. In which direction is he from the start?

(a) North-east (b) North-west (c) South-west (d) South-east

Q21 B is 3 m North of A, C is 4 m East of B, D is 7 m South of C and E is 4 m West of D. Where is E from A?

(a) 4 m North (b) 4 m South (c) 3 m South (d) 7 m South

Q22 In the same arrangement, what is the distance between A and D?

(a) 4 m (b) $4\sqrt{2}$ m (c) 5 m (d) 8 m

Q23 A car travels 12 km North, turns left and travels 9 km. What is the shortest distance from the start?

(a) 21 km (b) 15 km (c) 13 km (d) 3 km

Q24 Facing South-west, a man turns 135° anticlockwise and then 90° clockwise. Which way does he face?

(a) East (b) South (c) South-east (d) North

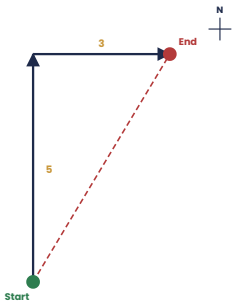
Q25 After turning right, right and then left, Sumit faces East. Which way was he facing at the start?

(a) North (b) South (c) West (d) East

8.6 Solutions

Q1 **Answer: (b) North-east**

Legs N5, E3: x = +3, y = +5, both positive → **North-east**.



Q2 **Answer: (b) 5 m**

$\sqrt{(3^2 + 4^2)} = \sqrt{25} =$ **5 m**.

Q3 **Answer: (b) East**

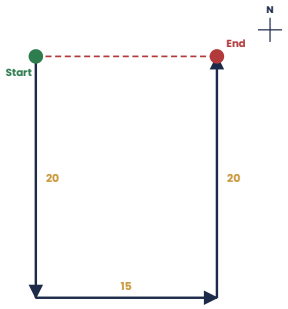
N → right E → right S → left **E**.

Q4 **Answer: (b) East**

180° + 90° = 270° (West); + 180° = 450° = 90°: **East**.

Q5 **Answer: (a) 15 m**

Facing S, left → E; facing E, left → N. Legs S20, E15, N20: x = 15, y = 0 → **15 m** (East).



Q6 **Answer: (b) West**

The morning sun is in the east, so the shadow falls to the **West**, behind him.

Q7 **Answer: (b) South-east**

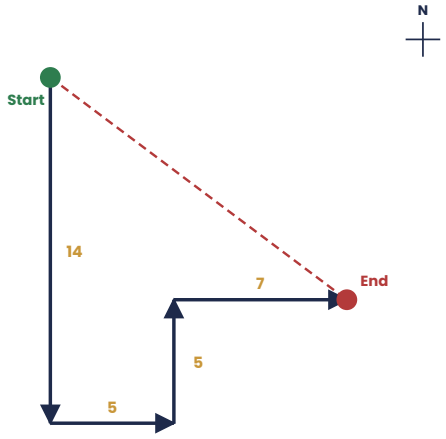
SE (135°) → N (0°) is a rotation of -135°. West: 270° - 135° = 135° = **South-east**.

Q8 **Answer: (b) 10 km**

Facing W, left → S. $\sqrt{(6^2 + 8^2)} =$ **10 km**.

Q9 **Answer: (b) 15 m**

x = 12, y = -9 → $\sqrt{(144 + 81)} = \sqrt{225} =$ **15 m**.

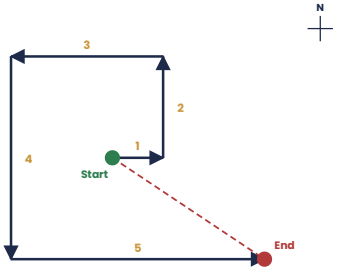


Q10 **Answer: (b) South-east**

x positive, y negative → **South-east**.

Q11 **Answer: (a) √13 km**

x = 1 - 3 + 5 = 3, y = 2 - 4 = -2 → √(9 + 4) = **√13 km** (South-east).



Q12 **Answer: (c) South-west**

270 + 45 = 315; + 180 = 495 = 135; - 270 = -135 = 225° = **South-west**.

Q13 **Answer: (b) South**

Evening shadow points east; it is on Sunil's right, so Sunil faces North and Ravi faces **South**.

Q14 **Answer: (b) East**

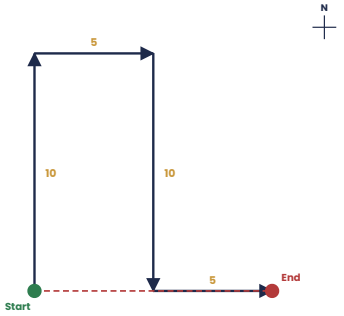
A and C are both 40 m from B at 45° below the horizontal, on opposite sides, so they are at the same height: C is due **East** of A.

Q15 **Answer: (b) 10 m**

Q is the right angle: √(6² + 8²) = **10 m**.

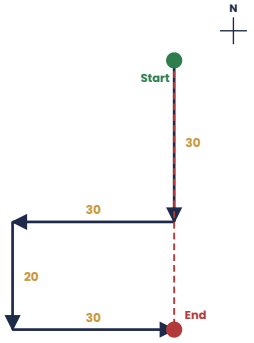
Q16 **Answer: (b) 10 m East**

Legs N10, E5, S10, E5 → x = 10, y = 0: **10 m East** of his house.



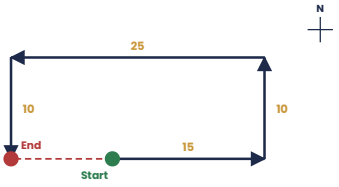
Q17 **Answer: (b) 50 m**

Legs S30, W30, S20, E30 → x = 0, y = −50: **50 m** (South).



Q18 **Answer: (a) 10 m West**

Legs E15, N10, W25, S10 → x = −10, y = 0: **10 m West** of the start.



Q19 **Answer: (b) North-east**

NE (45°) → W (270°) is a rotation of +225°. South: 180° + 225° = 405° = 45° = **North-east**.

Q20 **Answer: (b) North-west**

At 7 a.m. the shadow points west, so he walks West; right of West is North. x = −10, y = +10 → **North-west**.

Q21 **Answer: (b) 4 m South**

Coordinates: A(0, 0), B(0, 3), C(4, 3), D(4, -4), E(0, -4). E is **4 m South** of A.

Q22 **Answer: (b) 4√2 m**

D is (4, -4): √(16 + 16) = **4√2 m**, South-east of A.

Q23 **Answer: (b) 15 km**

Facing N, left → W. √(12² + 9²) = √225 = **15 km**.

Q24 **Answer: (b) South**

225° - 135° = 90° (East); + 90° = 180°: **South**.

Q25 **Answer: (a) North**

Right + right + left = one net right turn (+90°). So the start was East - 90° = **North**.

Convert every leg to an absolute direction (right = +90°, left = -90°), add east-west into x and north-south into y, then distance = √(x² + y²) and direction from the signs. For renamed directions apply one rotation; for shadows, morning shadows point west and evening shadows point east.

MODULE 09 · BEGINNER

Ranking, ordering and position

Minimum duration **1 day** | Level **Beginner**

Ranking questions give a person's position from one or both ends of a row, queue or class list and ask for the total, the position from the other end, the number of people between two persons, or what happens after an interchange. Ordering questions compare heights, weights or marks. This module gives the formulas, a row diagram for every case, the overlap trap, and 25 solved questions.

IN THIS MODULE

9.1	Basics	9.2	Two persons in a row
9.3	Interchange and shifting	9.4	Ordering by comparison
9.5	Practice questions	9.6	Solutions

9.1 Basics

9.1.1 Position from both ends

If a person is **L-th from the left** (or top, front) and **R-th from the right** (bottom, back), the person is counted twice, so

Total = L + R − 1 · Position from the other end = Total − Position + 1



Ravi is 7th from the left and 18 − 7 + 1 = 12th from the right; total 7 + 12 − 1 = 18

ONE PERSON IN A ROW

Given	Formula	Example
L and R	Total = L + R − 1	7th from left, 12th from right → 18
Total and L	R = Total − L + 1	40 students, 15th from top → 26th from bottom
Same position from both ends	Total = 2L − 1	7th from either end → 13

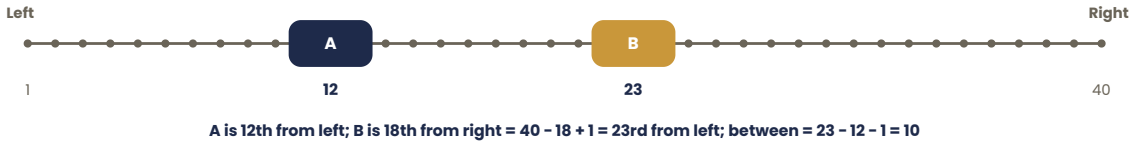
WATCH OUT

Ranks "from the top" and "from the bottom" in a class work exactly like left and right in a row. But if some students failed or were absent and the ranks are among those who passed, add them to the total at the end.

9.2 Two persons in a row

9.2.1 Persons between two people

Convert both positions to the **same end** (usually from the left). Then the number of persons between them is $|P_1 - P_2| - 1$.



A is 12th from left; B is 18th from right = 40 − 18 + 1 = 23rd from left; between = 23 − 12 − 1 = 10

TWO PERSONS IN A ROW OF 40

9.2.2 Total when the gap is given: non-overlap and overlap

If A is L-th from the left, B is R-th from the right and there are k persons between them, the total depends on whether A is to the left of B (no overlap) or their counts cross (overlap).

Case	Total	Example: L = 15, R = 12, k = 6
A left of B (no overlap): maximum	L + R + k	15 + 12 + 6 = 33
Counts cross (overlap): minimum	L + R − k − 2	15 + 12 − 6 − 2 = 19



no overlap: 15 + 6 + 12 = 33

MAXIMUM CASE



overlap: B (12th from right = 8th from left) is left of A; 6 between → total 19

MINIMUM CASE

TIP

If the question asks for "the total" without saying which, the answer is usually the non-overlap value; if it asks for the **minimum**, use L + R − k − 2.

9.3 Interchange and shifting

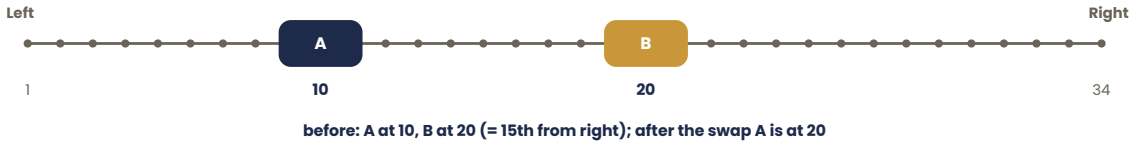
9.3.1 When two people swap places

After A and B interchange, A stands where B stood. So A's **new** position from one end equals B's **old** position from that end. Combine it with B's old position from the other end to get the total.

WORKED EXAMPLE

A is 10th from the left and B is 15th from the right. They interchange, and A becomes 20th from the left. How many are in the row?

A's new place is B's old place: 20th from the left and 15th from the right. Total = 20 + 15 − 1 = **34**.



before: A at 10, B at 20 (= 15th from right); after the swap A is at 20

INTERCHANGE

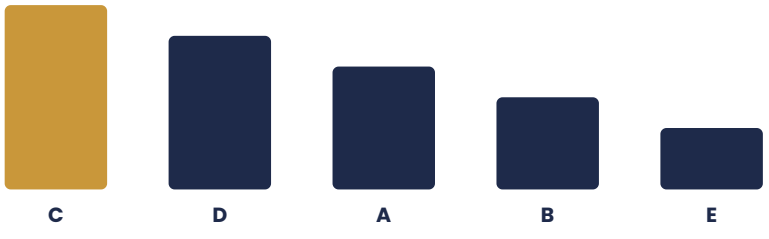
9.3.2 Shifting places

If A is 18th from the left and moves 4 places to the right, A is now 22nd from the left. If A is then 20th from the right, the total is 22 + 20 − 1 = **41**.

9.4 Ordering by comparison

9.4.1 Building the order

Statements like "A is taller than B but shorter than C" are written as a chain with > signs: C > A > B. Merge statements one at a time. Clues like "taller than only E" fix a position immediately (second from the bottom).



C > D > A > B > E

WRITING HEIGHTS AS AN ORDERED CHAIN

SHORTCUT TRICK

Use a vertical list: tallest at the top. Place absolute clues first ("tallest", "only one is shorter than B"), then relative clues. If a clue does not fit, keep two possible orders until another clue decides.

9.5 Practice questions

9.5.1 Basic level

Q1 Ravi is 7th from the left and 12th from the right in a row. How many people are in the row?

(a) 18 (b) 19 (c) 17 (d) 20

Q2 In a class of 40 students, Rita is 15th from the top. What is her rank from the bottom?

(a) 25th (b) 26th (c) 24th (d) 27th

Q3 In a row of 25 children, A is 10th from the right. What is A's position from the left?

(a) 15th (b) 16th (c) 14th (d) 17th

Q4 A is taller than B, B is taller than C and C is taller than D. Who is the shortest?

(a) A (b) B (c) C (d) D

Q5 Mohan is 9th from the front in a queue of 30. What is his position from the back?

(a) 21st (b) 22nd (c) 20th (d) 23rd

Q6 In a row of 30 boys, X is 14th from the left and Y is 10th from the right. How many boys are between X and Y?

(a) 5 (b) 6 (c) 7 (d) 8

Q7 P is taller than Q. R is taller than P. S is shorter than Q. Who is the tallest?

(a) P (b) Q (c) R (d) S

Q8 Kavita ranks 5th from the top and 26th from the bottom in her class. How many students are in the class?

(a) 31 (b) 30 (c) 29 (d) 32

9.5.2 Intermediate level

Q9 A is 10th from the left and B is 15th from the right in a row. When they interchange places, A becomes 20th from the left. How many people are in the row?

(a) 33 (b) 34 (c) 35 (d) 30

Q10 In the same row, what is B's position from the right after the interchange?

(a) 24th (b) 25th (c) 26th (d) 10th

Q11 In a row of 40, A is 12th from the left and B is 18th from the right. How many people are between them?

(a) 9 (b) 10 (c) 11 (d) 12

Q12 A is 15th from the left and B is 12th from the right, and there are 6 people between them (A is to the left of B). How many people are in the row?

(a) 31 (b) 33 (c) 27 (d) 19

Q13 A is 15th from the left and B is 12th from the right, and there are 6 people between them. What is the minimum number of people in the row?

(a) 17 (b) 19 (c) 21 (d) 33

Q14 A is heavier than B but lighter than C. D is heavier than C. E is the lightest. Who is the second heaviest?

(a) A **(b)** C **(c)** D **(d)** B

Q15 In a queue of 50, Amit is 10th from the front and Ravi is 25th from the back. Mohan stands exactly midway between them. What is Mohan's position from the front?

(a) 17th **(b)** 18th **(c)** 19th **(d)** 20th

Q16 Rohit scored more than Sunil but less than Mona. Mona scored less than Tina. Neha scored the least. Who is third from the top?

(a) Mona **(b)** Rohit **(c)** Sunil **(d)** Tina

9.5.3 Advanced level

Q17 In a row of 50 girls, X is 16th from the left. Z is 7 places to the right of X. What is Z's position from the right?

(a) 27th **(b)** 28th **(c)** 29th **(d)** 34th

Q18 A is 18th from the left in a row. He moves 4 places to the right and is then 20th from the right. How many are in the row?

(a) 39 **(b)** 41 **(c)** 40 **(d)** 42

Q19 Sita is 12th from the top in a class. Rita is 7 ranks below Sita and is 25th from the bottom. How many students are in the class?

(a) 42 **(b)** 43 **(c)** 44 **(d)** 36

Q20 A tree is 7th from either end of a row of trees. How many trees are in the row?

(a) 14 **(b)** 13 **(c)** 15 **(d)** 12

Q21 Among A, B, C, D, E, all of different heights, B is taller than only E. D is taller than A and B but shorter than C. Who is the second tallest?

(a) C **(b)** D **(c)** A **(d)** B

Q22 In the same group, how many people are shorter than A?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

Q23 In an exam, Ram is 11th from the top and 29th from the bottom among the students who passed. 6 students failed and 4 were absent. How many students are in the class?

(a) 39 **(b)** 45 **(c)** 49 **(d)** 43

Q24 A is 12th from the left and B is 9th from the right. They interchange, and A is now 17th from the left. How many are in the row?

(a) 24 **(b)** 25 **(c)** 26 **(d)** 29

Q25 A is 9th from the left and B is 14th from the right. After they interchange, A is 18th from the left. How many people are between A and B?

(a) 7 **(b)** 8 **(c)** 9 **(d)** 10

9.6 Solutions

Q1 **Answer: (a) 18**

Total = 7 + 12 − 1 = **18**.

Q2 **Answer: (b) 26th**

40 − 15 + 1 = **26th**.

Q3 **Answer: (b) 16th**

25 − 10 + 1 = **16th**.

Q4 **Answer: (d) D**

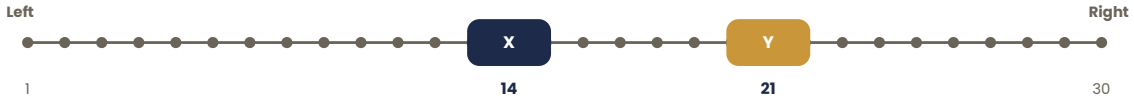
A > B > C > D, so **D** is the shortest.

Q5 **Answer: (b) 22nd**

30 − 9 + 1 = **22nd**.

Q6 **Answer: (b) 6**

Y is 30 − 10 + 1 = 21st from the left. Between = 21 − 14 − 1 = **6**.



Q7 **Answer: (c) R**

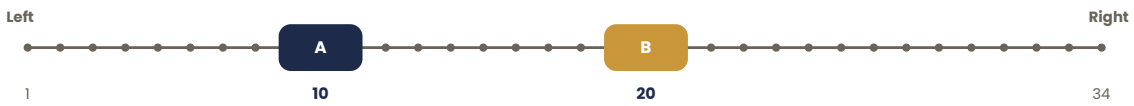
R > P > Q > S, so **R** is the tallest.

Q8 **Answer: (b) 30**

5 + 26 − 1 = **30**.

Q9 **Answer: (b) 34**

A's new place = B's old place: 20th from the left and 15th from the right. Total = 20 + 15 − 1 = **34**.



Q10 **Answer: (b) 25th**

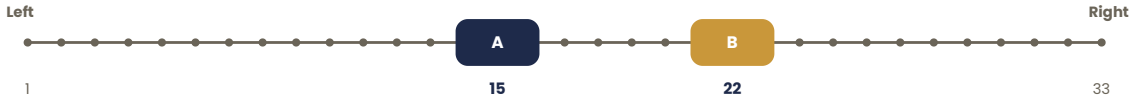
B now stands at A's old place, 10th from the left: 34 − 10 + 1 = **25th** from the right.

Q11 **Answer: (b) 10**

B is 40 − 18 + 1 = 23rd from the left. Between = 23 − 12 − 1 = **10**.

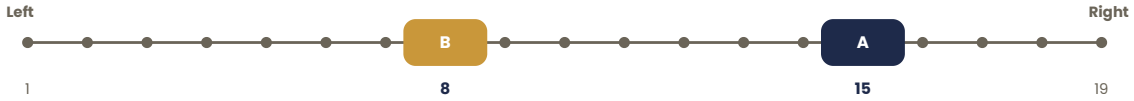
Q12 **Answer: (b) 33**

No overlap: 15 + 6 + 12 = **33**.



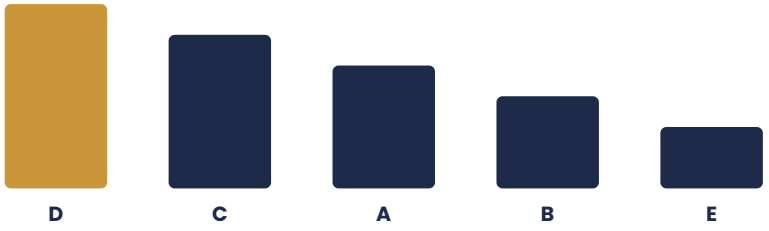
Q13 **Answer: (b) 19**

Overlap case: 15 + 12 − 6 − 2 = **19**. Check: B is then 8th from the left, A 15th, with 6 between.



Q14 **Answer: (b) C**

D > C > A > B > E. Second heaviest = **C**.



Q15 **Answer: (b) 18th**

Ravi is 50 − 25 + 1 = 26th from the front. Midway between 10 and 26 is (10 + 26) ÷ 2 = **18th**.

Q16 **Answer: (b) Rohit**

Tina > Mona > Rohit > Sunil > Neha. Third from the top = **Rohit**.

Q17 **Answer: (b) 28th**

Z is 16 + 7 = 23rd from the left, so 50 − 23 + 1 = **28th** from the right.

Q18 **Answer: (b) 41**

New position 22nd from the left and 20th from the right: 22 + 20 − 1 = **41**.

Q19 **Answer: (b) 43**

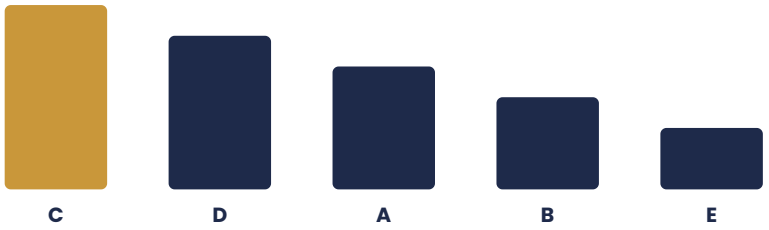
Rita is 12 + 7 = 19th from the top and 25th from the bottom: 19 + 25 − 1 = **43**.

Q20 **Answer: (b) 13**

2 × 7 − 1 = **13**.

Q21 **Answer: (b) D**

B is second from the bottom (only E is shorter). C > D > A, and A must be above B. Order: C > D > A > B > E. Second tallest = **D**.



Q22 **Answer: (b) 2**

Below A are B and E: **2**.

Q23 **Answer: (c) 49**

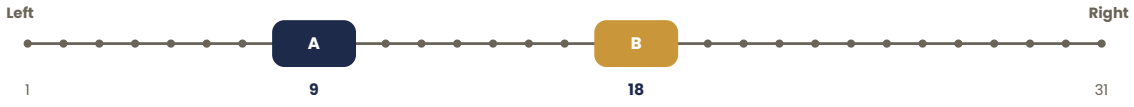
Passed = 11 + 29 − 1 = 39. Class = 39 + 6 + 4 = **49**.

Q24 **Answer: (b) 25**

A's new place = B's old place: 17th from the left and 9th from the right. Total = 17 + 9 − 1 = **25**.

Q25 **Answer: (b) 8**

Total = 18 + 14 − 1 = 31. Positions from the left are 9 and 18, so between = 18 − 9 − 1 = **8**.



Total = L + R − 1 and position from the other end = Total − position + 1. Convert both people to the same end before counting the gap. With a given gap, the total is L + R + k (no overlap) or L + R − k − 2 (overlap, minimum). After an interchange, a person's new position equals the other person's old position.

LEVEL 2

Intermediate level

Master alphabet tests, symbol operations, inequalities, syllogisms, linear and circular seating, puzzles, clocks, calendars and input-output questions

10 MODULES · 15 DAYS MINIMUM

Module 10 · Alphabet test	1 day
Module 11 · Mathematical operations	1 day
Module 12 · Inequalities	1 day
Module 13 · Syllogism	2 days
Module 14 · Linear seating arrangement	2 days
Module 15 · Circular seating arrangement	2 days
Module 16 · Puzzles	2 days
Module 17 · Clocks	1 day
Module 18 · Calendars	1 day
Module 19 · Input-output	2 days

MODULE 10 · INTERMEDIATE

Alphabet test

Minimum duration **1 day** | Level **Intermediate**

Alphabet-test questions work on the letters of a given word: letter pairs with the same gap as in the alphabet, dictionary order, rearranging letters, swapping positions, replacing letters by rule, and checking which words can be formed from a given word. This module gives a quick method for each type, pair diagrams, and 25 solved questions whose answers were checked by a program.

IN THIS MODULE

10.1	Letter pairs	10.2	Dictionary order
10.3	Rearranging and replacing letters	10.4	Forming words
10.5	Practice questions	10.6	Solutions

10.1 Letter pairs

10.1.1 Pairs with as many letters between them as in the alphabet

"How many pairs of letters in the word have as many letters between them in the word as in the alphabet?" For letters at word positions i and j, compare the **gap in the word** (j - i) with the **gap in the alphabet** (difference of their positions). Unless the question says "in the forward direction only", count pairs in both directions.



O (15) and N (14): adjacent in the word and in the alphabet → 1 pair

LETTER PAIRS IN MONKEY

SHORTCUT TRICK

Write the alphabet position under each letter. Then check neighbours (gap 1: positions differ by 1), next-but-one (gap 2: differ by 2), and so on. Stop when the word gap exceeds the largest position difference you can see; long gaps rarely match.

Word	Pairs (both directions)	Forward only
MONKEY	ON → 1	0
PRODUCTION	PI, RU, ON → 3	RU → 1
FRAGMENT	GE → 1	0

10.1.2 Letters in the same position as in the alphabet

A letter at word position k "is as far from the beginning of the word as in the alphabet" when it is the k-th letter of the alphabet. In BACKED: C is 3rd in the word and 3rd in the alphabet; E is 5th and 5th. Answer 2.

10.2 Dictionary order

10.2.1 Comparing words letter by letter

Compare the first letters; if equal, the second letters; and so on. A word that is a prefix of another comes first (ELEG before ELEGANT).

Words	Dictionary order
Apple, Apply, Appeal, Append	Appeal, Append, Apple, Apply
Elephant, Elegant, Element, Elevate, Elegy	Elegant, Elegy, Element, Elephant, Elevate

> **TIP**
> Line the words up one under another and underline the first column where they differ. Only that column decides their order.

10.3 Rearranging and replacing letters

10.3.1 Swapping positions

Write the positions 1, 2, 3 … above the word and perform the swaps exactly as stated, then read off the asked letter.

Instruction	Word	Result
swap 1st–2nd, 3rd–4th, 5th–6th, 7th–8th	ABSOLUTE	BAOSULET
reverse first five and last five (middle stays)	INSTRUCTION	RTSNIUNOITC
swap 1st–6th, 2nd–7th, … 5th–10th	COMPLAINTS	AINTSCOMPL
letters in alphabetical order	PRINCE	CEINPR

10.3.2 Replacing letters by a rule

Rules like "each vowel by the next letter and each consonant by the previous letter" are applied letter by letter. GOVERNMENT becomes F P U F Q M L F M S.

> **WATCH OUT**
> Positions "from the right" in the new word are counted in the **new** word, not the original. Write the new word out completely before counting.

10.4 Forming words

10.4.1 Which word can be formed?

A word can be formed from a given word only if every letter it needs is available **as many times** as it is needed. Count letters on both sides.

WORKED EXAMPLE

Which word cannot be formed from CONSTITUTION: COUNT, NOTION, STATION, SUIT?

CONSTITUTION has no A, so **STATION** cannot be formed. NOTION needs N twice, O twice, T once, I once: available.

10.5 Practice questions

10.5.1 Basic level

Q1 How many pairs of letters in MONKEY have as many letters between them in the word as in the alphabet?

(a) None (b) One (c) Two (d) Three

Q2 Arrange in dictionary order: Apple, Apply, Appeal, Append. Which word is third?

(a) Apple (b) Apply (c) Appeal (d) Append

Q3 If the letters of PRINCE are arranged in alphabetical order, which letter is 4th from the left?

(a) I (b) N (c) P (d) E

Q4 How many letters appear more than once in the word TRANSPORT?

(a) 1 (b) 2 (c) 3 (d) 4

Q5 Which word comes first in dictionary order: Cattle, Catch, Cater, Catalog?

(a) Cattle (b) Catch (c) Cater (d) Catalog

Q6 In ABSOLUTE, the 1st and 2nd letters are interchanged, the 3rd and 4th, the 5th and 6th, and the 7th and 8th. Which letter is now 7th from the left?

(a) T (b) E (c) L (d) U

Q7 Rearrange the letters U, N, K, T, R to form a meaningful word. What is its last letter?

(a) T (b) R (c) K (d) N

Q8 If each letter of MAJOR is replaced by the next letter of the alphabet, how many vowels are in the new word?

(a) 0 (b) 1 (c) 2 (d) 3

10.5.2 Intermediate level

Q9 How many pairs of letters in PRODUCTION have as many letters between them in the word as in the alphabet?

(a) 1 (b) 2 (c) 3 (d) 4

Q10 If the first five letters of INSTRUCTION are reversed and the last five letters are reversed (the middle letter stays), which letter is 4th from the right?

(a) I **(b)** O **(c)** T **(d)** N

Q11 Arrange in dictionary order: Elephant, Elegant, Element, Elevate, Elegy. Which word is fourth?

(a) Element **(b)** Elephant **(c)** Elevate **(d)** Elegy

Q12 If the letters of DEMOCRAT are arranged in alphabetical order, how many letters stay in the same position?

(a) None **(b)** One **(c)** Two **(d)** Three

Q13 How many letters in BACKED are in the same position in the word as in the alphabet?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 0

Q14 Arrange in dictionary order: Sincere, Simple, Sinner, Signal, Silent.

(a) Signal, Silent, Simple, Sincere, Sinner
(b) Signal, Simple, Silent, Sincere, Sinner
(c) Silent, Signal, Simple, Sinner, Sincere
(d) Signal, Silent, Simple, Sinner, Sincere

Q15 Which word cannot be formed from the letters of CONSTITUTION?

(a) COUNT **(b)** NOTION **(c)** STATION **(d)** SUIT

Q16 Which word cannot be formed from the letters of EXAMINATION?

(a) MAIN **(b)** NATION **(c)** TAXI **(d)** MATRON

10.5.3 Advanced level

Q17 If the letters of PERSONAL are arranged in alphabetical order, which letter is 3rd to the right of the letter 2nd from the left?

(a) N **(b)** O **(c)** P **(d)** L

Q18 In COMPLAINTS, the 1st and 6th letters are interchanged, then the 2nd and 7th, 3rd and 8th, 4th and 9th, and 5th and 10th. Which letter is 7th from the right?

(a) S **(b)** T **(c)** C **(d)** N

Q19 In GOVERNMENT, every vowel is replaced by the next letter and every consonant by the previous letter. What is the 5th letter of the new word?

(a) M **(b)** Q **(c)** S **(d)** P

Q20 Arrange in dictionary order: Necessary, Nectar, Neglect, Negative, Neighbour. Which word is in the middle?

(a) Nectar **(b)** Negative **(c)** Neglect **(d)** Necessary

Q21 How many pairs of letters in FRAGMENT have as many letters between them in the word as in the alphabet?

(a) None **(b)** One **(c)** Two **(d)** Three

Q22 If the letters of DESTINATION are written in reverse order, which letter is 4th to the right of the 7th letter from the left?

(a) E **(b)** D **(c)** S **(d)** T

Q23 How many pairs of letters in PRODUCTION have as many letters between them as in the alphabet, counting only the forward direction (the later letter is also later in the alphabet)?

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

Q24 In ABSOLUTE, after the swaps of question 6, how many letters are between B and T in the new word?

(a) 5 **(b)** 6 **(c)** 4 **(d)** 7

Q25 Which word can be formed from the letters of CONSTITUTION?

(a) STATION **(b)** CONTEST **(c)** INTUITION **(d)** COUNTS

10.6 Solutions

Q1 **Answer: (b) One**

Only O and N: adjacent in the word, and 15 - 14 = 1 in the alphabet. Answer **one**.



Q2 **Answer: (a) Apple**

Order: Appeal, Append, Apple, Apply (compare the 4th letters: E before L, then the 5th). Third = **Apple**.

Q3 **Answer: (b) N**

Alphabetical order: C, E, I, N, P, R. 4th = **N**.

Q4 **Answer: (b) 2**

T appears twice and R appears twice; all others once. Answer **2**.

Q5 **Answer: (d) Catalog**

All start with CAT; the 4th letters are T, C, E, A. A comes first: **Catalog**.

Q6 **Answer: (b) E**

New word: BAOSULET. 7th letter = **E**.

Q7 **Answer: (c) K**

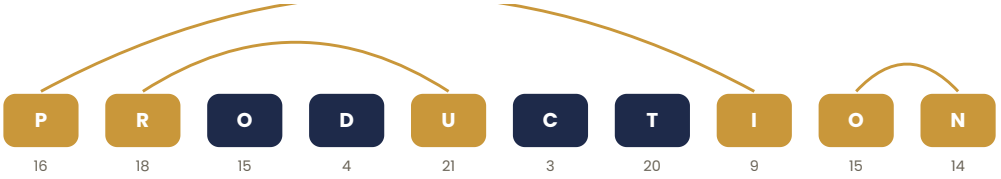
The word is TRUNK; its last letter is **K**.

Q8 **Answer: (a) 0**

MAJOR → NBKPS, which has no vowels: **0**.

Q9 **Answer: (c) 3**

P–I (positions 1 and 8, P16 - I9 = 7), R–U (positions 2 and 5, 21 - 18 = 3) and O–N (positions 9 and 10). Answer **3**.



Q10 **Answer: (b) O**

INSTR → RTSNI, U stays, CTION → NOITC. New word RTSNIUNOITC; 4th from the right = **O**.

Q11 **Answer: (b) Elephant**

Order: Elegant, Elegy, Element, Elephant, Elevate. Fourth = **Elephant**.

Q12 **Answer: (b) One**

Alphabetical: A C D E M O R T. Compare with D E M O C R A T: only T (8th) is unchanged. Answer **one**.

Q13 **Answer: (b) 2**

C is 3rd in both and E is 5th in both. Answer **2**.

Q14 **Answer: (a) Signal, Silent, Simple, Sincere, Sinner**

Third letters G, L, M, N, N decide the first three: Signal, Silent, Simple. For SINCERE and SINNER compare the 4th letters C and N: **Signal, Silent, Simple, Sincere, Sinner**.

Q15 **Answer: (c) STATION**

CONSTITUTION has no A, so **STATION** cannot be formed.

Q16 **Answer: (d) MATRON**

EXAMINATION has no R, so **MATRON** cannot be formed.

Q17 **Answer: (b) O**

Alphabetical: A E L N O P R S. 2nd from the left is E (position 2); 3rd to its right is position 5 = **O**.

Q18 **Answer: (b) T**

The two halves swap: AINTS + COMPL = AINTSCOMPL. 7th from the right = 4th from the left = **T**.

Q19 **Answer: (b) Q**

G→F, O→P, V→U, E→F, R→Q, N→M, M→L, E→F, N→M, T→S: FPUFQMLFMS. 5th letter = **Q**.

Q20 **Answer: (b) Negative**

Order: Necessary, Nectar, Negative, Neglect, Neighbour (NEGA before NEGL). Middle = **Negative**.

Q21 **Answer: (b) One**

G (7) at position 4 and E (5) at position 6: gap 2 in the word and 2 in the alphabet. No other pair works. Answer **one**.



Q22 **Answer: (b) D**

Reversed: NOITANITSED. 4th to the right of the 7th letter is the 11th letter = **D**.

Q23 **Answer: (b) 1**

Of the pairs PI, RU, ON only RU goes forward (R → U). Answer **1**.

Q24 **Answer: (b) 6**

New word BAOSULET: B is 1st and T is 8th, so 8 − 1 − 1 = **6** letters lie between them.

Q25 **Answer: (d) COUNTS**

STATION needs A and CONTEST needs E (neither is available); INTUITION needs three I's but CONSTITUTION has only two. **COUNTS** needs C, O, U, N, T, S, all available.

Write positions under the letters before you start. For pairs compare the word gap with the alphabet gap (both directions unless told otherwise). For dictionary order find the first differing column. Perform swaps and replacements on paper, then count in the new word. For word formation, check every letter and its count.

MODULE 11 · INTERMEDIATE

Mathematical operations

Minimum duration **1 day** | Level **Intermediate**

In these questions the usual arithmetic signs are replaced by other symbols or letters, or the signs of an equation are in the wrong places. You must substitute, interchange or fill in the operators and then evaluate correctly with BODMAS. This module covers the order of operations, symbol substitution, interchanging signs or numbers, missing operators and newly defined operations, with 25 solved questions checked by a program.

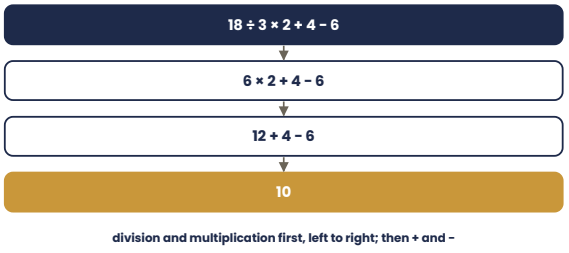
IN THIS MODULE

11.1 Order of operations
11.2 Symbol substitution
11.3 Interchanging signs and numbers
11.4 Missing operators and defined operations
11.5 Practice questions
11.6 Solutions

11.1 Order of operations

11.1.1 BODMAS

After the signs are replaced, the expression is evaluated in the normal order: **B**rackets, **O**rders (powers, roots), **D**ivision and **M**ultiplication (left to right), **A**ddition and **S**ubtraction (left to right).



EVALUATING WITH BODMAS

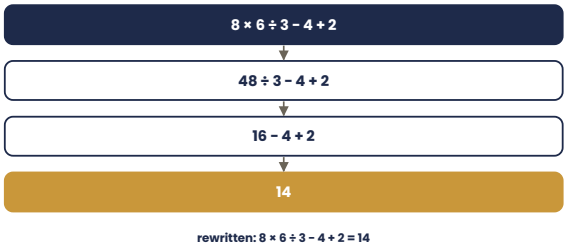
WATCH OUT
Division and multiplication have the **same** priority: work left to right. 18 ÷ 3 × 2 = 6 × 2 = 12, not 18 ÷ 6 = 3.

11.2 Symbol substitution

11.2.1 Rewrite first, then evaluate

"If + means ×, − means ÷, × means − and ÷ means +, find 8 + 6 − 3 × 4 ÷ 2." Rewrite the **whole** expression with the real signs in one step, then apply BODMAS. Never evaluate while substituting.

Given sign	+	−	×	÷
Means	×	÷	−	+



SUBSTITUTE EVERY SIGN, THEN EVALUATE

SHORTCUT TRICK
Write the substitution table in a single line above the expression (+ → ×, − → ÷, × → −, ÷ → +). Copying the expression with the new signs takes 10 seconds and removes almost all errors.

11.2.2 Letter codes for signs

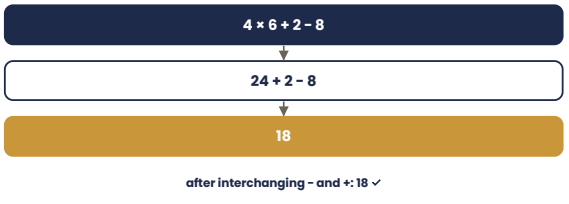
The same idea with letters: "P means +, Q means −, R means ×, S means ÷". 12 R 3 P 18 S 6 Q 4 = 12 × 3 + 18 ÷ 6 − 4 = 36 + 3 − 4 = **35**.

11.3 Interchanging signs and numbers

11.3.1 Which two signs must be interchanged?

Try each option: interchange the two signs everywhere in the equation and evaluate the left side. Only one option gives the stated right side.

WORKED EXAMPLE
Which two signs must be interchanged to make 4 × 6 − 2 + 8 = 18 correct?
Interchanging − and + gives 4 × 6 + 2 − 8 = 24 + 2 − 8 = 18 ✓. The other pairs give different values.



TESTING AN INTERCHANGE

TIP
Start with the option that changes a × or ÷, because those change the value the most; a quick estimate often rules out three options at once.

11.3.2 Interchanging numbers

"Which two numbers must be interchanged to make 6 × 4 + 2 = 16 correct?" Interchange 4 and 2: 6 × 2 + 4 = 16 ✓.

11.4 Missing operators and defined operations

11.4.1 Filling in the operators

For "12 ? 4 ? 3 ? 5 = 14", test the option sequences. Most can be rejected by a rough size check: if the result is small, there is probably a ÷ or -.

11.4.2 Newly defined operations

A rule such as **$a * b = a^2 + b$** defines a new operation. Substitute the numbers into the rule carefully, and evaluate inner brackets first: $(2 * 3) * 1 = (4 + 3) * 1 = 7 * 1 = 49 + 1 =$ **50**.

Given pattern	Rule	Apply
3 @ 4 = 25, 5 @ 12 = 169	$a^2 + b^2$	6 @ 8 = 100
8 * 3 = 55, 6 * 2 = 32	$a^2 - b^2$	7 * 4 = 33
5 $ 3 = 7	ab - (a + b)	4 $ 2 = 2

11.5 Practice questions

11.5.1 Basic level

Q1 If + means ×, - means ÷, × means - and ÷ means +, what is 8 + 6 - 3 × 4 ÷ 2?

(a) 12 **(b)** 14 **(c)** 16 **(d)** 10

Q2 If P means +, Q means -, R means × and S means ÷, what is 12 R 3 P 18 S 6 Q 4?

(a) 33 **(b)** 35 **(c)** 37 **(d)** 31

Q3 Using the signs of question 1, what is 15 - 3 × 4 + 2 ÷ 6?

(a) 3 **(b)** 5 **(c)** 7 **(d)** 1

Q4 Find the value of 18 ÷ 3 × 2 + 4 - 6.

(a) 10 **(b)** 1 **(c)** 7 **(d)** 13

Q5 If × means + and + means ×, what is 5 × 4 + 3?

(a) 27 **(b)** 17 **(c)** 12 **(d)** 23

Q6 If ÷ means - and - means ÷, what is 20 - 5 ÷ 2?

(a) 2 **(b)** 6 **(c)** 8 **(d)** 18

Q7 Which two signs must be interchanged to make 8 + 2 × 7 - 2 = 21 correct?

(a) + and × **(b)** + and - **(c)** × and - **(d)** × and ÷

Q8 Using the P, Q, R, S code of question 2, what is 20 S 4 R 3 P 5?

(a) 20 **(b)** 18 **(c)** 22 **(d)** 25

11.5.2 Intermediate level

Q9 Which two signs must be interchanged to make 4 × 6 − 2 + 8 = 18 correct?

(a) − and + **(b)** × and − **(c)** × and + **(d)** + and ÷

Q10 Which two signs must be interchanged to make 15 ÷ 3 + 6 × 2 = 48 correct?

(a) ÷ and + **(b)** ÷ and × **(c)** + and × **(d)** + and −

Q11 Choose the signs that make 12 ? 4 ? 3 ? 5 = 14 correct.

(a) ÷, ×, + **(b)** ×, ÷, + **(c)** +, ×, − **(d)** −, ×, +

Q12 Choose the signs that make 7 ? 4 ? 3 ? 2 = 31 correct.

(a) ×, +, × **(b)** +, ×, × **(c)** ×, ×, − **(d)** +, +, ×

Q13 Which two numbers must be interchanged to make 6 × 4 + 2 = 16 correct?

(a) 4 and 2 **(b)** 6 and 4
(c) 6 and 2 **(d)** No interchange is needed

Q14 If A means ÷, B means ×, C means + and D means −, what is 36 A 4 B 3 C 7 D 2?

(a) 30 **(b)** 32 **(c)** 34 **(d)** 28

Q15 If > means +, < means −, + means ÷ and − means ×, what is 36 + 4 − 3 > 2 < 5?

(a) 24 **(b)** 22 **(c)** 26 **(d)** 30

Q16 If ÷ and − are interchanged, which equation becomes correct?

(a) 10 ÷ 2 − 5 + 3 = 9 **(b)** 12 − 4 ÷ 2 × 3 = 3
(c) 20 − 5 ÷ 1 + 2 = 5 **(d)** 8 ÷ 4 − 2 × 1 = 3

11.5.3 Advanced level

Q17 If + and − are interchanged and the numbers 4 and 8 are interchanged, which equation becomes correct?

(a) 4 + 8 − 2 = 10 **(b)** 8 − 4 + 6 = 6 **(c)** 4 − 6 + 8 = 14 **(d)** 8 + 4 − 1 = 11

Q18 If $a * b = a^2 + b$, find $(2 * 3) * 1$.

(a) 49 **(b)** 50 **(c)** 51 **(d)** 10

Q19 If $a \$ b = ab - (a + b)$, find $(4 \$ 2) \$ 3$.

(a) 1 **(b)** 2 **(c)** 3 **(d)** 0

Q20 If 8 * 3 = 55 and 6 * 2 = 32, find 7 * 4.

(a) 33 **(b)** 65 **(c)** 45 **(d)** 28

Q21 If 2 # 3 = 13, 3 # 4 = 25 and 4 # 5 = 41, find 5 # 6.

(a) 55 **(b)** 61 **(c)** 66 **(d)** 71

Q22 Choose the signs that make 18 ? 6 ? 4 ? 2 = 20 correct.

(a) ÷, ×, + **(b)** −, +, × **(c)** +, −, × **(d)** ×, ÷, −

Q23 Choose the signs that make 9 ? 3 ? 4 ? 6 = 6 correct.

(a) ÷, ×, − **(b)** −, ×, ÷ **(c)** +, −, × **(d)** ×, ÷, −

Q24 Which two signs must be interchanged to make 18 + 3 × 2 ÷ 4 = 16 correct?

(a) + and × **(b)** + and ÷ **(c)** × and ÷ **(d)** + and −

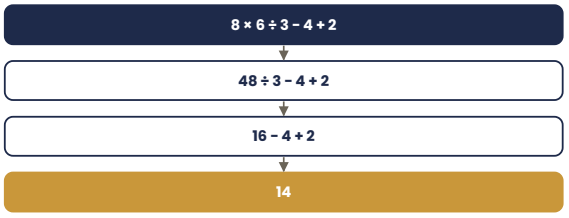
Q25 If L means +, M means −, N means × and P means ÷, what is 28 N 4 L 64 P 8 M 30?

(a) 86 **(b)** 90 **(c)** 94 **(d)** 80

11.6 Solutions

Q1 **Answer: (b) 14**

Rewrite: 8 × 6 ÷ 3 − 4 + 2 = 48 ÷ 3 − 4 + 2 = 16 − 4 + 2 = **14**.



Q2 **Answer: (b) 35**

12 × 3 + 18 ÷ 6 − 4 = 36 + 3 − 4 = **35**.

Q3 **Answer: (a) 3**

Rewrite: 15 ÷ 3 − 4 × 2 + 6 = 5 − 8 + 6 = **3**.

Q4 **Answer: (a) 10**

18 ÷ 3 = 6; 6 × 2 = 12; 12 + 4 − 6 = **10**.

Q5 **Answer: (b) 17**

Rewrite: 5 + 4 × 3 = 5 + 12 = **17**.

Q6 **Answer: (a) 2**

Rewrite: 20 ÷ 5 − 2 = 4 − 2 = **2**.

Q7 **Answer: (a) + and ×**

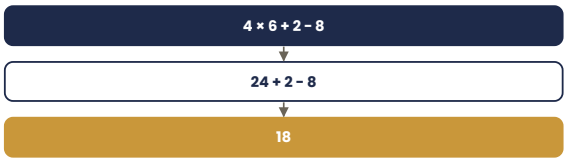
Interchanging + and × gives 8 × 2 + 7 − 2 = 16 + 7 − 2 = 21 ✓.

Q8 **Answer: (a) 20**

20 ÷ 4 × 3 + 5 = 5 × 3 + 5 = **20**.

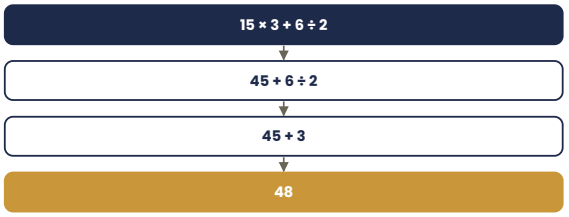
Q9 **Answer: (a) − and +**

Interchanging − and +: 4 × 6 + 2 − 8 = 18 ✓.



Q10 **Answer: (b) ÷ and ×**

Interchanging ÷ and ×: 15 × 3 + 6 ÷ 2 = 45 + 3 = 48 ✓.



Q11 **Answer: (a) ÷, ×, +**

12 ÷ 4 × 3 + 5 = 3 × 3 + 5 = **14** ✓. (The others give 12 × 4 ÷ 3 + 5 = 21, 12 + 4 × 3 − 5 = 19 and 12 − 4 × 3 + 5 = 5.)

Q12 **Answer: (b) +, ×, ×**

7 + 4 × 3 × 2 = 7 + 24 = **31** ✓.

Q13 **Answer: (a) 4 and 2**

Interchanging 4 and 2: 6 × 2 + 4 = 12 + 4 = 16 ✓.

Q14 **Answer: (b) 32**

36 ÷ 4 × 3 + 7 − 2 = 9 × 3 + 5 = **32**.

Q15 **Answer: (a) 24**

Rewrite: 36 ÷ 4 × 3 + 2 − 5 = 27 + 2 − 5 = **24**.

Q16 **Answer: (c) 20 − 5 ÷ 1 + 2 = 5**

Option (c) becomes 20 ÷ 5 − 1 + 2 = 4 − 1 + 2 = 5 ✓. The others: (a) 10 − 2 ÷ 5 + 3 = 12.6, (b) 12 ÷ 4 − 2 × 3 = −3, (d) 8 − 4 ÷ 2 × 1 = 6.

Q17 **Answer: (b) 8 − 4 + 6 = 6**

Option (b) becomes 4 + 8 − 6 = 6 ✓. The others: (a) 8 − 4 + 2 = 6, (c) 8 + 6 − 4 = 10, (d) 4 − 8 + 1 = −3.

Q18 **Answer: (b) 50**

2 ∗ 3 = 4 + 3 = 7; 7 ∗ 1 = 49 + 1 = **50**.

Q19 **Answer: (a) 1**

4 $ 2 = 8 − 6 = 2; 2 $ 3 = 6 − 5 = **1**.

Q20 **Answer: (a) 33**

Rule $a^2 - b^2$: 64 − 9 = 55, 36 − 4 = 32. So 49 − 16 = **33**.

Q21 **Answer: (b) 61**

Rule $a^2 + b^2$: 4 + 9 = 13, 9 + 16 = 25, 16 + 25 = 41. So 25 + 36 = **61**.

Q22 **Answer: (b) −, +, ×**

18 − 6 + 4 × 2 = 18 − 6 + 8 = **20** ✓.

Q23 **Answer: (a) ÷, ×, −**

9 ÷ 3 × 4 − 6 = 12 − 6 = **6** ✓.

Q24 **Answer: (b) + and ÷**

Interchanging + and ÷: 18 ÷ 3 × 2 + 4 = 12 + 4 = 16 ✓.

Q25 **Answer: (b) 90**

28 × 4 + 64 ÷ 8 − 30 = 112 + 8 − 30 = **90**.

Rewrite the whole expression with the real signs before evaluating, then apply BODMAS (÷ and × left to right, then + and − left to right). For interchange questions, apply the swap everywhere and test each option; for missing operators, use a size estimate to reject options quickly; for defined operations, find the rule from all given examples.

MODULE 12 · INTERMEDIATE

Inequalities

Minimum duration **1 day** | Level **Intermediate**

Inequality questions give relations such as A ≥ B > C = D and ask which conclusions definitely follow. The skill is joining relations along a chain correctly, recognising when no relation can be drawn, and spotting the either-or case. This module covers direct and combined inequalities, either-or conclusions and the coded inequalities used in bank exams, with 25 solved questions verified by a program that tests every possible ordering.

IN THIS MODULE

12.1 Basics
12.2 Combining relations
12.3 Either-or conclusions
12.4 Coded inequalities
12.5 Practice questions
12.6 Solutions

12.1 Basics

12.1.1 Symbols and what "follows" means

Symbol	Meaning	Symbol	Meaning
A > B	A is greater than B	A < B	A is less than B
A ≥ B	A is greater than or equal to B	A ≤ B	A is less than or equal to B
A = B	A equals B		

A conclusion **follows** only if it is true in **every** situation allowed by the statements. If it is true in some situations and false in others, it does not follow.

12.1.2 Reading a chain

Write the statements as one chain when they share letters: A ≥ B, B > C, C = D becomes **A ≥ B > C = D**. To compare two letters, walk along the chain between them.



from A to D: ≥, >, = → all point the same way, at least one strict → A > D

WALKING ALONG A CHAIN

12.2 Combining relations

12.2.1 The combining rules

Signs on the path from X to Y	Result	Example
all =	X = Y	A = B = C → A = C
only ≥ and = (no >)	X ≥ Y	A ≥ B = C → A ≥ C
≥ / = with at least one >	X > Y	A ≥ B > C → A > C
signs point in opposite directions (> and <, ≥ and ≤ …)	no relation	A > B < C → A ? C

SHORTCUT TRICK
Look only at the signs **between** the two letters. If any sign points the other way, stop: no definite relation. Otherwise the strongest sign wins (> beats ≥ beats =).

WATCH OUT
A ≥ B does **not** mean A > B. A conclusion "A > B" from A ≥ B does not follow, because A = B is possible.

12.3 Either-or conclusions

12.3.1 When neither follows alone but one must be true

If two conclusions involve the **same two letters**, neither follows on its own, and together they cover every possibility (one is always true), the answer is "Either I or II follows". The common pairs are:

Known relation	Conclusion I	Conclusion II
X ≥ Y (or no relation)	X > Y	X = Y
no relation	X ≥ Y	X < Y
no relation	X > Y	X ≤ Y

WORKED EXAMPLE
Statement: W ≥ X ≥ Y. I: W > Y. II: W = Y.
W ≥ Y is certain, so W is either greater than Y or equal to it. Neither conclusion is certain, but one of them must be true: **either I or II follows**.

TIP
Either-or never applies to two conclusions about different pairs of letters, and never when one of the conclusions already follows.

12.4 Coded inequalities

12.4.1 Decode first

Bank exams replace the signs by symbols. Write the code table once, translate every statement and conclusion, and then solve as usual. This module uses:

Code	@	#	$	%	&
Means	>	≥	=	≤	<

WORKED EXAMPLE

A @ B # C $ D. I: A @ D. II: B # D.

Decoded: A > B ≥ C = D. I: A > D (path >, ≥, =) ✓. II: B ≥ D (path ≥, =) ✓. **Both follow**.

12.5 Practice questions

12.5.1 Basic level

Q1 Statements: A > B ≥ C. Conclusions: I. A > C II. B > C

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q2 Statements: P ≥ Q = R. Conclusions: I. P ≥ R II. R > P

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q3 Statements: X < Y ≤ Z. Conclusions: I. Z > X II. Z ≥ Y

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q4 Statements: M = N > O. Conclusions: I. M > O II. O < N

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q5 Statements: A ≥ B, B ≤ C. Conclusions: I. A > C II. A ≤ C

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q6 Statements: P > Q, Q > R, R ≥ S. Conclusions: I. P > S II. Q ≥ S

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q7 Statements: D ≤ E < F = G. Conclusions: I. G > D II. E = G

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q8 Statements: K < L, L > M. Conclusions: I. K < M II. M < K

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

12.5.2 Intermediate level

Q9 Statements: A ≥ B > C ≤ D. Conclusions: I. A > C II. D > B

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q10 Statements: P ≤ Q = R ≥ S. Conclusions: I. P ≤ R II. S ≤ Q

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q11 Statements: J > K ≥ L, L = M. Conclusions: I. K > M II. K = M

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q12 Statements: W ≥ X ≥ Y. Conclusions: I. W > Y II. W = Y

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q13 Statements: A < B ≤ C, C > D. Conclusions: I. A < D II. B > D

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q14 Statements: Q ≥ R = S < T. Conclusions: I. Q ≥ S II. T > R

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q15 Statements: E > F ≤ G = H. Conclusions: I. H ≥ F II. E > G

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q16 Statements: L ≤ M < N, O > N. Conclusions: I. O > L II. M < O

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

12.5.3 Advanced level

Q17 Statements: A @ B # C $ D. Conclusions: I. A @ D II. B # D (@ means >, # ≥, $ =, % ≤, & <)

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q18 Statements: P % Q & R $ S. Conclusions: I. P & S II. Q $ S (@ means >, # ≥, $ =, % ≤, & <)

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q19 Statements: M # N, N % O, O & P. Conclusions: I. M @ P II. P @ N (@ means >, # ≥, $ =, % ≤, & <)

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q20 Statements: X $ Y @ Z, Z # W. Conclusions: I. X @ W II. Y $ W (@ means >, # ≥, $ =, % ≤, & <)

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q21 Statements: A & B, B # C, C @ D. Conclusions: I. A @ D II. B @ D (@ means >, # ≥, $ =, % ≤, & <)

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q22 Statements: F # G, G $ H, H # I. Conclusions: I. F @ I II. F $ I (@ means >, # ≥, $ =, % ≤, & <)

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q23 Statements: T % U, U & V, V $ W. Conclusions: I. W @ T II. T $ W (@ means >, # ≥, $ =, % ≤, & <)

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q24 Statements: K @ L, L % M, N # M. Conclusions: I. K @ M II. N # L (@ means >, # ≥, $ =, % ≤, & <)

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q25 Statements: R # S, T % S, T @ U. Conclusions: I. R # T II. U & S (@ means >, # ≥, $ =, % ≤, & <)

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

12.6 Solutions

Q1 **Answer: (a) Only I follows**
Chain: A > B ≥ C. I (A > C) follows; II (B > C) is not certain. Answer: **Only I follows**.

Q2 **Answer: (a) Only I follows**
Chain: P ≥ Q = R. I (P ≥ R) follows; II (R > P) is definitely false. Answer: **Only I follows**.

Q3 **Answer: (e) Both I and II follow**
Chain: X < Y ≤ Z. I (Z > X) follows; II (Z ≥ Y) follows. Answer: **Both I and II follow**.

Q4 **Answer: (e) Both I and II follow**
Chain: M = N > O. I (M > O) follows; II (O < N) follows. Answer: **Both I and II follow**.

Q5 **Answer: (c) Either I or II follows**
Chain: A ≥ B, B ≤ C. I (A > C) is not certain; II (A ≤ C) is not certain. The two conclusions are about the same pair, and one of them is always true, so either I or II follows. Answer: **Either I or II follows**.

Q6 **Answer: (e) Both I and II follow**
Chain: P > Q, Q > R, R ≥ S. I (P > S) follows; II (Q ≥ S) follows. Answer: **Both I and II follow**.

Q7 **Answer: (a) Only I follows**
Chain: D ≤ E < F = G. I (G > D) follows; II (E = G) is definitely false. Answer: **Only I follows**.

Q8 **Answer: (d) Neither I nor II follows**
Chain: K < L, L > M. I (K < M) is not certain; II (M < K) is not certain. The two conclusions can both be false together, so this is not an either-or case. Answer: **Neither I nor II follows**.

Q9 **Answer: (a) Only I follows**
Chain: A ≥ B > C ≤ D. I (A > C) follows; II (D > B) is not certain. Answer: **Only I follows**.



Q10 **Answer: (e) Both I and II follow**

Chain: P ≤ Q = R ≥ S. I (P ≤ R) follows; II (S ≤ Q) follows. Answer: **Both I and II follow**.



Q11 **Answer: (c) Either I or II follows**

Chain: J > K ≥ L, L = M. I (K > M) is not certain; II (K = M) is not certain. The two conclusions are about the same pair, and one of them is always true, so either I or II follows. Answer: **Either I or II follows**.



Q12 **Answer: (c) Either I or II follows**

Chain: W ≥ X ≥ Y. I (W > Y) is not certain; II (W = Y) is not certain. The two conclusions are about the same pair, and one of them is always true, so either I or II follows. Answer: **Either I or II follows**.



Q13 **Answer: (d) Neither I nor II follows**

Chain: A < B ≤ C, C > D. I (A < D) is not certain; II (B > D) is not certain. Answer: **Neither I nor II follows**.



Q14 **Answer: (e) Both I and II follow**

Chain: Q ≥ R = S < T. I (Q ≥ S) follows; II (T > R) follows. Answer: **Both I and II follow**.



Q15 **Answer: (a) Only I follows**

Chain: E > F ≤ G = H. I (H ≥ F) follows; II (E > G) is not certain. Answer: **Only I follows**.



Q16 **Answer: (e) Both I and II follow**

Chain: L ≤ M < N, O > N. I (O > L) follows; II (M < O) follows. Answer: **Both I and II follow**.



Q17 **Answer: (e) Both I and II follow**

Decoded: A > B ≥ C = D; I: A > D; II: B ≥ D. Chain: A > B ≥ C = D. I (A > D) follows; II (B ≥ D) follows. Answer: **Both I and II follow**.



Q18 **Answer: (a) Only I follows**

Decoded: P ≤ Q < R = S; I: P < S; II: Q = S. Chain: P ≤ Q < R = S. I (P < S) follows; II (Q = S) is definitely false. Answer: **Only I follows**.



Q19 **Answer: (b) Only II follows**

Decoded: M ≥ N, N ≤ O, O < P; I: M > P; II: P > N. Chain: M ≥ N, N ≤ O, O < P. I (M > P) is not certain; II (P > N) follows. Answer: **Only II follows**.



Q20 **Answer: (a) Only I follows**

Decoded: X = Y > Z, Z ≥ W; I: X > W; II: Y = W. Chain: X = Y > Z, Z ≥ W. I (X > W) follows; II (Y = W) is definitely false. Answer: **Only I follows**.



Q21 **Answer: (b) Only II follows**

Decoded: A < B, B ≥ C, C > D; I: A > D; II: B > D. Chain: A < B, B ≥ C, C > D. I (A > D) is not certain; II (B > D) follows. Answer: **Only II follows**.



Q22 **Answer: (c) Either I or II follows**

Decoded: F ≥ G, G = H, H ≥ I; I: F > I; II: F = I. Chain: F ≥ G, G = H, H ≥ I. I (F > I) is not certain; II (F = I) is not certain. The two conclusions are about the same pair, and one of them is always true, so either I or II follows. Answer: **Either I or II follows**.



Q23 **Answer: (a) Only I follows**

Decoded: T ≤ U, U < V, V = W; I: W > T; II: T = W. Chain: T ≤ U, U < V, V = W. I (W > T) follows; II (T = W) is definitely false. Answer: **Only I follows**.



Q24 **Answer: (b) Only II follows**

Decoded: K > L, L ≤ M, N ≥ M; I: K > M; II: N ≥ L. Chain: K > L, L ≤ M, N ≥ M. I (K > M) is not certain; II (N ≥ L) follows. Answer: **Only II follows**.



Q25 **Answer: (e) Both I and II follow**

Decoded: R ≥ S, T ≤ S, T > U; I: R ≥ T; II: U < S. Chain: R ≥ S, T ≤ S, T > U. I (R ≥ T) follows; II (U < S) follows. Answer: **Both I and II follow**.



A conclusion follows only if it holds in every case. Walk along the chain between the two letters: any sign pointing the other way means no relation; otherwise > beats ≥ beats =. Either-or applies only to two conclusions on the same pair that are each uncertain but together always true. In coded questions, decode everything first.

MODULE 13 · INTERMEDIATE

Syllogism

Minimum duration **2 days** | Level **Intermediate**

A syllogism gives two or three statements about groups ("All pens are books", "Some books are cups") and asks which conclusions must follow, however true or false the statements are in real life. This module covers the four standard statements and their Venn diagrams, the method of checking every possible diagram, quick combination rules, complementary (either-or) pairs, possibility conclusions and "only a few" and "only" statements, followed by 25 solved questions checked by a program that tests every possible diagram.

IN THIS MODULE

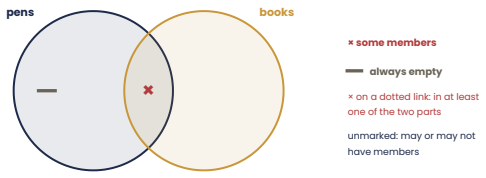
13.1 Basics
13.2 Method and quick rules
13.3 Either-or pairs
13.4 Possibility conclusions
13.5 "Only a few" and "only"
13.6 Practice questions
13.7 Solutions

13.1 Basics

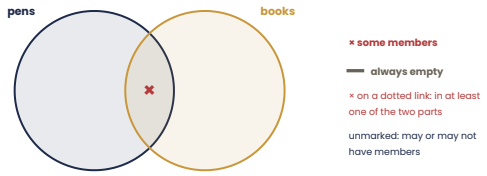
13.1.1 The four standard statements

Treat every statement as true, even if it is absurd in real life ("All cats are tables"). Use only what the statements say. The four standard forms are:

Statement	Meaning	Diagram
All A are B	every A is inside B	circle A inside circle B
No A are B	A and B do not overlap	separate circles
Some A are B	at least one A is a B	overlapping circles, cross in the overlap
Some A are not B	at least one A is outside B	cross in the part of A outside B



ALL PENS ARE BOOKS: THE PART OF PENS OUTSIDE BOOKS IS ALWAYS EMPTY



SOME PENS ARE BOOKS: THE OVERLAP IS CERTAINLY OCCUPIED

WATCH OUT
"Some" means **at least one**, possibly all. "Some A are B" does not imply "Some A are not B".

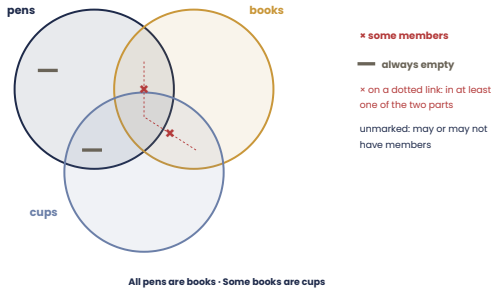
13.1.2 Conversions that always work

From	Always follows	Does not follow
All A are B	Some A are B; Some B are A	All B are A
No A are B	No B are A; Some A are not B	—
Some A are B	Some B are A	All A are B
Some A are not B	(nothing new)	Some B are not A

13.2 Method and quick rules

13.2.1 Draw the basic diagram, then test every case

Draw the **basic** diagram that shows only what the statements force: dashes for regions that must be empty and crosses for regions that must have members. A conclusion **follows** only if it is true in **every** way of filling the unmarked regions.



A BASIC DIAGRAM WITH THREE GROUPS

WORKED EXAMPLE
All pens are books. Some books are cups. Does "Some pens are cups" follow?
The cross is in books ∩ cups, but it may be outside pens. Pens and cups may or may not overlap, so the conclusion **does not follow**.

13.2.2 Quick combination rules (for two statements A–B, B–C)

Statement 1 (A–B)	Statement 2 (B–C)	Conclusion (A–C)
All	All	All A are C
All	No	No A are C
Some	All	Some A are C
Some	No	Some A are not C
No	All (C–B direction: All C are B)	No A are C
Some	Some	no definite conclusion
No	No	no definite conclusion
All	Some	no definite conclusion

SHORTCUT TRICK
The middle term (B) must be fully covered (an **All** statement with B as subject, or a No statement) for any definite conclusion. Two "Some" statements never give a definite conclusion between A and C.

13.3 Either-or pairs

13.3.1 Complementary conclusions

If neither conclusion follows, check whether they form a **complementary pair** about the same two groups: one of them must be true in every case.

Pair	Why complementary
Some A are B / No A are B	either the overlap has members or it does not
All A are B / Some A are not B	either every A is in B or at least one is outside

WORKED EXAMPLE
Some boys are girls. Some girls are men. I: Some boys are men. II: No boys are men.
Neither is certain, but boys and men either overlap or they do not: **either I or II follows**.

13.4 Possibility conclusions

13.4.1 "It is possible that …"

A possibility conclusion follows if there is **at least one** diagram, consistent with the statements, in which it is true. It fails only if the statements make it impossible.

Statements	Possibility	Result
Some A are B, All B are C	all A are C	possible: follows

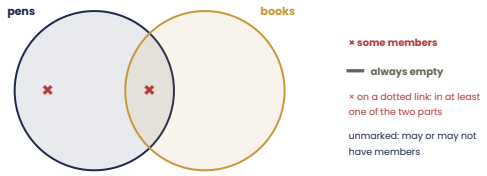
Statements	Possibility	Result
All X are Y, No Y are Z	some Z are X	impossible: X is inside Y, which has no Z

TIP
For a possibility, try to draw a diagram that makes it true. If the statements force a dash in the region you need, it is impossible; otherwise it follows.

13.5 "Only a few" and "only"

13.5.1 Two newer statement types

Statement	Means	Consequence
Only a few A are B	Some A are B **and** some A are not B	"All A are B" is impossible
Only A are B	All B are A	every B is inside A



ONLY A FEW PENS ARE BOOKS: TWO CROSSES

13.6 Practice questions

13.6.1 Basic level

Q1 Statements: All pens are books. All books are cups. Conclusions: I. All pens are cups. II. Some cups are pens.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q2 Statements: Some cats are dogs. All dogs are rats. Conclusions: I. Some cats are rats. II. All rats are dogs.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q3 Statements: No fans are bulbs. All bulbs are lamps. Conclusions: I. No fans are lamps. II. Some lamps are not fans.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q4 Statements: All roses are flowers. No flowers are trees. Conclusions: I. No roses are trees. II. Some trees are roses.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q5 Statements: Some boys are girls. Some girls are men. Conclusions: I. Some boys are men. II. No boys are men.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q6 Statements: All cars are buses. Some buses are vans. Conclusions: I. Some cars are vans. II. Some vans are buses.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q7 Statements: No apples are mangoes. No mangoes are grapes. Conclusions: I. No apples are grapes. II. Some apples are grapes.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q8 Statements: Some keys are locks. No locks are doors. Conclusions: I. Some keys are not doors. II. Some doors are keys.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

13.6.2 Intermediate level

Q9 Statements: Some tables are chairs. All chairs are desks. Conclusions: I. Some desks are tables. II. It is possible that all tables are desks.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q10 Statements: All stars are moons. No moons are suns. Conclusions: I. No stars are suns. II. It is possible that some suns are stars.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q11 Statements: Only a few pens are books. All books are copies. Conclusions: I. Some pens are copies. II. Some pens are not books.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q12 Statements: Only cups are plates. Some cups are bowls. Conclusions: I. All plates are cups. II. Some bowls are plates.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q13 Statements: No lions are tigers. Some tigers are bears. Conclusions: I. Some bears are not lions. II. It is possible that some lions are bears.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q14 Statements: All mangoes are fruits. Some fruits are not seeds. Conclusions: I. All mangoes are seeds. II. Some mangoes are not seeds.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q15 Statements: Some pencils are erasers. Some erasers are rulers. Conclusions: I. Some pencils are rulers. II. It is possible that all pencils are rulers.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q16 Statements: All doctors are singers. All doctors are dancers. Conclusions: I. Some singers are dancers. II. Some dancers are not singers.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

13.6.3 Advanced level

Q17 Statements: All apples are balls. Some balls are cats. No cats are dogs. Conclusions: I. Some balls are not dogs. II. It is possible that no apples are dogs.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q18 Statements: Some cards are files. All files are boxes. No boxes are walls. Conclusions: I. Some cards are not walls. II. Some files are walls.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q19 Statements: No pins are quills. All quills are rods. Some rods are sticks. Conclusions: I. Some rods are not pins. II. It is possible that some sticks are quills.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q20 Statements: All books are pens. All pens are tables. No tables are chairs. Conclusions: I. No books are chairs. II. Some chairs are pens.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q21 Statements: Some ants are bees. Some bees are cows. Some cows are deer. Conclusions: I. Some ants are deer. II. No ants are deer.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q22 Statements: Only a few kites are lamps. All lamps are mats. No mats are nets. Conclusions: I. Some kites are not nets. II. It is possible that all kites are mats.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q23 Statements: All ants are bats. No bats are cats. Some cats are dogs. Conclusions: I. Some ants are dogs. II. Some dogs are not ants.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q24 Statements: Some maps are notes. No notes are oils. All oils are paints. Conclusions: I. Some paints are not notes. II. Some maps are not oils.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

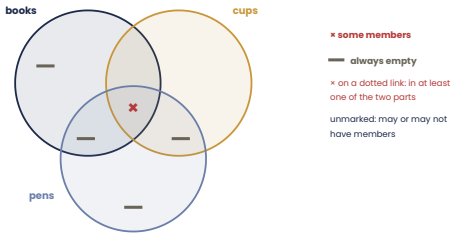
Q25 Statements: All rings are stones. Some stones are tools. All tools are units. Conclusions: I. Some stones are units. II. Some rings are units.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

13.7 Solutions

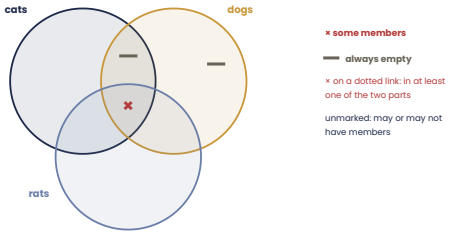
Q1 **Answer: (e) Both I and II follow**

All + All gives All pens are cups, and since pens exist inside cups, some cups are pens. Answer: **Both I and II follow**.



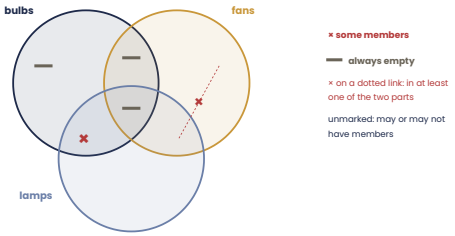
Q2 **Answer: (a) Only I follows**

Some + All: the cats that are dogs are also rats, so I follows. All dogs are rats cannot be reversed, so II does not follow. Answer: **Only I follows**.



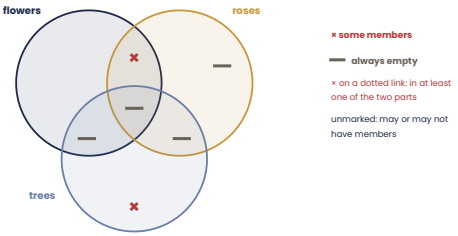
Q3 **Answer: (b) Only II follows**

Fans may overlap the part of lamps outside bulbs, so I is not certain. The bulbs are lamps and are not fans, so some lamps are not fans: II follows. Answer: **Only II follows**.



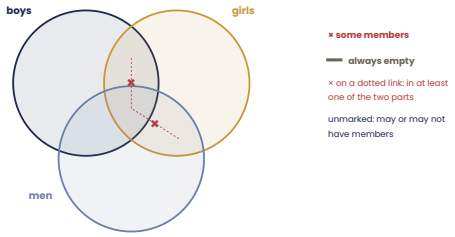
Q4 **Answer: (a) Only I follows**

All + No gives No roses are trees (I). II contradicts I, so it does not follow. Answer: **Only I follows**.



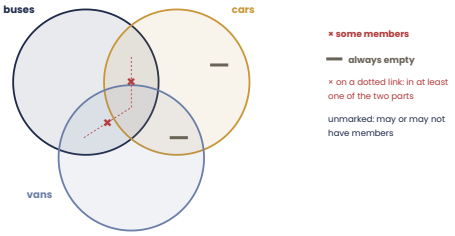
Q5 **Answer: (c) Either I or II follows**

Some + Some gives no definite conclusion, but I and II form a complementary pair about boys and men. Answer: **Either I or II follows**.



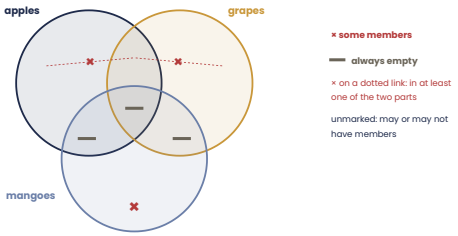
Q6 **Answer: (b) Only II follows**

The vans in buses may lie outside cars, so I is uncertain. Some buses are vans converts to some vans are buses: II follows. Answer: **Only II follows**.



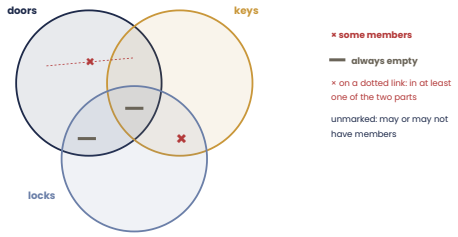
Q7 **Answer: (c) Either I or II follows**

No + No gives nothing definite about apples and grapes, but I and II are complementary. Answer: **Either I or II follows**.



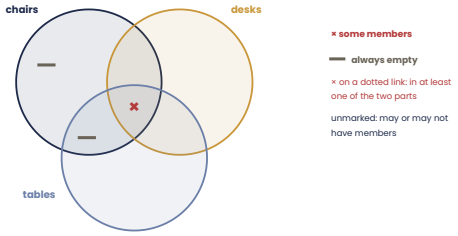
Q8 **Answer: (a) Only I follows**

Some + No: the keys that are locks are not doors, so I follows. Doors and keys may or may not overlap, so II is uncertain. Answer: **Only I follows**.



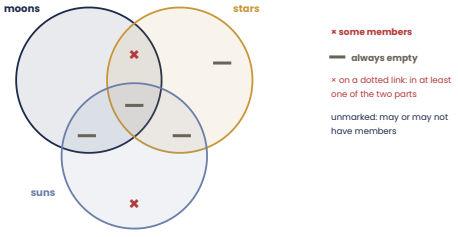
Q9 **Answer: (e) Both I and II follow**

Some tables are desks (through chairs), so some desks are tables: I follows. Nothing stops every table from being a desk, so the possibility II follows. Answer: **Both I and II follow**.



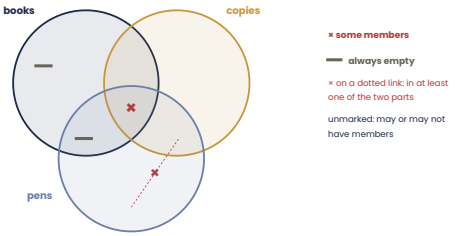
Q10 **Answer: (a) Only I follows**

All + No gives No stars are suns, so I follows and the possibility II is impossible. Answer: **Only I follows**.



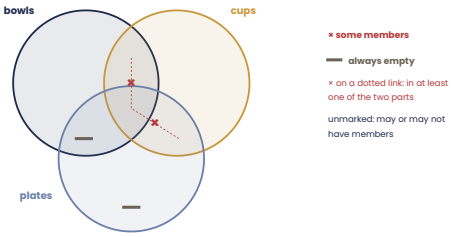
Q11 **Answer: (e) Both I and II follow**

"Only a few pens are books" means some pens are books (so they are copies: I) and some pens are not books (II). Answer: **Both I and II follow**.



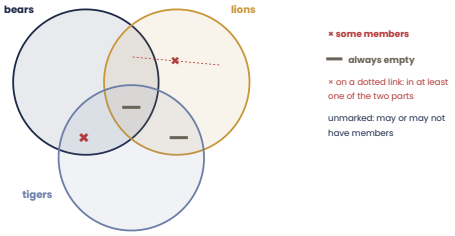
Q12 **Answer: (a) Only I follows**

"Only cups are plates" means all plates are cups, so I follows. The bowls among cups may not be plates: II is uncertain. Answer: **Only I follows**.



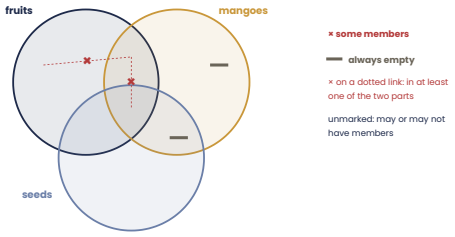
Q13 **Answer: (e) Both I and II follow**

The bears that are tigers are not lions, so I follows. Lions may overlap the other bears, so the possibility II follows. Answer: **Both I and II follow**.



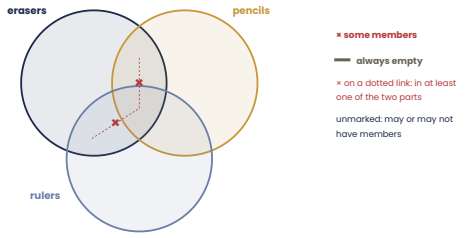
Q14 **Answer: (c) Either I or II follows**

The fruits outside seeds may or may not be mangoes, so neither is certain, but All / Some-not about mangoes and seeds is a complementary pair. Answer: **Either I or II follows**.



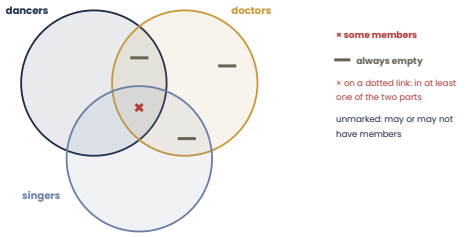
Q15 **Answer: (b) Only II follows**

Some + Some gives no definite conclusion, so I does not follow. Every pencil could be a ruler without breaking any statement, so the possibility II follows. Answer: **Only II follows**.



Q16 **Answer: (a) Only I follows**

The doctors are both singers and dancers, so some singers are dancers (I). Dancers may all be singers, so II is uncertain. Answer: **Only I follows**.



Q17 **Answer: (e) Both I and II follow**

The balls that are cats are not dogs, so I follows. Apples and dogs are not linked by any definite statement, so "no apples are dogs" is possible: II follows. Answer: **Both I and II follow**.

Q18 **Answer: (a) Only I follows**

The cards that are files are boxes and therefore not walls: I follows. Files are inside boxes and no box is a wall, so II is definitely false. Answer: **Only I follows**.

Q19 **Answer: (e) Both I and II follow**

Quills are rods but not pins, so some rods are not pins (I). The sticks among rods may include quills, so the possibility II follows. Answer: **Both I and II follow**.

Q20 **Answer: (a) Only I follows**

Books are inside pens, which are inside tables, and no table is a chair: I follows. Pens are tables, so no chair is a pen: II is false. Answer: **Only I follows**.

Q21 **Answer: (c) Either I or II follows**

Only Some statements: nothing definite about ants and deer, but I and II are complementary. Answer: **Either I or II follows**.

Q22 **Answer: (e) Both I and II follow**

Only a few kites are lamps, so some kites are lamps; these are mats and therefore not nets: I follows. The kites that are not lamps may still be mats, so all kites being mats is possible: II follows. Answer: **Both I and II follow**.

Q23 **Answer: (b) Only II follows**

Ants and dogs may or may not overlap (the dogs that are cats are certainly not ants, but other dogs may be), so I is uncertain. The dogs that are cats are not bats and so not ants: II follows. Answer: **Only II follows**.

Q24 **Answer: (e) Both I and II follow**

Oils exist, are paints and are not notes, so some paints are not notes (I). The maps that are notes are not oils (II). Answer: **Both I and II follow**.

Q25 **Answer: (a) Only I follows**

The stones that are tools are units, so I follows. Those stones may lie outside rings, so II is uncertain. Answer: **Only I follows**.

Assume the statements are true and draw the basic diagram: dashes where a region must be empty, crosses where it must have members. A conclusion follows only if it is true in every possible diagram; a possibility follows if at least one diagram allows it. Some + Some and No + No give nothing definite; then check for a complementary pair (Some/No or All/Some-not on the same two groups).

MODULE 14 · INTERMEDIATE

Linear seating arrangement

Minimum duration **2 days** | Level **Intermediate**

Linear arrangements place people in one or two rows using clues about ends, neighbours and distances. The difficulty is keeping left and right straight when people face different directions, and keeping track of cases until a clue eliminates them. This module covers the left-right rule, single rows facing north or south, the step-by-step case method, and double rows facing each other, followed by 25 solved questions; every arrangement was confirmed unique by a program.

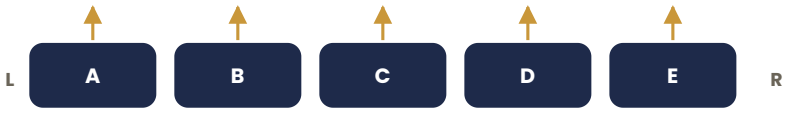
IN THIS MODULE

14.1 Basics
14.2 Single row: the case method
14.3 People facing south
14.4 Double rows
14.5 Practice questions
14.6 Solutions

14.1 Basics

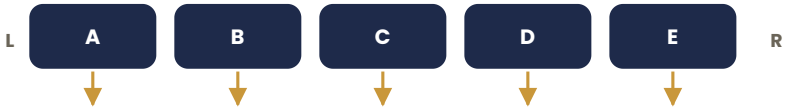
14.1.1 Left and right depend on the facing direction

Draw the row as you see it on paper, with positions numbered from **your** left. If the people face **North** (away from you, up the page), their left and right are the same as yours. If they face **South** (towards you), their left is your right and their right is your left.



facing North: A is at their left end; B is to the immediate right of A

ROW FACING NORTH: THEIR LEFT = YOUR LEFT



facing South: E is at their left end; A is to the immediate right of B

ROW FACING SOUTH: THEIR LEFT = YOUR RIGHT

SHORTCUT TRICK

For a south-facing row, simply turn every “right” in the question into “your left” before placing anyone. Do this once while reading, not while solving.

14.1.2 Clue vocabulary

Clue	Meaning (row facing North)
A is to the immediate right of B	A is in the next seat to the right of B
A is second to the right of B	one person between, A on the right
A is to the right of B	A is anywhere to the right of B
A sits between B and C	A is somewhere between them (immediately between if stated)
A is at an end / extreme end	first or last seat
A and B are immediate neighbours	adjacent seats, order unknown

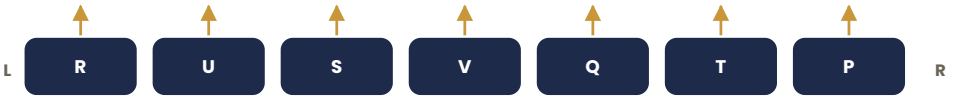
WATCH OUT
"To the right of" without "immediate" does not mean the next seat. Keep such clues for later; they only eliminate cases.

14.2 Single row: the case method

14.2.1 Place definite clues first, then branch

Order the clues by how much they fix: (1) fixed seats (middle, ends), (2) clues that link to an already placed person, (3) the rest. When a clue allows two placements, draw both cases side by side and continue until one breaks.

WORKED EXAMPLE
Seven people P, Q, R, S, T, U, V sit in a row facing North. V sits in the middle. P sits third to the right of V. R sits at one of the ends. Q sits second to the left of P. T sits between Q and P. U is an immediate neighbour of R.
V is 4th. P is 7th (right end), so R is 1st (left end). Q is 5th and T is 6th. U is next to R, so U is 2nd, and S takes the remaining seat, 3rd. Order: **R U S V Q T P**.



Set A (questions 11 to 15)

THE SOLVED ROW

TIP
Write position numbers 1 to n under a blank row before you start. Placing letters on numbered seats is faster than redrawing arrows.

14.3 People facing south

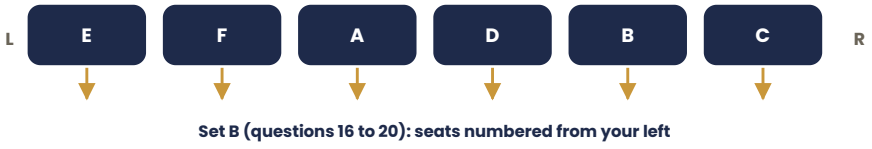
14.3.1 Convert, then solve

For a south-facing row, keep the row in your own coordinates and convert each clue: “immediate right of X” becomes “one seat to **your left** of X”.

WORKED EXAMPLE

Six people A to F sit in a row facing South. C sits at an extreme end. B sits to the immediate right of C. A sits third to the right of C. D sits between A and B. E does not sit next to A.

If C were at your left end, “immediate right of C” (your left) would be off the row, so C is at your right end (seat 6). B is seat 5, A is seat 3, D is seat 4. E is not next to A, so E is seat 1 and F seat 2. As you see them: **E F A D B C**.



THE SOLVED SOUTH-FACING ROW

14.4 Double rows

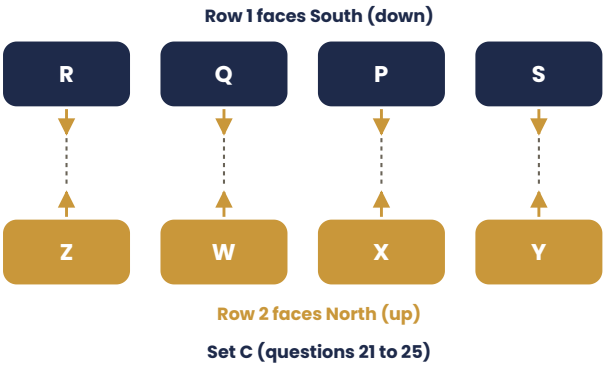
14.4.1 Two rows facing each other

In double-row questions, Row 1 usually faces South and Row 2 faces North, so every person faces someone in the other row. Draw both rows in your coordinates and apply the left-right rule separately to each row.

WORKED EXAMPLE

P, Q, R, S sit in Row 1 facing South; W, X, Y, Z sit in Row 2 facing North; each person faces one person of the other row. S sits at an end of Row 1 and faces Y. W faces the person to the immediate right of P. X sits second to the right of Z. P sits second to the left of R. Z sits at an extreme end.

X is second to the right of Z (Row 2 faces North: your right), so Z, X are seats 1, 3 or 2, 4. W faces the person to P's immediate right, which is one seat to your left of P, so W is directly left of P's column. Z is at an end, so Z is seat 1 and X seat 3. P sits second to the left of R (Row 1 faces South: two seats to your right), so R and P are seats 1 and 3 or 2 and 4; S is at an end and Y faces S, so S and Y are in column 4, W in column 2 and P in column 3. That leaves one arrangement: Row 1 **R Q P S**, Row 2 **Z W X Y**.



THE SOLVED DOUBLE ROW

SHORTCUT TRICK
In a double row, a person and the person facing them are in the same column. "Diagonally opposite" means the opposite end of the other row.

14.5 Practice questions

14.5.1 Basic level

Q1 Five friends A, B, C, D, E sit in a row facing North. C is in the middle, A is at the left end, B is to the immediate right of C and E is to the immediate left of C. Who is at the right end?

(a) B **(b)** D **(c)** E **(d)** A

Q2 In a row facing North, P is to the left of Q, R is to the right of Q and S is to the right of R. There are only four people. Who is second from the left?

(a) P **(b)** Q **(c)** R **(d)** S

Q3 Four people sit in a row facing South in the order W, X, Y, Z as you look at them from your left. Who is at their extreme left?

(a) W **(b)** X **(c)** Y **(d)** Z

Q4 Five people sit in a row facing North. M is second from the left and N is second from the right. How many people sit between M and N?

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

Q5 A, B, C, D, E sit in a row facing North. B sits exactly between A and C. D sits to the immediate right of C. E is at the left end and A is to the immediate right of E. What is the order from the left?

(a) E, A, B, C, D **(b)** E, B, A, C, D **(c)** A, E, B, C, D **(d)** E, A, C, B, D

Q6 Seven people sit in a row and X sits exactly in the middle. What is X's position from the right end?

(a) 3rd **(b)** 4th
(c) 5th **(d)** Cannot be determined

Q7 In a row facing North, R sits third to the right of S, and T sits to the immediate left of R. Who sits second to the right of S?

(a) R **(b)** T
(c) Nobody **(d)** Cannot be determined

Q8 Five people P, Q, R, S, T sit in a row facing North. P is at the left end. Q sits second to the right of P. R sits to the immediate right of Q. T does not sit next to P. Who sits between P and Q?

(a) S **(b)** T **(c)** R **(d)** Nobody

Q9 People sit in a row facing South. A is to the immediate right of B. As you look at the row from the front, on which side of B is A?

(a) Left **(b)** Right
(c) Behind **(d)** Cannot be determined

Q10 Six people sit in a row facing North. M is at the left end and N at the right end. O sits second to the right of M. P sits to the immediate left of N. Q sits between O and P. R sits next to M. Who sits fourth from the left?

(a) O **(b)** P **(c)** Q **(d)** R

14.5.2 Intermediate level

Q11 Set A (14.2.1): who sits third to the left of T?

(a) U **(b)** S **(c)** V **(d)** R

Q12 Set A: how many people sit between U and Q?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

Q13 Set A: who sits at the extreme left end?

(a) P **(b)** U **(c)** R **(d)** S

Q14 Set A: which of these pairs are immediate neighbours?

(a) S and Q **(b)** V and T **(c)** U and S **(d)** R and S

Q15 Set A: if all seven were seated in alphabetical order from the left, how many would keep their present seats?

(a) None **(b)** One **(c)** Two **(d)** Three

Q16 Set B (14.3.1): who sits at the extreme right end of the row (their right)?

(a) C **(b)** E **(c)** F **(d)** B

Q17 Set B: who sits to the immediate left of A?

(a) F **(b)** D **(c)** B **(d)** E

Q18 Set B: how many people sit between F and B?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

Q19 Set B: who sits second to the right of D?

(a) A **(b)** F **(c)** B **(d)** C

Q20 Set B: who sits exactly between C and D?

(a) A **(b)** B **(c)** E **(d)** F

14.5.3 Advanced level

Q21 Set C (14.4.1): who faces P?

(a) W (b) X (c) Y (d) Z

Q22 Set C: who sits diagonally opposite R?

(a) Z (b) W (c) X (d) Y

Q23 Set C: who sits to the immediate left of W?

(a) Z (b) X (c) Y (d) Nobody

Q24 Set C: who faces the person sitting to the immediate left of P?

(a) W (b) X (c) Y (d) Z

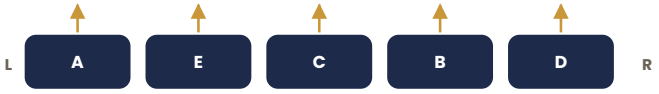
Q25 Set C: which statement is true?

(a) Q faces X (b) R sits at an end of Row 1
(c) W sits at an end of Row 2 (d) S faces Z

14.6 Solutions

Q1 **Answer: (b) D**

Order: A, E, C, B, D. The right end is **D**.



Q2 **Answer: (b) Q**

The only order is P, Q, R, S. Second from the left = **Q**.

Q3 **Answer: (d) Z**

Facing South, their left is your right. Your right end is **Z**.



Q4 **Answer: (b) 1**

M is seat 2 and N is seat 4, so **1** person sits between them.

Q5 **Answer: (a) E, A, B, C, D**

E is seat 1 and A seat 2. B is between A and C, so B 3 and C 4; D is right of C, seat 5. Order **E, A, B, C, D**.

Q6 **Answer: (b) 4th**

The middle of 7 seats is seat 4 from either end: **4th**.

Q7 **Answer: (b) T**

If S is at seat s, R is at s + 3 and T at s + 2. Second to the right of S is **T**.

Q8 **Answer: (a) S**

P 1, Q 3, R 4. T is not next to P, so T is 5 and S is 2. Between P and Q sits **S**.



Q9 **Answer: (a) Left**

Facing South, their right is your left, so A appears on your **left** of B.

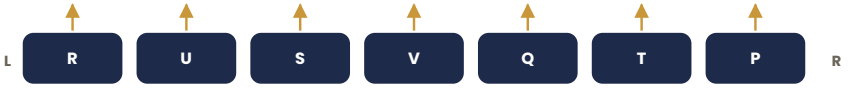
Q10 **Answer: (c) Q**

M 1, R 2, O 3, P 5, N 6, so Q is 4. Fourth from the left = **Q**.



Q11 **Answer: (b) S**

Order R U S V Q T P. T is seat 6; third to its left is seat 3 = **S**.



Q12 **Answer: (b) 2**

U is seat 2 and Q seat 5: S and V sit between them, **2** people.

Q13 **Answer: (c) R**

Seat 1 is **R**.

Q14 **Answer: (c) U and S**

U (2) and S (3) sit next to each other. The other pairs are farther apart. Answer **U and S**.

Q15 **Answer: (a) None**

Alphabetical: P Q R S T U V. Present: R U S V Q T P. No seat matches: **none**.

Q16 **Answer: (b) E**

They face South, so their right end is your left end, seat 1: **E**.



Q17 **Answer: (b) D**

Their left is your right. The seat to your right of A (seat 3) is seat 4 = **D**.

Q18 **Answer: (b) 2**

F is seat 2 and B seat 5 (in your coordinates): A and D sit between, **2** people.

Q19 **Answer: (b) F**

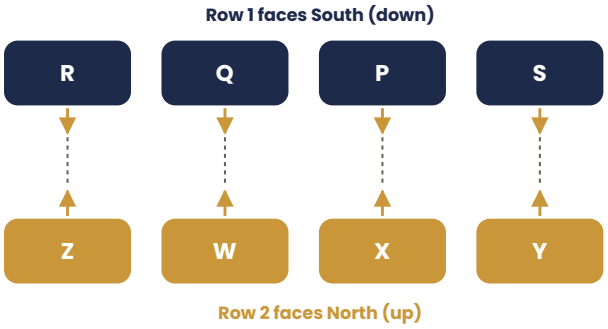
Their right is your left. D is seat 4; two seats to your left is seat 2 = **F**.

Q20 **Answer: (b) B**

C is seat 6 and D seat 4; seat 5 between them is **B**.

Q21 **Answer: (b) X**

Row 1: R Q P S; Row 2: Z W X Y. P is in column 3, facing **X**.



Q22 **Answer: (d) Y**

R is at the left end of Row 1; diagonally opposite is the right end of Row 2: **Y**.

Q23 **Answer: (a) Z**

Row 2 faces North, so left is your left. W is in column 2; to its left is **Z**.

Q24 **Answer: (c) Y**

P faces South, so P's left is your right: S (column 4). S faces **Y**.

Q25 **Answer: (b) R sits at an end of Row 1**

Row 1 is R Q P S, so **R sits at an end of Row 1**. Q faces W, W is in column 2 and S faces Y.

Draw the row in your own coordinates and number the seats. For people facing North, their left is yours; for people facing South, swap left and right. Place fixed seats first, then linked clues, and branch into cases when a clue allows two positions. In double rows, a person and the one facing them share a column.

MODULE 15 · INTERMEDIATE

Circular seating arrangement

Minimum duration **2 days** | Level **Intermediate**

In circular arrangements people sit around a round (or square) table, facing the centre, facing outside, or a mix of both. There are no ends, so you fix one person and place everyone relative to them. This module covers the left-right rule for each facing direction, opposite seats, the step-by-step method, mixed facing and square tables, followed by 25 solved questions; every arrangement was confirmed unique by a program.

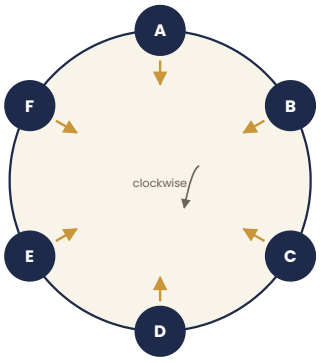
IN THIS MODULE

15.1 Basics
15.2 Everyone facing the centre
15.3 Facing outside and mixed facing
15.4 Square tables
15.5 Practice questions
15.6 Solutions

15.1 Basics

15.1.1 Left and right around a circle

Look at the table from above. For a person **facing the centre**, their left hand points **clockwise** and their right hand anticlockwise. For a person **facing outside**, it is the reverse: left is anticlockwise, right is clockwise.



facing the centre: B is to the immediate left of A, F to the immediate right

EVERYONE FACING THE CENTRE (LEFT = CLOCKWISE)

Facing	Left (immediate, second …)	Right
Centre	clockwise	anticlockwise
Outside	anticlockwise	clockwise

SHORTCUT TRICK

Imagine yourself sitting at the bottom of the circle facing up (towards the centre): your left is the left side of the page, which is the clockwise direction from the bottom. For outward-facing people, flip it.

15.1.2 Opposite seats and counting

With n equally spaced seats (n even), the person opposite seat k is at seat k + n/2. “Second to the left” means two seats away in the left direction, with one person in between. Between two people there are two arcs; questions specify the direction (“counting clockwise from A”) or ask for the shorter arc.

15.2 Everyone facing the centre

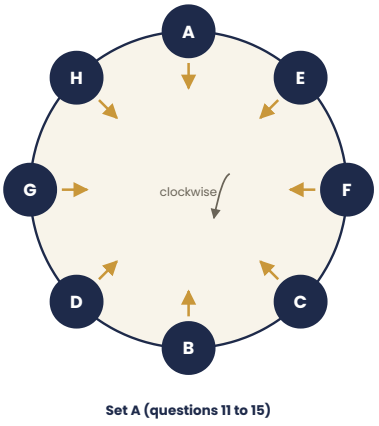
15.2.1 Fix one person, then place relative to them

There is no end to start from, so fix the first person named in a definite clue at the top of the circle and place everyone else relative to them. Opposite and “second to the left/right” clues fix seats immediately; neighbour clues often need two cases.

WORKED EXAMPLE

Eight people A to H sit around a circle facing the centre. B sits opposite A. C sits to the immediate right of B and D to the immediate left of B. G sits second to the left of B. E is an immediate neighbour of A. F sits between E and C.

Put A at the top; B is at the bottom. Right of B is anticlockwise, left is clockwise, so C and D sit on either side of B and G is one seat beyond D. E is next to A, and F must sit between E and C, which is only possible on the C side: A, E, F, C, B, D, G, H clockwise.



THE SOLVED CIRCLE

WATCH OUT

Do not forget that the circle wraps around: the seat after the last one is the first. “Immediate neighbour of A” has two candidates.

15.3 Facing outside and mixed facing

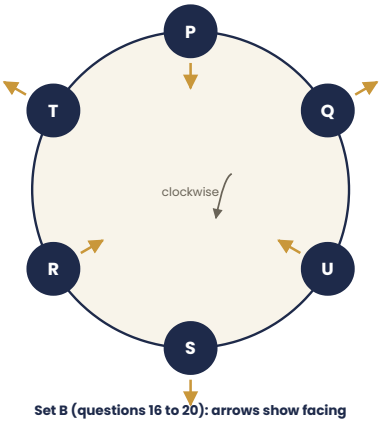
15.3.1 Apply the rule person by person

When some people face the centre and others face outside, decide each person's left and right separately from the direction **that person** faces. Draw an arrow for every person as soon as you know their facing.

WORKED EXAMPLE

Six people P, Q, R, S, T, U sit around a circle; people facing the centre and facing outside alternate. P faces the centre. Q sits to the immediate left of P. S sits opposite P. U sits to the immediate right of Q. T is an immediate neighbour of P.

P faces in, so P's left is clockwise: Q is the next seat clockwise and faces out (alternating). Q's right is clockwise (Q faces out), so U is next. S is opposite P. T is next to P on the other side, and R takes the last seat. Clockwise: P, Q, U, S, R, T.



MIXED FACING

15.4 Square tables

15.4.1 Corners and middles

At a square table with 8 seats, four people sit at the corners and four at the middles of the sides. Treat it exactly like a circle of 8 (seat k + 4 is opposite), but read clues such as “sits at a corner” or “sits at the middle of a side” as alternate seats.

TIP

Number the 8 seats clockwise starting from a corner: corners are the even seats 0, 2, 4, 6 and side middles the odd seats.

15.5 Practice questions

15.5.1 Basic level

Q1 In a circle where everyone faces the centre, in which direction around the circle is a person's immediate left neighbour?

(a) Clockwise **(b)** Anticlockwise
(c) Opposite **(d)** It depends on the number of people

Q2 In a circle where everyone faces outside, in which direction is a person's immediate left neighbour?

(a) Clockwise **(b)** Anticlockwise
(c) Opposite **(d)** It depends on the number of people

Q3 Eight people sit equally spaced around a circle. How many people sit between A and the person opposite A, counting in one direction?

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

Q4 A, B, C, D, E, F sit around a circle facing the centre, clockwise in that order. Who is to the immediate left of A?

(a) B **(b)** F **(c)** C **(d)** E

Q5 In the same circle, who is second to the right of A?

(a) C **(b)** D **(c)** E **(d)** F

Q6 In the same circle, who sits opposite C?

(a) E **(b)** F **(c)** A **(d)** D

Q7 P, Q, R, S, T sit around a round table facing the centre, clockwise in that order. Who sits between R and T (shorter arc)?

(a) Q **(b)** S **(c)** P **(d)** Nobody

Q8 A, B, C, D sit around a square table, one on each side, facing the centre. A sits opposite C, and B sits to the immediate left of A. Who sits to the immediate right of A?

(a) B **(b)** C **(c)** D **(d)** Nobody

Q9 In a circle facing the centre, M sits third to the left of N. Where does N sit relative to M?

(a) Third to the left of M **(b)** Third to the right of M
(c) Opposite M **(d)** Second to the right of M

Q10 A, B, C, D, E, F sit around a circle facing outside, clockwise in that order. Who is to the immediate left of A?

(a) B **(b)** F **(c)** C **(d)** E

15.5.2 Intermediate level

Q11 Set A (15.2.1): who sits opposite E?

(a) B **(b)** D **(c)** G **(d)** C

Q12 Set A: who sits second to the left of H?

(a) A **(b)** E **(c)** G **(d)** D

Q13 Set A: counting clockwise from A, how many people sit between A and C?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

Q14 Set A: who sits to the immediate right of A?

(a) E **(b)** H **(c)** G **(d)** B

Q15 Set A: which pair sits opposite each other?

(a) F and G **(b)** E and B **(c)** C and D **(d)** A and H

Q16 Set B (15.3.1): who sits to the immediate left of S?

(a) U **(b)** R **(c)** Q **(d)** T

Q17 Set B: who sits to the immediate right of T?

(a) R **(b)** P **(c)** S **(d)** Q

Q18 Set B: who sits opposite U?

(a) P **(b)** R **(c)** T **(d)** Q

Q19 Set B: how many people face outside?

(a) 2 **(b)** 3 **(c)** 4 **(d)** 1

Q20 Set B: which of these faces the centre?

(a) Q **(b)** S **(c)** R **(d)** T

15.5.3 Advanced level

Q21 Eight people J to Q sit around a circle; people facing the centre and outside alternate, and J faces the centre. Q sits opposite J. K sits to the immediate right of Q. M sits second to the left of K. L sits to the immediate left of Q. N is not a neighbour of J. P sits to the immediate right of J. O sits to the immediate right of M. Who sits opposite M?

(a) K **(b)** L **(c)** N **(d)** P

Q22 In the same arrangement, who sits to the immediate left of N?

(a) L **(b)** P **(c)** Q **(d)** J

Q23 In the same arrangement, counting clockwise from K, how many people sit between K and N?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

Q24 In the same arrangement, who sits second to the right of P?

(a) J **(b)** M **(c)** N **(d)** O

Q25 In the same arrangement, which of these people faces outside?

(a) O **(b)** N **(c)** L **(d)** Q

15.6 Solutions

Q1 **Answer: (a) Clockwise**

Facing the centre, the left hand points **clockwise**.

Q2 **Answer: (b) Anticlockwise**

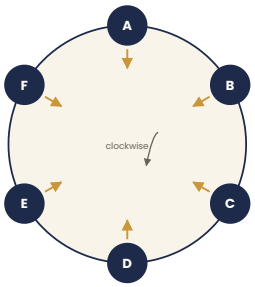
Facing outside reverses the rule: left is **anticlockwise**.

Q3 **Answer: (b) 3**

The opposite seat is 4 seats away, so **3** people lie between them on each side.

Q4 **Answer: (a) B**

Left is clockwise: the next seat clockwise from A is **B**.



Q5 **Answer: (c) E**

Right is anticlockwise: F is first, **E** second.

Q6 **Answer: (b) F**

With 6 seats, opposite is 3 seats away: C → D → E → **F**.

Q7 **Answer: (b) S**

Clockwise R, S, T: **S** sits between them.

Q8 **Answer: (c) D**

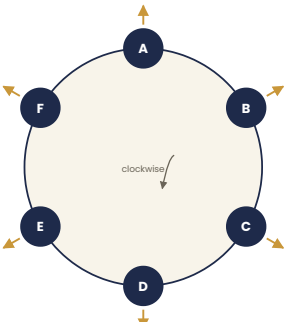
Left of A is clockwise, so B is next clockwise, then C (opposite), then D. A's right (anticlockwise) is **D**.

Q9 **Answer: (b) Third to the right of M**

M is three seats clockwise from N, so N is three seats anticlockwise from M: **third to the right of M**.

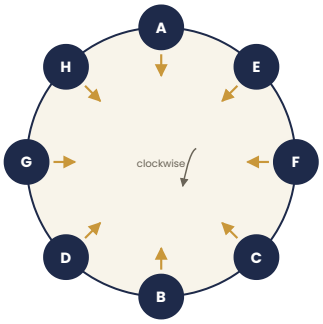
Q10 **Answer: (b) F**

Facing outside, left is anticlockwise: the seat before A, which is **F**.



Q11 **Answer: (b) D**

Clockwise A, E, F, C, B, D, G, H. E is seat 1; opposite is seat 5 = **D**.



Q12 **Answer: (b) E**

Left is clockwise: from H, A is first and **E** second.

Q13 **Answer: (b) 2**

Clockwise A, E, F, C: E and F, so **2**.

Q14 **Answer: (b) H**

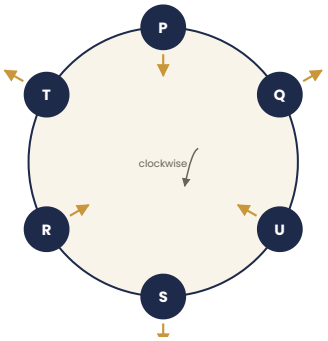
Right is anticlockwise: the seat before A is **H**.

Q15 **Answer: (a) F and G**

F is seat 2 and G seat 6: **F and G** are opposite.

Q16 **Answer: (a) U**

Clockwise P(in), Q(out), U(in), S(out), R(in), T(out). S faces outside, so S's left is anticlockwise: **U**.



Q17 **Answer: (b) P**

T faces outside, so right is clockwise: after T comes **P**.

Q18 **Answer: (c) T**

U is seat 2; seat 5 is **T**.

Q19 **Answer: (b) 3**

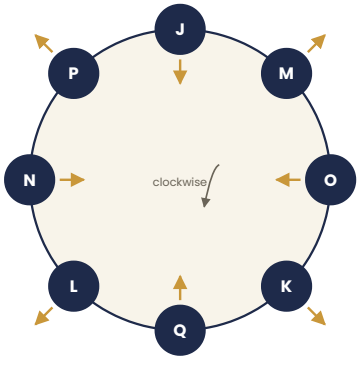
Q, S and T face outside: **3**.

Q20 **Answer: (c) R**

P, U and R face the centre; of the options only **R**.

Q21 **Answer: (b) L**

J is seat 0 (in), Q seat 4 (in). Q's right is anticlockwise → K seat 3 (out). K faces out, so K's left is anticlockwise: M is two seats back, seat 1. L is Q's left (clockwise), seat 5. P is J's right, seat 7. M faces out, so M's right is clockwise: O is seat 2, and N seat 6. Clockwise: J, M, O, K, Q, L, N, P. Opposite M (seat 1) is seat 5 = **L**.



Q22 **Answer: (b) P**

N (seat 6) faces the centre, so its left is clockwise: **P**.

Q23 **Answer: (b) 2**

Clockwise K, Q, L, N: **2** people (Q and L).

Q24 **Answer: (b) M**

P (seat 7) faces outside, so right is clockwise: J first, **M** second.

Q25 **Answer: (c) L**

Odd seats face outside: M, K, L, P. Of the options, **L**.

Fix one person at the top and place everyone relative to them. Facing the centre, left is clockwise; facing outside, left is anticlockwise; in mixed arrangements apply the rule to each person separately. With n seats, the opposite seat is n/2 away. Keep two cases when a neighbour clue is ambiguous, and let a later clue eliminate one.

MODULE 16 · INTERMEDIATE

Puzzles

Minimum duration **2 days** | Level **Intermediate**

Puzzles combine several attributes (person, floor, day, colour, subject, pet) and a list of clues, and ask questions about the final arrangement. The skill is choosing the right table, entering definite clues first and keeping cases until a clue breaks them. This module covers floor puzzles, scheduling puzzles and matching grids, with a solved table for every set and 25 solved questions; every set was confirmed unique by a program.

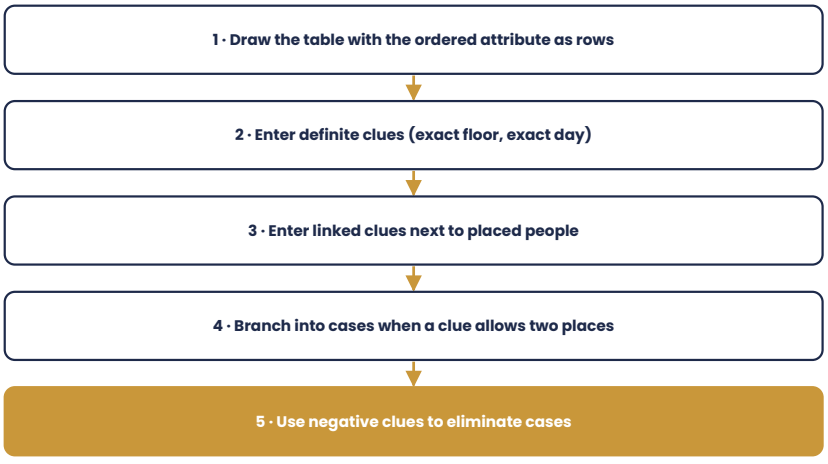
IN THIS MODULE

16.1 Basics
16.2 Floor puzzles
16.3 Scheduling puzzles
16.4 Matching grids
16.5 Practice questions
16.6 Solutions

16.1 Basics

16.1.1 Choose the table first

Puzzle type	Rows	Columns
Floor puzzle	floors (top floor at the top)	person, then other attributes
Scheduling	days, months or time slots in order	person, subject, …
Matching grid	people	attributes (no natural order)



THE PUZZLE METHOD

SHORTCUT TRICK

Read all clues once before writing anything and tick the definite ones. A negative clue ("F does not live on floor 3") is usually the one that breaks a tie at the end; keep it for last.

16.2 Floor puzzles

16.2.1 A six-floor building

Six persons A to F live on floors 1 (ground) to 6 (top), one on each floor, and each likes a different colour: Red, Blue, Green, Black, White, Pink.

1. B lives on the lowest floor.
2. Two persons live between B and C.
3. A lives immediately above B.
4. D lives above F.
5. E lives on an odd-numbered floor.
6. F does not live on floor 3.
7. The person on floor 4 likes Green.
8. The person who likes Red lives immediately below E.
9. D likes Blue.
10. The person who likes White lives above the person who likes Black, and B likes neither White nor Black.

Solution. B is on floor 1 and A on floor 2; C is three floors above B, on floor 4. Floors 3, 5, 6 remain for D, E, F. F is not on 3 and D is above F, so F is on 5 and D on 6, leaving E on 3. Colours: floor 4 (C) likes Green; Red is immediately below E, so A likes Red; D likes Blue. B, E, F share Pink, Black, White; B likes neither White nor Black, so B likes Pink; White is above Black, so F likes White and E Black.

Floor	Person	Colour
6	D	Blue
5	F	White
4	C	Green
3	E	Black
2	A	Red
1	B	Pink

Floor set (questions 11 to 15)

THE SOLVED FLOOR PUZZLE

WATCH OUT
"Two persons live between B and C" does not say who is above. Here B is on the lowest floor, so C must be above; in general keep both cases.

16.3 Scheduling puzzles

16.3.1 Five lectures in a week

Five teachers P, Q, R, S, T each give one lecture on a different day from Monday to Friday, each on a different subject: Maths, Physics, Chemistry, Biology, English.

1. R lectures exactly in the middle of the week.
2. Q lectures on the last day.
3. T lectures on the day immediately after R.
4. S does not lecture on Monday.
5. P teaches Maths.
6. The Chemistry lecture is on the day immediately after the Maths lecture.
7. English is taught on the day immediately before Physics.
8. Biology is not taught on Friday.

Solution. R is on Wednesday, T on Thursday, Q on Friday. S is not on Monday, so P is on Monday and S on Tuesday. P teaches Maths, so Chemistry is on Tuesday (S). English is immediately before Physics: either Wednesday–Thursday or Thursday–Friday. Biology is not on Friday, so it must be on Wednesday, giving English on Thursday (T) and Physics on Friday (Q).

Day	Teacher	Subject
Mon	P	Maths
Tue	S	Chemistry
Wed	R	Biology
Thu	T	English
Fri	Q	Physics

Schedule set (questions 16 to 20)

THE SOLVED SCHEDULE

16.4 Matching grids

16.4.1 People and attributes with no order

When nothing is ordered (no floors or days), use a grid with people as rows and mark ✓ and ×. Each row and each column has exactly one ✓.

Four friends, Amit, Bina, Chetan and Divya, each like a different fruit (Apple, Banana, Cherry, Grape) and own a different pet (Cat, Dog, Fish, Parrot).

1. Amit owns the dog.
2. The person who likes Apple owns the parrot.
3. Bina likes neither Apple nor Cherry.
4. Divya owns the fish.
5. Chetan does not own the cat.
6. The person who owns the cat likes Grapes.
7. Amit does not like Cherry.

Solution. Amit has the dog and Divya the fish, so Bina and Chetan have the cat and the parrot; Chetan does not have the cat, so Chetan has the parrot and Bina the cat. The parrot owner likes Apple (Chetan) and the cat owner likes Grapes (Bina). Amit does not like Cherry, so Amit likes Banana and Divya Cherry.

Person	Fruit	Pet
Amit	Banana	Dog
Bina	Grape	Cat
Chetan	Apple	Parrot
Divya	Cherry	Fish

Grid set (questions 21 to 25)

THE SOLVED MATCHING GRID

TIP

In a grid, every ✓ lets you put × in the rest of its row and column. When a row has only one empty cell left, it must be the ✓.

16.5 Practice questions

16.5.1 Basic level

Q1 Three boxes are stacked. The red box is above the blue box, and the green box is below the blue box. Which box is on top?

(a) Red **(b)** Blue
(c) Green **(d)** Cannot be determined

Q2 A, B, C, D live on floors 1 to 4, one per floor. A lives above B, C lives below B, and D lives on the top floor. Who lives on floor 2?

(a) A **(b)** B **(c)** C **(d)** D

Q3 M, N, O, P, Q each receive one call on a different day from Monday to Friday. M's call is on Wednesday, N's the day after M's, O's the day before M's, and P's on Monday. Whose call is on Friday?

(a) N **(b)** O **(c)** P **(d)** Q

Q4 A, B, C each own one car: red, blue or white. A's car is not red and B's car is blue. Who owns the red car?

(a) A **(b)** B
(c) C **(d)** Cannot be determined

Q5 In a five-storey building, X lives on floor 3, Y directly above X and Z on the top floor. Who lives directly below Z?

(a) X **(b)** Y
(c) Nobody **(d)** Cannot be determined

Q6 P, Q, R, S were born in January, April, July and October, one each. P was born in April. Q was born after P but not in October. R was born before P. Who was born in October?

(a) P **(b)** Q **(c)** R **(d)** S

Q7 P, Q, R each play one sport: cricket, hockey or tennis. P plays neither cricket nor tennis, and Q does not play cricket. Who plays cricket?

(a) P (b) Q
(c) R (d) Cannot be determined

Q8 A, B, C, D, E take an exam on different days from Monday to Friday. A takes it after B but before C. D takes it on Monday and E on Friday. Who takes it on Wednesday?

(a) A (b) B (c) C (d) D

Q9 A building has four floors (1 to 4), one family on each. The Sharmas live above the Guptas but below the Vermas, and the Raos live on floor 1. Who lives on floor 3?

(a) Guptas (b) Sharmas (c) Vermas (d) Raos

Q10 Six boxes P, Q, R, S, T, U are stacked one above another. Only Q is above P. Which box is second from the top?

(a) Q (b) P
(c) R (d) Cannot be determined

16.5.2 Intermediate level

Q11 Floor set (16.2.1): who lives on the top floor?

(a) C (b) D (c) E (d) F

Q12 Floor set: which colour does E like?

(a) White (b) Black (c) Red (d) Green

Q13 Floor set: how many persons live between A and F?

(a) 1 (b) 2 (c) 3 (d) 4

Q14 Floor set: who likes White?

(a) B (b) D (c) E (d) F

Q15 Floor set: which combination is correct?

(a) C – floor 4 – Green (b) A – floor 3 – Red
(c) E – floor 5 – Black (d) B – floor 2 – Pink

Q16 Schedule set (16.3.1): who teaches Biology?

(a) P (b) R (c) T (d) Q

Q17 Schedule set: on which day is the Physics lecture?

(a) Wednesday (b) Thursday (c) Friday (d) Tuesday

Q18 Schedule set: which subject does S teach?

(a) Maths (b) Chemistry (c) English (d) Biology

Q19 Schedule set: how many lectures are there between Maths and English?

(a) 1 (b) 2 (c) 3 (d) 0

Q20 Schedule set: who lectures on the day immediately before T?

(a) S (b) R (c) Q (d) P

16.5.3 Advanced level

Q21 Grid set (16.4.1): which fruit does Amit like?

(a) Apple (b) Banana (c) Cherry (d) Grape

Q22 Grid set: who owns the parrot?

(a) Amit (b) Bina (c) Chetan (d) Divya

Q23 Grid set: which fruit does the fish owner like?

(a) Apple (b) Banana (c) Cherry (d) Grape

Q24 Grid set: which pair is correctly matched?

(a) Bina – Apple (b) Divya – Cat (c) Bina – Grape (d) Amit – Parrot

Q25 Grid set: if clue 7 (Amit does not like Cherry) were removed, which fact could no longer be determined?

(a) Who owns the cat (b) Who likes Apple
(c) Which fruit Divya likes (d) Who owns the fish

16.6 Solutions

Q1 **Answer: (a) Red**

Order from the top: red, blue, green. The **red** box is on top.

Q2 **Answer: (b) B**

D is on 4; A > B > C fills floors 3, 2, 1. Floor 2 is **B**.

Floor	Person
4	D
3	A
2	B
1	C

Q3 **Answer: (d) Q**

P Mon, O Tue, M Wed, N Thu, so Friday is **Q**.

Q4 **Answer: (c) C**

B has blue; A is not red, so A has white and **C** has red.

Q5 **Answer: (b) Y**

Y is on floor 4 and Z on floor 5, so directly below Z is **Y**.

Q6 **Answer: (d) S**

Q is after April but not October, so Q is July. R is before April, so January. October is **S**.

Q7 **Answer: (c) R**

P plays hockey; Q is not cricket, so Q plays tennis and **R** plays cricket.

Q8 **Answer: (a) A**

B, A, C fill Tuesday to Thursday in that order, so Wednesday is **A**.

Q9 **Answer: (b) Sharmas**

Raos 1, Guptas 2, Sharmas 3, Vermas 4. Floor 3 is the **Sharmas**.

Q10 **Answer: (b) P**

Only Q is above P, so Q is on top and **P** is second.

Q11 **Answer: (b) D**

From the solved table, floor 6 is **D**.

Floor	Person	Colour
6	D	Blue
5	F	White
4	C	Green
3	E	Black
2	A	Red
1	B	Pink

Q12 **Answer: (b) Black**

E (floor 3) likes **Black**.

Q13 **Answer: (b) 2**

A is on floor 2 and F on floor 5: E and C live between, **2** persons.

Q14 **Answer: (d) F**

White is on floor 5: **F**.

Q15 **Answer: (a) C – floor 4 – Green**

C – floor 4 – Green matches the table; A is on 2, E on 3, B on 1.

Q16 **Answer: (b) R**

Biology is on Wednesday, taught by **R**.

Day	Teacher	Subject
Mon	P	Maths
Tue	S	Chemistry
Wed	R	Biology
Thu	T	English
Fri	Q	Physics

Q17 **Answer: (c) Friday**

Physics is taught by Q on **Friday**.

Q18 **Answer: (b) Chemistry**

S lectures on Tuesday, the day after Maths: **Chemistry**.

Q19 **Answer: (b) 2**

Maths is Monday and English Thursday; Tuesday and Wednesday lie between: **2**.

Q20 **Answer: (b) R**

T is on Thursday; Wednesday is **R**.

Q21 **Answer: (b) Banana**

Amit has the dog; Apple goes with the parrot and Grape with the cat, and Amit does not like Cherry, so Amit likes **Banana**.

Person	Fruit	Pet
Amit	Banana	Dog
Bina	Grape	Cat
Chetan	Apple	Parrot
Divya	Cherry	Fish

Q22 **Answer: (c) Chetan**

Bina and Chetan share the cat and the parrot; Chetan does not own the cat, so **Chetan** owns the parrot.

Q23 **Answer: (c) Cherry**

Divya owns the fish and likes **Cherry**.

Q24 **Answer: (c) Bina – Grape**

Bina owns the cat, and the cat owner likes Grapes: **Bina – Grape**.

Q25 **Answer: (c) Which fruit Divya likes**

Without clue 7, Amit and Divya could share Banana and Cherry either way, so **which fruit Divya likes** becomes uncertain. The pets and the Apple and Grape owners do not depend on clue 7.

Pick the table first: floors or days as ordered rows, a ✓/× grid when nothing is ordered. Enter definite clues, then clues linked to placed people, branch into cases when needed, and use negative clues to eliminate cases. Always re-check the final table against every clue before answering.

MODULE 17 · INTERMEDIATE

Clocks

Minimum duration **1 day** | Level **Intermediate**

Clock questions ask for the angle between the hands, the times when the hands coincide, are opposite or at right angles, mirror images of a clock, and the time shown by a clock that gains or loses. All of them follow from two speeds: the minute hand moves 6° per minute and the hour hand 0.5° per minute. This module derives the formulas, draws the clock for each example, and gives 25 solved questions checked with exact fractions.

IN THIS MODULE

17.1 Basics and the angle formula
17.2 Coinciding, opposite and right-angle hands
17.3 Mirror images
17.4 Fast and slow clocks
17.5 Practice questions
17.6 Solutions

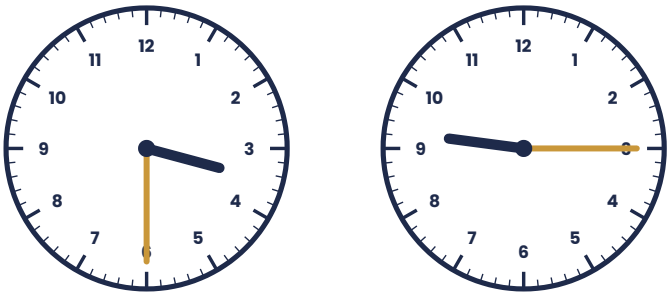
17.1 Basics and the angle formula

17.1.1 Speeds of the hands

Hand	Full turn	Speed
Minute hand	360° in 60 min	6° per minute
Hour hand	360° in 12 h = 720 min	0.5° per minute (30° per hour)
Relative speed		minute hand gains 5.5° per minute

At H hours and M minutes, the hour hand is at 30H + 0.5M degrees from 12 and the minute hand at 6M degrees. Their difference gives the formula:

Angle between the hands = | 30H − 5.5M | (if the result is above 180°, subtract it from 360° for the smaller angle)



USING THE ANGLE FORMULA

WATCH OUT
At 3:30 the hour hand is not on the 3: it has moved halfway to the 4. Forgetting the hour hand's movement is the most common error.

17.2 Coinciding, opposite and right-angle hands

17.2.1 Solving 30H - 5.5M = ± angle

To find when the hands are θ apart between H and H + 1 o'clock, solve $|30H - 5.5M| = \theta$ for M between 0 and 60. This gives $M = (2/11)(30H \pm \theta)$.

Position	θ	Between 2 and 3	Per 12 hours	Per day
Coincide	0°	10 10/11 min past 2	11	22
Opposite	180°	43 7/11 min past 2	11	22
Right angle	90°	27 3/11 min past 2	22	44

SHORTCUT TRICK
The hands coincide every **65 5/11 minutes** (720/11). Between 2 and 3 they coincide at 2 × 65 5/11 - 120 = 10 10/11 minutes past 2. In general: coincide at (60H/11) minutes past H.



HANDS COINCIDING BETWEEN 2 AND 3

TIP
The hands coincide only 11 times in 12 hours (not 12), because between 11 and 1 they coincide only once, at 12:00. Likewise they are at right angles 22 times, not 24.

17.3 Mirror images

17.3.1 The 11:60 rule

In a mirror placed vertically beside the clock, the image of time t shows **12:00 - t**. Write 12:00 as 11:60 to subtract easily: the mirror image of 4:20 is 11:60 - 4:20 = **7:40**. The same rule works backwards: if the mirror shows 8:15, the real time is 11:60 - 8:15 = 3:45.



A TIME AND ITS MIRROR IMAGE

For a **water image** (reflection below the clock), exams use the rule 18:30 - t, for example 18:30 - 3:40 = 14:50 → 2:50. This is an exam convention; mirror questions are far more common.

17.4 Fast and slow clocks

17.4.1 Gaining and losing time

If a clock gains g minutes per hour, after t hours it shows t × g minutes more than the true time. To find when a clock that loses time shows the right time again, it must lose a full 12 hours (720 minutes).

WORKED EXAMPLE

The hands of a faulty clock coincide every 64 minutes of true time. How much does it gain per day?
In a correct clock they coincide every 720/11 = 65 5/11 minutes. The faulty clock does the work of 65 5/11 clock-minutes in 64 true minutes, gaining 16/11 minutes. Per day: (1440 ÷ 64) × 16/11 = 360/11 = **32 8/11 minutes**.

17.5 Practice questions

17.5.1 Basic level

Q1 What is the angle between the hands at 3:00?

(a) 60° **(b)** 90° **(c)** 120° **(d)** 180°

Q2 What is the angle between the hands at 3:30?

(a) 75° **(b)** 90° **(c)** 60° **(d)** 105°

Q3 What is the smaller angle between the hands at 9:15?

(a) 172.5° **(b)** 187.5° **(c)** 180° **(d)** 165°

Q4 A clock shows 4:20. What time does its mirror image show?

(a) 7:40 **(b)** 8:40 **(c)** 7:20 **(d)** 8:20

Q5 How many times do the hands of a clock coincide in 12 hours?

(a) 10 **(b)** 11 **(c)** 12 **(d)** 24

Q6 What is the angle between the hands at 6:00?

(a) 90° **(b)** 120° **(c)** 150° **(d)** 180°

Q7 Through how many degrees does the minute hand turn in 20 minutes?

(a) 60° **(b)** 100° **(c)** 120° **(d)** 240°

Q8 Through how many degrees does the hour hand turn in 1 hour 20 minutes?

(a) 30° **(b)** 40° **(c)** 45° **(d)** 80°

17.5.2 Intermediate level

Q9 What is the angle between the hands at 7:20?

(a) 90° **(b)** 100° **(c)** 110° **(d)** 120°

Q10 At what time between 2 and 3 o'clock do the hands coincide?

(a) 10 10/11 min past 2 **(b)** 10 min past 2
(c) 11 min past 2 **(d)** 10 5/11 min past 2

Q11 At what time between 4 and 5 o'clock are the hands opposite each other?

(a) 54 6/11 min past 4 **(b)** 50 min past 4
(c) 54 min past 4 **(d)** 45 5/11 min past 4

Q12 A clock shows 2:35. What time does its mirror image show?

(a) 9:25 **(b)** 10:25 **(c)** 9:35 **(d)** 10:35

Q13 After 3:00, when are the hands first at a right angle again?

(a) 32 8/11 min past 3 **(b)** 30 min past 3
(c) 32 min past 3 **(d)** 33 3/11 min past 3

Q14 A clock gains 5 minutes every hour. It is set right at 8 a.m. What time does it show at 2 p.m. (true time)?

(a) 2:25 p.m. **(b)** 2:30 p.m. **(c)** 2:35 p.m. **(d)** 2:05 p.m.

Q15 What is the angle between the hands at 10:25?

(a) 162.5° **(b)** 157.5° **(c)** 167.5° **(d)** 150°

Q16 The mirror image of a clock shows 8:15. What is the actual time?

(a) 3:45 **(b)** 4:45 **(c)** 3:15 **(d)** 4:15

17.5.3 Advanced level

Q17 The hands of a faulty clock coincide every 64 minutes of true time. How much does the clock gain per day?

(a) 32 8/11 min **(b)** 30 min **(c)** 36 min **(d)** 16/11 min

Q18 Between 5 and 6 o'clock, when are the hands first 60° apart?

(a) 16 4/11 min past 5 **(b)** 38 2/11 min past 5
(c) 15 min past 5 **(d)** 20 min past 5

Q19 How many times are the hands of a clock at right angles in a day?

(a) 22 **(b)** 24 **(c)** 44 **(d)** 48

Q20 A watch loses 2 minutes every hour. It is set right at noon on Monday. After how many days will it next show the correct time?

(a) 12 **(b)** 15 **(c)** 30 **(d)** 6

Q21 At what time between 7 and 8 o'clock are the hands in a straight line pointing in opposite directions?

(a) 5 5/11 min past 7 **(b)** 5 min past 7
(c) 6 min past 7 **(d)** 54 6/11 min past 7

Q22 What is the angle between the hands at 12:20?

(a) 100° **(b)** 110° **(c)** 120° **(d)** 120.5°

Q23 The mirror image of a clock shows 7:10. What is the actual time?

(a) 4:50 **(b)** 5:50 **(c)** 4:10 **(d)** 5:10

Q24 What is the reflex angle between the hands at 5:20?

(a) 320° **(b)** 300° **(c)** 40° **(d)** 310°

Q25 A clock gains 10 minutes in 24 hours. It is set right at noon on Monday. What time will it show at noon on Wednesday?

(a) 12:10 p.m. **(b)** 12:20 p.m. **(c)** 11:40 a.m. **(d)** 12:30 p.m.

17.6 Solutions

Q1 **Answer: (b) 90°**

$|30 \times 3 - 0| = \mathbf{90°}$.



Q2 **Answer: (a) 75°**

$|90 - 5.5 \times 30| = |90 - 165| = \mathbf{75°}$.



Q3 **Answer: (a) 172.5°**

|270 - 82.5| = 187.5°, which is above 180°, so the smaller angle is 360 - 187.5 = **172.5°**.

Q4 **Answer: (a) 7:40**

11:60 - 4:20 = **7:40**.

Q5 **Answer: (b) 11**

Every 720/11 minutes, so **11** times in 12 hours.

Q6 **Answer: (d) 180°**

|180 - 0| = **180°**.

Q7 **Answer: (c) 120°**

20 × 6° = **120°**.

Q8 **Answer: (b) 40°**

80 minutes × 0.5° = **40°**.

Q9 **Answer: (b) 100°**

|210 - 110| = **100°**.



Q10 **Answer: (a) 10 10/11 min past 2**

30 × 2 = 5.5M → M = 60/5.5 = 120/11 = **10 10/11 minutes past 2**.

Q11 **Answer: (a) 54 6/11 min past 4**

5.5M - 120 = 180 → M = 300/5.5 = 600/11 = **54 6/11 minutes past 4**.

Q12 **Answer: (a) 9:25**

11:60 - 2:35 = **9:25**.

Q13 **Answer: (a) 32 8/11 min past 3**

At 3:00 they are already at 90°. Next: 5.5M - 90 = 90 → M = 180/5.5 = 360/11 = **32 8/11 minutes past 3**.

Q14 **Answer: (b) 2:30 p.m.**

6 hours × 5 minutes = 30 minutes gained: **2:30 p.m.**

Q15 **Answer: (a) 162.5°**

|300 − 137.5| = **162.5°**.

Q16 **Answer: (a) 3:45**

11:60 − 8:15 = **3:45**.

Q17 **Answer: (a) 32 8/11 min**

Normal gap 720/11 = 65 5/11 min, so the clock gains 16/11 min every 64 min. Per day: 1440/64 × 16/11 = **32 8/11 minutes**.

Q18 **Answer: (a) 16 4/11 min past 5**

|150 − 5.5M| = 60 → M = 90/5.5 = 16 4/11 or 210/5.5 = 38 2/11. The first is **16 4/11 minutes past 5**.



Q19 **Answer: (c) 44**

22 times in 12 hours, so **44** times in 24 hours.

Q20 **Answer: (b) 15**

It must lose 12 hours = 720 minutes: 720 ÷ 2 = 360 hours = **15 days**.

Q21 **Answer: (a) 5 5/11 min past 7**

210 − 5.5M = 180 → M = 30/5.5 = 60/11 = **5 5/11 minutes past 7**.

Q22 **Answer: (b) 110°**

|0 − 5.5 × 20| = **110°**.

Q23 **Answer: (a) 4:50**

11:60 − 7:10 = **4:50**.

Q24 **Answer: (a) 320°**

|150 − 110| = 40°, so the reflex angle is 360 − 40 = **320°**.

Q25 **Answer: (b) 12:20 p.m.**

48 hours = 2 days, so it gains 20 minutes: **12:20 p.m.**

Minute hand 6°/min, hour hand 0.5°/min, angle = |30H − 5.5M| (use 360° minus it for the smaller angle when over 180°). Solve |30H − 5.5M| = θ for coinciding (0°), opposite (180°) and right-angle (90°) times. Mirror image = 11:60 − time. A losing or gaining clock is right again after it is 12 hours off.

MODULE 18 · INTERMEDIATE

Calendars

Minimum duration **1 day** | Level **Intermediate**

Calendar questions ask for the day of the week of a date, the day after a number of days, leap years, and years with identical calendars. Everything rests on one idea, odd days: the days left over after removing complete weeks. This module covers odd days, leap-year rules, the odd-day method and a fast code method for any date, and repeating calendars, with 25 solved questions checked against a real calendar.

IN THIS MODULE

18.1 Basics
18.2 Odd days
18.3 Day of the week for any date
18.4 Repeating calendars
18.5 Practice questions
18.6 Solutions

18.1 Basics

18.1.1 Leap years

An ordinary year has 365 days and a leap year 366 (February has 29 days). A year is a leap year if it is divisible by 4, **except** century years, which must be divisible by 400.

Year	Divisible by 4?	Century year?	Leap year?
2024	yes	no	yes
2026	no	no	no
1900	yes	yes (not divisible by 400)	no
2000	yes	yes (divisible by 400)	yes

WATCH OUT
1700, 1800, 1900 and 2100 are **not** leap years even though they are divisible by 4.

18.2 Odd days

18.2.1 Days left after complete weeks

Odd days are the remainder when a number of days is divided by 7. Moving forward by n days moves the weekday forward by n mod 7.

Period	Days	Odd days
Ordinary year	365 = 52 weeks + 1	1
Leap year	366 = 52 weeks + 2	2
100 years	76 ordinary + 24 leap	(76 + 48) mod 7 = 5
200 years		3
300 years		1
400 years		0

WORKED EXAMPLE

Today is Monday. What day will it be after 61 days?

61 = 8 × 7 + 5, so move 5 days forward from Monday: **Saturday**.



MOVING FORWARD BY ODD DAYS

SHORTCUT TRICK

Because 400 years have 0 odd days, the calendar repeats every 400 years. The last day of a century can only be Monday, Wednesday, Friday or Sunday (5, 3, 1, 0 odd days), never Tuesday, Thursday or Saturday.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Odd days	3	0 (1 in leap)	3	2	3	2	3	3	2	3	2	3

18.3 Day of the week for any date

18.3.1 The code method

Add four codes and take the remainder on division by 7: **date + month code + year code + century code**, then subtract 1 for January or February of a leap year. Remainder 0 = Sunday, 1 = Monday, … 6 = Saturday.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Code	0	3	3	6	1	4	6	2	5	0	3	5

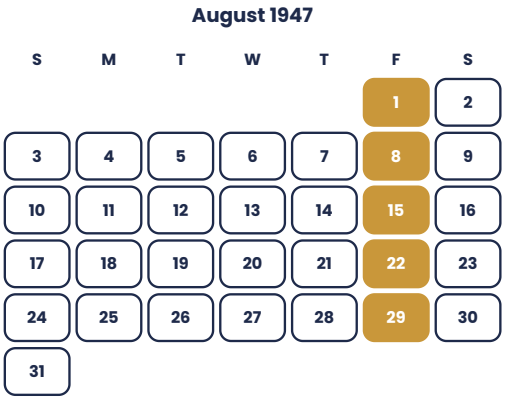
Century	1700s	1800s	1900s	2000s
Code	4	2	0	6

Year code = last two digits (YY) + ⌊YY ÷ 4⌋.

WORKED EXAMPLE

15 August 1947

Date 15 + month code (Aug) 2 + year code 47 + 11 = 58 + century code 0 = 75. 75 mod 7 = 5 → **Friday**.



AUGUST 1947: THE 15TH IS A FRIDAY

TIP

This program-checked method works for every date from 1700 to 2099. For questions like "if 1 January is a Monday, what day is …", counting odd days from the given date is faster.

18.4 Repeating calendars

18.4.1 Years with the same calendar

Two years have identical calendars if they start on the same weekday and are both leap or both ordinary. Add up the odd days year by year from the given year until the total is a multiple of 7 and the year type matches.

Year	Same calendar	Rule of thumb
2026 (ordinary)	2037	year after a leap year: + 11
2023 (ordinary)	2034	two years after a leap year: + 11
2024 (leap)	2052	leap year: + 28

18.5 Practice questions

18.5.1 Basic level

Q1 How many odd days are there in an ordinary year?

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

Q2 How many odd days are there in a leap year?

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

Q3 Is 1900 a leap year?

(a) Yes (b) No
(c) Only in some countries (d) Cannot be determined

Q4 Today is Monday. What day will it be after 61 days?

(a) Friday (b) Saturday (c) Sunday (d) Thursday

Q5 How many odd days are there in 100 years?

(a) 3 (b) 4 (c) 5 (d) 6

Q6 In an ordinary year 1 January is a Sunday. What day is 31 December?

(a) Saturday (b) Sunday (c) Monday (d) Friday

Q7 5 March is a Tuesday. What day is 26 March of the same year?

(a) Monday (b) Tuesday (c) Wednesday (d) Sunday

Q8 Which of these is a leap year?

(a) 2100 (b) 2000 (c) 1900 (d) 2022

18.5.2 Intermediate level

Q9 What day of the week was 15 August 1947?

(a) Thursday (b) Friday (c) Saturday (d) Sunday

Q10 What day of the week was 26 January 1950?

(a) Wednesday (b) Thursday (c) Friday (d) Monday

Q11 1 January 2025 was a Wednesday. What day was 1 January 2026?

(a) Wednesday (b) Thursday (c) Friday (d) Tuesday

Q12 1 January 2024 was a Monday. What day was 1 January 2025?

(a) Tuesday (b) Wednesday (c) Thursday (d) Monday

Q13 Which year will have the same calendar as 2026?

(a) 2032 (b) 2036 (c) 2037 (d) 2043

Q14 Which year will have the same calendar as 2024?

(a) 2048 (b) 2052 (c) 2056 (d) 2030

Q15 How many Sundays are there in February 2026?

(a) 3 (b) 4 (c) 5 (d) 6

Q16 What day of the week was 2 October 1869?

(a) Friday (b) Saturday (c) Sunday (d) Monday

18.5.3 Advanced level

Q17 8 February 2005 was a Tuesday. What day was 8 February 2004?

(a) Monday (b) Sunday (c) Saturday (d) Wednesday

Q18 On which dates of March 2027 will Saturdays fall?

(a) 6, 13, 20, 27 **(b)** 5, 12, 19, 26
(c) 7, 14, 21, 28 **(d)** 1, 8, 15, 22, 29

Q19 What day of the week was 1 January 2001?

(a) Sunday **(b)** Monday **(c)** Tuesday **(d)** Saturday

Q20 The last day of a century cannot be which day?

(a) Monday **(b)** Tuesday **(c)** Wednesday **(d)** Friday

Q21 What day of the week will 31 December 2099 be?

(a) Wednesday **(b)** Thursday **(c)** Friday **(d)** Sunday

Q22 15 March 2026 is a Sunday. What day will 15 March 2027 be?

(a) Sunday **(b)** Monday **(c)** Tuesday **(d)** Saturday

Q23 How many times does the 13th of a month fall on a Friday in 2026?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

Q24 What day of the week was 1 January 1900?

(a) Sunday **(b)** Monday **(c)** Tuesday **(d)** Saturday

Q25 Which is the next year after 2023 with exactly the same calendar?

(a) 2028 **(b)** 2029 **(c)** 2034 **(d)** 2051

18.6 Solutions

Q1 **Answer: (b) 1**

365 = 52 × 7 + 1: **1** odd day.

Q2 **Answer: (c) 2**

366 = 52 × 7 + 2: **2** odd days.

Q3 **Answer: (b) No**

It is a century year not divisible by 400: **no**.

Q4 **Answer: (b) Saturday**

61 mod 7 = 5; Monday + 5 = **Saturday**.



Q5 **Answer: (c) 5**

100 years = 76 ordinary + 24 leap years: 76 + 48 = 124 = 17 × 7 + 5, so **5**.

Q6 **Answer: (b) Sunday**

31 December is 364 days later, exactly 52 weeks: **Sunday**.

Q7 **Answer: (b) Tuesday**

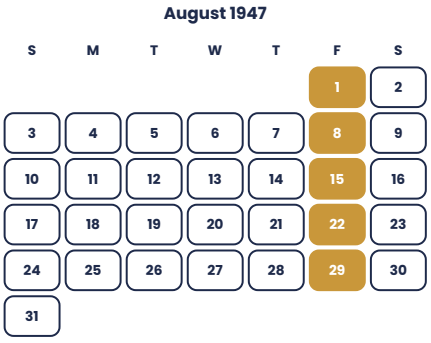
21 days later, exactly 3 weeks: **Tuesday**.

Q8 **Answer: (b) 2000**

2000 is divisible by 400. 2100 and 1900 are century years not divisible by 400, and 2022 is not divisible by 4.

Q9 **Answer: (b) Friday**

15 + 2 (Aug) + 47 + 11 + 0 (1900s) = 75; 75 mod 7 = 5 → **Friday**.



Q10 **Answer: (b) Thursday**

26 + 0 (Jan) + 50 + 12 + 0 = 88; 88 mod 7 = 4 → **Thursday** (1950 is not a leap year, so no correction).

Q11 **Answer: (b) Thursday**

2025 is an ordinary year: 1 odd day. Wednesday + 1 = **Thursday**.

Q12 **Answer: (b) Wednesday**

2024 is a leap year: 2 odd days. Monday + 2 = **Wednesday**.

Q13 **Answer: (c) 2037**

Odd days from 2026: 2026 1, 2027 1, 2028 2, 2029 1, 2030 1, 2031 1, 2032 2, 2033 1, 2034 1, 2035 1, 2036 2 → total 14, a multiple of 7, and 2037 is ordinary like 2026: **2037**.

Q14 **Answer: (b) 2052**

For a leap year the calendar repeats after 28 years (when no century year interrupts): **2052**.

Q15 **Answer: (b) 4**

February 2026 has 28 days, exactly 4 weeks, so every weekday occurs **4** times.

February 2026

S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28

Q16 **Answer: (b) Saturday**

2 + 0 (Oct) + 69 + 17 + 2 (1800s) = 90; 90 mod 7 = 6 → **Saturday**.

Q17 **Answer: (b) Sunday**

The year from 8 Feb 2004 to 8 Feb 2005 contains 29 Feb 2004, so it has 366 days = 2 odd days. Going back 2 days from Tuesday: **Sunday**.

Q18 **Answer: (a) 6, 13, 20, 27**

1 March 2027: 1 + 3 + 27 + 6 + 6 = 43; 43 mod 7 = 1 → Monday. The first Saturday is the 6th, so **6, 13, 20, 27**.

March 2027

S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31			

Q19 **Answer: (b) Monday**

Code method: 1 (date) + 0 (Jan) + 1 (YY = 01, plus 01 ÷ 4 = 0) + 6 (2000s) = 8; 8 mod 7 = 1 → **Monday**. 2001 is not a leap year, so no correction.

Q20 **Answer: (b) Tuesday**

100, 200, 300, 400 years give 5, 3, 1, 0 odd days: Friday, Wednesday, Monday, Sunday. It can never be **Tuesday** (or Thursday or Saturday).

Q21 **Answer: (b) Thursday**

31 + 5 (Dec) + 99 + 24 + 6 = 165; 165 mod 7 = 4 → **Thursday**.

Q22 **Answer: (b) Monday**

From March 2026 to March 2027 there is no 29 February, so 365 days = 1 odd day: **Monday**.

Q23 **Answer: (c) 3**

1 Jan 2026 is a Thursday. The 13ths fall on Friday in February, March and November 2026: **3** times.

Q24 **Answer: (b) Monday**

Code method: 1 (date) + 0 (Jan) + 0 (YY = 00) + 0 (1900s) = 1 → **Monday**. 1900 is not a leap year (century not divisible by 400), so no correction.

Q25 **Answer: (c) 2034**

Odd days 2023 1, 2024 2, 2025 1, 2026 1, 2027 1, 2028 2, 2029 1, 2030 1, 2031 1, 2032 2, 2033 1 → 14, and 2034 is ordinary: **2034**.

Odd days = days mod 7: 1 per ordinary year, 2 per leap year, 5 per 100 years, 0 per 400 years. Century years are leap only if divisible by 400. For any date use date + month code + year code (YY + YY/4) + century code, minus 1 for Jan–Feb of a leap year, mod 7 (0 = Sunday). Calendars repeat when the year type matches and the odd days total a multiple of 7.

MODULE 19 · INTERMEDIATE

Input-output

Minimum duration **2 days** | Level **Intermediate**

An input-output (machine) question shows a line of words and numbers and how a machine rearranges or changes it step by step. You must find the rule from the example and apply it to a new input. This module covers sorting machines (numbers, words, and mixed lines), operation machines and flowcharts, with a step diagram for every example and 25 solved questions generated and checked by a program.

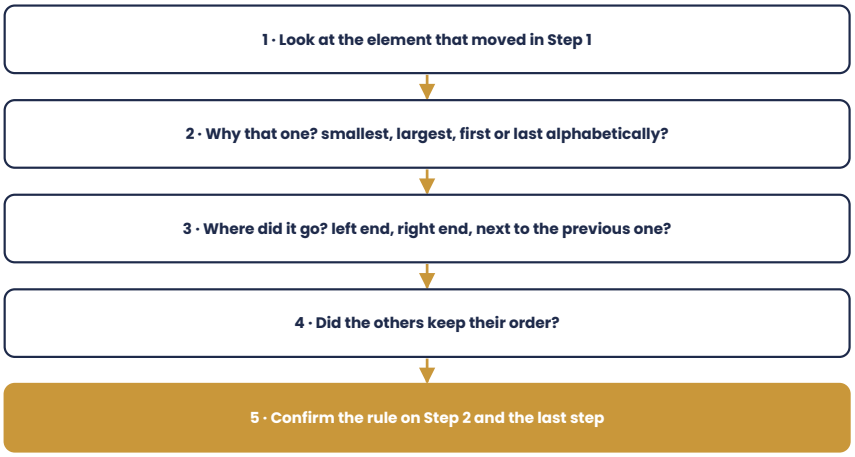
IN THIS MODULE

19.1 Basics
19.2 Sorting machines
19.3 Mixed words-and-numbers machines
19.4 Operation machines and flowcharts
19.5 Practice questions
19.6 Solutions

19.1 Basics

19.1.1 How to find the rule

Compare the input with Step 1, then Step 1 with Step 2, and ask: **which element moved, and where did it go?** In sorting machines one element (or one word and one number) settles at an end each step, and the rest shift without changing their order. In operation machines every element changes by the same rule.



FINDING THE MACHINE RULE

WATCH OUT

Do not assume the rest of the line is rearranged at every step. In standard machines the unmoved elements keep their relative order, which is what lets you write any step directly.

19.2 Sorting machines

19.2.1 Numbers in ascending order from the left

Rule A: in each step, the smallest number not yet arranged moves to the left, next to the numbers already arranged; the others shift right without changing order.

Input	67	23	91	15	48	34
Step 1	15	67	23	91	48	34
Step 2	15	23	67	91	48	34
Step 3	15	23	34	67	91	48
Step 4	15	23	34	48	67	91

Rule A: 4 steps (gold = the number placed in this step)

A NUMBER-SORTING MACHINE

SHORTCUT TRICK

To write Step k directly: the k smallest numbers in order, followed by the remaining numbers in their **original** order. No need to write the steps in between.

19.2.2 Words in alphabetical order

Rule B: the same machine for words: in each step the alphabetically first remaining word moves to the left block.

Input	tiger	apple	mango	dog	zebra	cat
Step 1	apple	tiger	mango	dog	zebra	cat
Step 2	apple	cat	tiger	mango	dog	zebra
Step 3	apple	cat	dog	tiger	mango	zebra
Step 4	apple	cat	dog	mango	tiger	zebra

Rule B: 4 steps

A WORD-SORTING MACHINE

TIP

If an element is already in its correct place, that step may show no change for it, and the machine finishes early. Count the steps in which something actually moves.

19.3 Mixed words-and-numbers machines

19.3.1 One word and one number per step

Rule C: in each step, the alphabetically first remaining word moves to the left block, and the largest remaining number moves to the right block (each new number settles just left of the numbers already placed). The rest keep their order.

Input	fan	27	bat	81	cup	45	ant	63
Step 1	ant	fan	27	bat	cup	45	63	81
Step 2	ant	bat	fan	27	cup	45	63	81
Step 3	ant	bat	cup	fan	27	45	63	81

Rule C: 3 steps

A MIXED MACHINE

Rule E (used in questions 21 to 25): in each step, the smallest remaining number moves to the left block and the alphabetically **last** remaining word moves to the right block.

Input	pen	52	arc	17	map	33
Step 1	17	52	arc	map	33	pen
Step 2	17	33	52	arc	map	pen

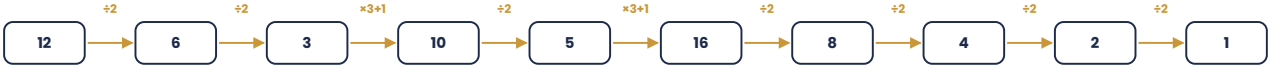
Rule E on a sample input

ANOTHER MIXED MACHINE

19.4 Operation machines and flowcharts

19.4.1 Every element changes by a rule

Some machines change values instead of positions, for example "if the number is even, halve it; if odd, multiply by 3 and add 1". Apply the rule step by step and keep a count.



12 reaches 1 in 9 steps

AN OPERATION MACHINE (THE 3N + 1 RULE)

19.4.2 Flowcharts with a loop

A flowchart repeats a box until a condition is met. Track the variable in a small table: one row per pass through the loop.

Pass	x before	x = 2x + 1	x > 50?
1	3	7	no
2	7	15	no
3	15	31	no
4	31	63	yes → stop

19.5 Practice questions

19.5.1 Basic level

Q1 Rule A (19.2.1), input 67 23 91 15 48 34. Which is Step 2?

(a) 15 23 67 91 48 34 **(b)** 15 23 34 67 91 48
(c) 15 67 23 91 48 34 **(d)** 15 23 34 48 91 67

Q2 Rule A, input 67 23 91 15 48 34. How many steps are needed to complete the arrangement?

(a) 3 **(b)** 4 **(c)** 5 **(d)** 6

Q3 Rule A, input 67 23 91 15 48 34. Which number is 4th from the left in Step 3?

(a) 67 **(b)** 48 **(c)** 91 **(d)** 34

Q4 Rule B (19.2.2), input tiger apple mango dog zebra cat. Which is Step 1?

(a) apple tiger mango dog zebra cat **(b)** apple cat tiger mango dog zebra
(c) apple tiger mango dog cat zebra **(d)** tiger apple mango dog zebra cat

Q5 Rule B, input tiger apple mango dog zebra cat. How many steps are needed?

(a) 4 **(b)** 5 **(c)** 6 **(d)** 3

Q6 Rule B, input tiger apple mango dog zebra cat. Which word is 3rd from the right in Step 2?

(a) mango **(b)** dog **(c)** zebra **(d)** tiger

Q7 A machine applies the rule "if even, halve it; if odd, multiply by 3 and add 1". How many steps does it take to turn 12 into 1?

(a) 8 **(b)** 9 **(c)** 10 **(d)** 7

Q8 Using the same rule, how many steps does it take to turn 7 into 1?

(a) 14 **(b)** 15 **(c)** 16 **(d)** 17

Q9 A flowchart starts with x = 3 and repeats "x = 2x + 1" until x is greater than 50. What is the final value of x?

(a) 31 **(b)** 63 **(c)** 127 **(d)** 51

Q10 In the same flowchart, how many times is the box "x = 2x + 1" executed?

(a) 3 **(b)** 4 **(c)** 5 **(d)** 6

19.5.2 Intermediate level

Q11 Rule C (19.3.1), input fan 27 bat 81 cup 45 ant 63. Which is Step 1?

(a) ant fan 27 bat cup 45 63 81 **(b)** ant fan bat 27 cup 45 63 81
(c) fan 27 bat cup 45 ant 63 81

Q12 Rule C, input fan 27 bat 81 cup 45 ant 63. Which is Step 2?

(a) ant bat fan 27 cup 45 63 81 **(b)** ant bat cup fan 27 45 63 81
(c) ant bat fan cup 27 45 63 81 **(d)** ant bat fan 27 cup 63 45 81

Q13 Rule C, input fan 27 bat 81 cup 45 ant 63. How many steps are needed?

(a) 3 **(b)** 4 **(c)** 5 **(d)** 6

Q14 Rule C, input fan 27 bat 81 cup 45 ant 63. In Step 2, which element is 4th from the right?

(a) cup **(b)** 45 **(c)** 27 **(d)** fan

Q15 Rule C, input fan 27 bat 81 cup 45 ant 63. In the final step, which element is exactly in the middle-left (4th from the left)?

(a) fan **(b)** cup **(c)** 27 **(d)** bat

Q16 Rule A, input 56 12 89 34 71 20. Which is Step 3?

(a) 12 20 34 56 89 71 **(b)** 12 20 34 89 56 71
(c) 12 20 56 34 89 71 **(d)** 12 20 34 71 56 89

Q17 Rule B, input pear lime fig date kiwi. Which is Step 2?

(a) date fig pear lime kiwi **(b)** date fig pear kiwi lime
(c) date fig lime kiwi pear **(d)** date pear lime fig kiwi

Q18 Rule B, input pear lime fig date kiwi. How many steps are needed?

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

Q19 Rule A: Step 2 of some input is "11 19 64 37 52 28". What is Step 4?

(a) 11 19 28 37 64 52 **(b)** 11 19 28 37 52 64
(c) 11 19 28 64 37 52 **(d)** 11 19 37 28 64 52

Q20 Rule A: if the input has 7 numbers and none is already in its final place, what is the largest possible number of steps?

(a) 5 **(b)** 6 **(c)** 7 **(d)** 8

19.5.3 Advanced level

Q21 Rule E (19.3.1), input kite 39 bell 14 rose 62 frog 25. Which is Step 1?

(a) 14 kite 39 bell 62 frog 25 rose **(b)** 14 kite 39 bell rose 62 frog 25
(c) kite 14 39 bell 62 frog 25 rose **(d)** 14 39 kite bell 62 frog 25 rose

Q22 Rule E, input kite 39 bell 14 rose 62 frog 25. How many steps are needed?

(a) 3 **(b)** 4 **(c)** 5 **(d)** 6

Q23 Rule E, input kite 39 bell 14 rose 62 frog 25. Which is the final arrangement?

(a) 14 25 39 62 bell frog kite rose **(b)** 14 25 39 62 rose kite frog bell

(c) 62 39 25 14 bell frog kite rose **(d)** bell frog kite rose 14 25 39 62

Q24 Rule E, input kite 39 bell 14 rose 62 frog 25. In Step 2, which element is 3rd from the left?

(a) 39 **(b)** bell **(c)** frog **(d)** kite

Q25 Rule E, input kite 39 bell 14 rose 62 frog 25. In Step 3, how many elements lie between "bell" and 62?

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

19.6 Solutions

Q1 **Answer: (a) 15 23 67 91 48 34**

Step 1 moves 15 (smallest) to the left; Step 2 moves 23 next to it; the rest keep their order: **15 23 67 91 48 34**.

Input	67	23	91	15	48	34
Step 1	15	67	23	91	48	34
Step 2	15	23	67	91	48	34

Q2 **Answer: (b) 4**

Steps move 15, 23, 34 and 48 in turn; after Step 4 the line is 15 23 34 48 67 91 and 67, 91 are already in order. Answer **4**.

Q3 **Answer: (a) 67**

Step 3: 15 23 34 67 91 48. 4th from the left = **67**.

Q4 **Answer: (a) apple tiger mango dog zebra cat**

"apple" is first alphabetically; it is already second, so it moves to the left end: **apple tiger mango dog zebra cat**.

Q5 **Answer: (a) 4**

The steps place apple, cat, dog, mango; after that tiger and zebra are already in order, so the machine stops after **4** steps.

Input	tiger	apple	mango	dog	zebra	cat
Step 1	**apple**	tiger	mango	dog	zebra	cat
Step 2	apple	**cat**	tiger	mango	dog	zebra
Step 3	apple	cat	**dog**	tiger	mango	zebra
Step 4	apple	cat	dog	**mango**	tiger	zebra

Q6 **Answer: (a) mango**

Step 2: apple cat tiger mango dog zebra. 3rd from the right = **mango**.

Q7 **Answer: (b) 9**

12 → 6 → 3 → 10 → 5 → 16 → 8 → 4 → 2 → 1: **9** steps.

Q8 **Answer: (c) 16**

7 → 22 → 11 → 34 → 17 → 52 → 26 → 13 → 40 → 20 → 10 → 5 → 16 → 8 → 4 → 2 → 1: **16** steps.

Q9 **Answer: (b) 63**

x goes 3 → 7 → 15 → 31 → 63, and 63 > 50, so it stops at **63**.

Q10 **Answer: (b) 4**

It runs for 7, 15, 31 and 63: **4** times.

Q11 **Answer: (a) ant fan 27 bat cup 45 63 81**

"ant" moves to the left end and 81 (largest) to the right end; the rest keep their order: **ant fan 27 bat cup 45 63 81**.

Input	fan	27	bat	81	cup	45	ant	63
Step 1	**ant**	fan	27	bat	cup	45	63	**81**

Q12 **Answer: (a) ant bat fan 27 cup 45 63 81**

"bat" joins the left block and 63 settles just left of 81: **ant bat fan 27 cup 45 63 81**.

Q13 **Answer: (a) 3**

One word and one number settle per step; with four of each the final line ant bat cup fan 27 45 63 81 appears after Step **3**.

Q14 **Answer: (a) cup**

Step 2: ant bat fan 27 cup 45 63 81. 4th from the right = **cup**.

Q15 **Answer: (a) fan**

Final: ant bat cup fan 27 45 63 81. 4th from the left = **fan**.

Q16 **Answer: (a) 12 20 34 56 89 71**

The three smallest (12, 20, 34) in order, then the rest in original order: **12 20 34 56 89 71**.

Q17 **Answer: (a) date fig pear lime kiwi**

The two first words alphabetically (date, fig) in order, then the rest in original order: **date fig pear lime kiwi**.

Q18 **Answer: (c) 4**

Steps: date pear lime fig kiwi; date fig pear lime kiwi; date fig kiwi pear lime; date fig kiwi lime pear. Answer **4**.

Input	pear	lime	fig	date	kiwi
Step 1	**date**	pear	lime	fig	kiwi
Step 2	date	**fig**	pear	lime	kiwi
Step 3	date	fig	**kiwi**	pear	lime
Step 4	date	fig	kiwi	**lime**	pear

Q19 **Answer: (a) 11 19 28 37 64 52**

Step 3 moves 28 next to 19: 11 19 28 64 37 52. Step 4 moves 37: **11 19 28 37 64 52**.

Q20 **Answer: (b) 6**

Each step fixes one number; after 6 numbers are placed the 7th is automatically in place: at most **6** steps.

Q21 **Answer: (a) 14 kite 39 bell 62 frog 25 rose**

14 (smallest number) moves to the left end and "rose" (last alphabetically) to the right end: **14 kite 39 bell 62 frog 25 rose**.

Input	kite	39	bell	14	rose	62	frog	25
Step 1	**14**	kite	39	bell	62	frog	25	**rose**
Step 2	14	**25**	39	bell	62	frog	**kite**	rose
Step 3	14	25	39	**62**	bell	frog	kite	rose

Q22 **Answer: (a) 3**

One number and one word settle per step; the final line 14 25 39 62 bell frog kite rose appears after Step **3**.

Q23 **Answer: (a) 14 25 39 62 bell frog kite rose**

Numbers ascending on the left, words in alphabetical order on the right (the last word was placed first, at the right end): **14 25 39 62 bell frog kite rose**.

Q24 **Answer: (a) 39**

Step 2: 14 25 39 bell 62 frog kite rose. 3rd from the left = **39**.

Q25 **Answer: (a) 0**

Step 3: 14 25 39 62 bell frog kite rose. Elements between bell and 62: **0**.

Find which element moved between consecutive steps, why it was chosen (smallest, largest, alphabetical first or last) and where it went; the others keep their order. Then any step k can be written directly: the k placed elements in order plus the rest in their original order. For operation machines and flowcharts, keep a table with one row per step.

LEVEL 3

Advanced level

Handle statements and assumptions, conclusions and arguments, cause and effect, data sufficiency, critical reasoning, formal logic and advanced puzzles under exam timing

8 MODULES · 14 DAYS MINIMUM

Module 20 · Statement and assumptions	2 days
Module 21 · Conclusions, arguments and courses of action	2 days
Module 22 · Cause and effect, decision making	2 days
Module 23 · Data sufficiency	2 days
Module 24 · Critical reasoning	2 days
Module 25 · Formal logic	2 days
Module 26 · Advanced puzzles	1 day
Module 27 · Exam strategy and mock practice	1 day

MODULE 20 · ADVANCED

Statement and assumptions

Minimum duration **2 days** | Level **Advanced**

An assumption is something the speaker takes for granted but does not say. Statement-assumption questions give a statement (an advertisement, notice, advice, decision or claim) and ask which assumptions are implicit in it. This module explains what counts as an assumption, the negation test, the patterns for each type of statement, the traps (extreme words, new information, restated facts), and gives 25 solved questions.

IN THIS MODULE

20.1 Basics
20.2 Testing an assumption
20.3 Statement types
20.4 Traps
20.5 Practice questions
20.6 Solutions

20.1 Basics

20.1.1 What an assumption is

An **assumption** is an unstated idea that the speaker must believe for the statement to make sense. It is not stated in the statement (that would be a fact), and it is not a result of the statement (that would be a conclusion). It is a hidden **premise**: a belief that comes before the statement.



AN ASSUMPTION COMES BEFORE THE STATEMENT; A CONCLUSION COMES AFTER IT

WORKED EXAMPLE

"Use XYZ toothpaste for whiter teeth."

The advertiser assumes that people **want** whiter teeth (otherwise the advertisement is pointless). The advertiser does **not** need to assume that other toothpastes cannot whiten teeth.

Answer choice	Use when
Only I is implicit	I is necessary, II is not
Only II is implicit	II is necessary, I is not
Either I or II is implicit	neither alone is necessary, but one of the two must be assumed (rare)
Neither I nor II is implicit	neither is necessary
Both I and II are implicit	both are necessary

20.2 Testing an assumption

20.2.1 The negation test

Negate the proposed assumption. If the statement then becomes pointless or self-contradictory, the assumption is implicit. If the statement still makes sense, it is not.

Statement	Proposed assumption	Negation	Implicit?
Please travel light by train.	Heavy luggage causes inconvenience.	Heavy luggage causes no inconvenience → the request is pointless.	Yes
Please travel light by train.	Passengers read notices.	Passengers never read notices → putting up the notice is pointless.	Yes
Use XYZ toothpaste for whiter teeth.	Other toothpastes cannot whiten teeth.	Other toothpastes can whiten teeth → the advertisement still makes sense.	No

SHORTCUT TRICK
Ask: "Would the speaker still say this if the assumption were false?" If yes, the assumption is not implicit.

20.3 Statement types

20.3.1 What each kind of statement assumes

Statement type	Usually implicit	Usually not implicit
Advertisement	people want the benefit; the advertisement will attract buyers	competitors are bad; the product is the only one
Notice or appeal	people will read and respond; the problem is real	everyone will obey; people never follow rules
Advice	the advice will help; the person can follow it	it is the only way
Decision or policy	the step will achieve its aim; it is feasible	it will solve the problem completely; there are no side effects
Claim with "despite", "even though"	the normal expectation is the opposite	the reason for the surprise

TIP
Notices, appeals and advertisements almost always assume that people will notice them and that they will have some effect. These assumptions are implicit even though they look obvious.

20.4 Traps

20.4.1 Extreme words, new information and restated facts

- **Extreme words** (only, all, never, always, completely, best) usually make an assumption too strong: "Only people with computer skills get jobs" is not assumed by "Learn computer skills to get a job".
- **New information** the speaker does not need ("Ram missed the class") is not an assumption, even if it is possible.
- **Restated facts** already in the statement are not assumptions; they are given.
- **Consequences** ("the shop will make a loss") are not assumptions; assumptions come before the statement, not after it.

> **WATCH OUT**
> An assumption does not have to be true in real life, and a true fact is not automatically an assumption. Judge only what the speaker must believe.

20.5 Practice questions

20.5.1 Basic level

Q1 Statement: Advertisement: "Use XYZ toothpaste for whiter teeth."
Assumptions: I. People want whiter teeth. II. Other toothpastes cannot whiten teeth.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q2 Statement: Notice by the railways: "Please travel light."
Assumptions: I. Heavy luggage causes inconvenience to passengers. II. Passengers read such notices.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q3 Statement: The school has decided to start evening classes for working adults.
Assumptions: I. There are working adults who want to study. II. All working adults are free in the evening.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q4 Statement: Ravi to Sita: "Take the metro and you will reach the office on time."
Assumptions: I. The metro runs on schedule. II. Sita wants to reach the office on time.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q5 Statement: The government has raised the price of petrol by 5%.
Assumptions: I. People will stop using vehicles. II. The increase will bring in more revenue.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q6 Statement: Notice in a bus: "Do not lean out of the window of a moving bus."
Assumptions: I. Leaning out of a moving bus is dangerous. II. Passengers never follow instructions.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q7 Statement: Admission is open to candidates who scored at least 60% in Class XII.
Assumptions: I. Candidates with lower marks will not apply. II. Class XII marks indicate the ability to cope with the course.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q8 Statement: Shop notice: "Buy one shirt, get one free, this week only."
Assumptions: I. The offer will attract more customers. II. The shop will make a loss this week.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

20.5.2 Intermediate level

Q9 Statement: To reduce traffic jams, the city will build a flyover at the main junction.
Assumptions: I. The flyover will help reduce traffic jams. II. The city has the resources to build the flyover.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q10 Statement: "If you want to get a job, learn computer skills."
Assumptions: I. Only people with computer skills get jobs. II. Computer skills help in getting a job.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q11 Statement: The company has asked its employees to work from home on Fridays.
Assumptions: I. Employees have the facilities to work from home. II. Working from home will reduce productivity.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q12 Statement: The library will remain closed on Sunday for maintenance.
Assumptions: I. Maintenance work cannot be done on any other day. II. The library is normally open on Sundays.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q13 Statement: The doctor advised Mr Rao to walk for 30 minutes every day.
Assumptions: I. Walking is good for Mr Rao's health. II. Mr Rao can spare 30 minutes a day.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q14 Statement: To attract more tourists, the state will set up information desks at railway stations.
Assumptions: I. Tourists need information when they arrive. II. Many tourists arrive by train.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q15 Statement: The shop sells fresh vegetables every morning.
Assumptions: I. Vegetables sold in the evening are not fresh. II. People buy vegetables only in the morning.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q16 Statement: Ram asked Shyam to bring his notebook so that Ram could copy the notes.
Assumptions: I. Ram does not own a notebook. II. Shyam is Ram's best friend.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

20.5.3 Advanced level

Q17 Statement: The government has advised people not to travel to the hill region during the monsoon because of landslides.
Assumptions: I. Landslides are likely in the hill region during the monsoon. II. People will pay attention to the advice.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q18 Statement: Despite the increase in fares, the number of passengers on the airline has gone up.
Assumptions: I. Fare increases usually reduce the number of passengers. II. The airline now offers better service.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q19 Statement: Only those who have a valid pass will be allowed inside the stadium.
Assumptions: I. Some people may try to enter without a pass. II. Passes have been issued to people.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q20 Statement: The bank will open branches in all villages with a population of more than 5,000.
Assumptions: I. Villages with fewer than 5,000 people do not need banking services. II. People in such villages need banking services.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q21 Statement: Advertisement: "Learn French in 30 days with our online course."
Assumptions: I. People want to learn French quickly. II. It is impossible to learn French in less than 30 days.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q22 Statement: The municipal corporation has asked residents to store water because the supply will be cut for two days.
Assumptions: I. Residents have the means to store water. II. Residents will not need water during these two days.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q23 Statement: To improve results, the principal decided to give extra classes to weak students.
Assumptions: I. Extra classes will help weak students improve. II. Weak students will attend the extra classes.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q24 Statement: A newspaper reported that the river is polluted and asked people not to bathe in it. Assumptions: I. Bathing in polluted water can harm health. II. The river was never clean.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

Q25 Statement: Mohan to his friend: "Let us take the 8 a.m. bus to reach the exam centre in time." Assumptions: I. There is no other way to reach the exam centre. II. The 8 a.m. bus will reach the exam centre in time.

(a) Only I is implicit
(b) Only II is implicit
(c) Either I or II is implicit
(d) Neither I nor II is implicit
(e) Both I and II are implicit

20.6 Solutions

Q1 **Answer: (a) Only I is implicit**

I is implicit: the advertisement is pointless if nobody wants whiter teeth. II is not needed; XYZ can be one of many good toothpastes. Answer: **Only I is implicit**.

Q2 **Answer: (e) Both I and II are implicit**

Without I the request has no reason; without II the notice is pointless. Both are implicit. Answer: **Both I and II are implicit**.

Q3 **Answer: (a) Only I is implicit**

I is the reason for the decision. II is extreme ("all"); the school only needs some adults to be free. Answer: **Only I is implicit**.

Q4 **Answer: (e) Both I and II are implicit**

Ravi's advice assumes the metro is punctual (I) and that reaching on time matters to Sita (II). Answer: **Both I and II are implicit**.

Q5 **Answer: (b) Only II is implicit**

A price increase assumes some benefit, such as revenue (II). "Stop using vehicles" is extreme and not needed (I). Answer: **Only II is implicit**.

Q6 **Answer: (a) Only I is implicit**

I is the reason for the notice. II contradicts the purpose of a notice. Answer: **Only I is implicit**.

Q7 **Answer: (b) Only II is implicit**

The cut-off assumes marks reflect ability (II). The rule does not need I; it exists precisely because some may apply. Answer: **Only II is implicit**.

Q8 **Answer: (a) Only I is implicit**

An offer assumes it will attract buyers (I). II is a possible consequence, not an assumption. Answer: **Only I is implicit**.

Q9 **Answer: (e) Both I and II are implicit**

A decision assumes it will work (I) and that it is feasible (II). Answer: **Both I and II are implicit**.

Q10 **Answer: (b) Only II is implicit**

The advice assumes the skills help (II). "Only" in I is too strong. Answer: **Only II is implicit**.

Q11 **Answer: (a) Only I is implicit**

The instruction assumes it is possible (I). II would argue against the decision, so it is not assumed. Answer: **Only I is implicit**.

Q12 **Answer: (b) Only II is implicit**

Announcing a Sunday closure only makes sense if it is usually open on Sundays (II). I is too strong; Sunday may just be convenient. Answer: **Only II is implicit**.

Q13 **Answer: (e) Both I and II are implicit**

Advice assumes it is beneficial (I) and possible to follow (II). Answer: **Both I and II are implicit**.

Q14 **Answer: (e) Both I and II are implicit**

The desks are useful only if tourists need information (I) and come through railway stations (II). Answer: **Both I and II are implicit**.

Q15 **Answer: (d) Neither I nor II is implicit**

The statement describes the shop's practice. It says nothing about evening vegetables (I), and "only" in II is extreme. Neither is implicit. Answer: **Neither I nor II is implicit**.

Q16 **Answer: (d) Neither I nor II is implicit**

The request assumes Shyam's notebook has the notes, but neither I nor II is needed for the request to make sense. Answer: **Neither I nor II is implicit**.

Q17 **Answer: (e) Both I and II are implicit**

The advice assumes the danger is real (I) and that people will heed it (II). Answer: **Both I and II are implicit**.

Q18 **Answer: (a) Only I is implicit**

"Despite" shows the speaker expects higher fares to reduce passengers (I). II is one possible reason, not something the speaker must believe. Answer: **Only I is implicit**.

Q19 **Answer: (e) Both I and II are implicit**

The rule assumes that some may try to enter without a pass (I) and that passes exist (II). Answer: **Both I and II are implicit**.

Q20 **Answer: (b) Only II is implicit**

The decision assumes a need in those villages (II). It says nothing about smaller villages (I). Answer: **Only II is implicit**.

Q21 **Answer: (a) Only I is implicit**

The selling point is speed, so the advertiser assumes people want to learn quickly (I). II is not needed. Answer: **Only I is implicit**.

Q22 **Answer: (a) Only I is implicit**

The request assumes residents can store water (I). II contradicts the reason for the request. Answer: **Only I is implicit**.

Q23 **Answer: (e) Both I and II are implicit**

The decision assumes the classes help (I) and that the students will come (II). Answer: **Both I and II are implicit**.

Q24 **Answer: (a) Only I is implicit**

The appeal assumes polluted water is harmful (I). II is about the past and not needed. Answer: **Only I is implicit**.

Q25 **Answer: (b) Only II is implicit**

The suggestion assumes the bus will get them there in time (II). It does not assume the bus is the only way (I). Answer: **Only II is implicit**.

An assumption is an unstated belief the speaker must hold for the statement to make sense. Use the negation test: if negating it makes the statement pointless, it is implicit. Advertisements, notices, advice and decisions assume that they will work and can be acted upon. Reject extreme words, unnecessary new information, restated facts and consequences.

MODULE 21 · ADVANCED

Conclusions, arguments and courses of action

Minimum duration **2 days** | Level **Advanced**

These verbal-reasoning formats all test judgement about what a statement supports: which conclusions definitely follow, which arguments are strong, which actions are sensible responses to a problem, and how true an inference from a passage is. This module gives a clear test for each format, the typical traps, and 25 solved questions with the reasoning written out.

IN THIS MODULE

21.1 Statement and conclusions
21.2 Strong and weak arguments
21.3 Courses of action
21.4 Inferences from a passage
21.5 Practice questions
21.6 Solutions

21.1 Statement and conclusions

21.1.1 A conclusion must follow from the statement alone

A conclusion follows only if it can be drawn **directly from the statement**, without outside knowledge and without adding assumptions. Facts that are generally true but not supported by the statement do not follow.



A CONCLUSION COMES AFTER THE STATEMENT

Conclusion type	Follows?
Restates or combines the given facts	Yes
A regular pattern described in the statement continuing	Usually yes
Reverses a one-way statement (“closed on Mondays” → “today is Monday”)	No
Uses extreme words or outside facts	No
Two complementary conclusions, neither certain	Either

WATCH OUT
“The shop is closed on Mondays; the shop is closed today” does not mean today is Monday: the shop may also close on other days.

21.2 Strong and weak arguments

21.2.1 What makes an argument strong

A strong argument is **relevant** to the question, **important** (it touches a significant consequence) and **directly connected** to the yes or no it supports. An argument is weak if it is trivial, emotional, based on an unproven comparison, ambiguous, or does not actually support its side.

Weak because	Example (Should mobile phones be banned in schools?)
Does not support its side	"No, students use them to play games."
Trivial or superficial	"Yes, phones are expensive."
Unproven comparison	"No, other countries allow them."
Emotional or extreme	"Yes, phones destroy the young generation."

SHORTCUT TRICK
Read the argument with its "Yes" or "No" in front and ask: does this reason, if true, really make a serious case for that answer? If you have to stretch it, it is weak.

21.3 Courses of action

21.3.1 A course of action must be practical and proportionate

A course of action follows if it is **practical**, would **reduce or solve the problem**, and is **proportionate** (not extreme). Banning, closing or punishing everyone are usually too extreme.

Problem	Follows	Does not follow
Rise in night accidents on a highway	Increase night patrolling and speed checks	Close the highway at night
Food poisoning after a feast	Test food samples from the caterer	Ban all weddings

21.4 Inferences from a passage

21.4.1 Five grades of truth

Grade	Meaning
Definitely true	directly stated or follows necessarily
Probably true	likely, given the passage, but not certain
Data inadequate	the passage says nothing either way
Probably false	unlikely, given the passage
Definitely false	contradicts the passage

PASSAGE FOR QUESTIONS 22 TO 25

In the last five years, the number of electric two-wheelers sold in the city has increased four times, mainly because of government subsidies and rising petrol prices. However, the number of public charging stations has grown much more slowly, and many buyers charge their vehicles at home.

TIP

Answer only from the passage. Your own knowledge about electric vehicles must not change the grade.

21.5 Practice questions

21.5.1 Basic level

Q1 Statement: All the students of Class X passed in Mathematics. Ravi is a student of Class X.
Conclusions: I. Ravi passed in Mathematics. II. Ravi is good at all subjects.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q2 Statement: The shop is closed on Mondays. Today the shop is closed.
Conclusions: I. Today is Monday. II. The shop is closed on some days.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q3 Statement: Prices of vegetables rise every year during the monsoon.
Conclusions: I. Vegetable prices will rise during this year's monsoon. II. Vegetables are cheap in summer.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q4 Statement: Smoking is injurious to health.
Conclusions: I. Non-smokers are always healthy. II. Smokers may suffer health problems.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q5 Statement: The meeting will start either at 10 a.m. or at 11 a.m.
Conclusions: I. The meeting will start at 10 a.m. II. The meeting will start at 11 a.m.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q6 Statement: Some doctors are rich. Ravi is a doctor.
Conclusions: I. Ravi is rich. II. Ravi is not rich.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q7 Statement: This book is meant for beginners and covers basic topics only.
Conclusions: I. Advanced learners will not find this book useful at all. II. Beginners can learn basic topics from this book.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q8 Statement: The train was late because of fog.
Conclusions: I. Fog can delay trains. II. The train is usually on time.

(a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

21.5.2 Intermediate level

Q9 Question: Should mobile phones be banned in schools?
Arguments: I. Yes, they distract students from their studies. II. No, students use them to play games.

(a) Only argument I is strong
(b) Only argument II is strong
(c) Either I or II is strong
(d) Neither I nor II is strong
(e) Both I and II are strong

Q10 Question: Should voting be made compulsory?
Arguments: I. Yes, it will increase participation in democracy. II. No, it would infringe on personal freedom.

(a) Only argument I is strong
(b) Only argument II is strong
(c) Either I or II is strong
(d) Neither I nor II is strong
(e) Both I and II are strong

Q11 Question: Should all roads be made toll-free?
Arguments: I. Yes, people do not like paying tolls. II. No, toll money is needed to build and maintain roads.

(a) Only argument I is strong
(b) Only argument II is strong
(c) Either I or II is strong
(d) Neither I nor II is strong
(e) Both I and II are strong

Q12 Question: Should students wear school uniforms?
Arguments: I. Yes, uniforms reduce visible differences between rich and poor students. II. No, all schools in Western countries have abolished uniforms.

(a) Only argument I is strong
(b) Only argument II is strong
(c) Either I or II is strong
(d) Neither I nor II is strong
(e) Both I and II are strong

Q13 Question: Should junk food be banned near schools?
Arguments: I. Yes, it harms children's health. II. No, junk food tastes good.

(a) Only argument I is strong
(b) Only argument II is strong
(c) Either I or II is strong
(d) Neither I nor II is strong
(e) Both I and II are strong

Q14 Question: Should school examinations be abolished?
Arguments: I. Yes, students feel stressed before examinations. II. No, examinations are needed to assess what students have learnt.

(a) Only argument I is strong
(b) Only argument II is strong
(c) Either I or II is strong
(d) Neither I nor II is strong
(e) Both I and II are strong

Q15 Problem: Many students in a village school fail in Mathematics because there is no Mathematics teacher.
Courses of action: I. The government should appoint a Mathematics teacher in the school. II. The village school should be closed.

(a) Only course I follows
(b) Only course II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q16 Problem: There has been a sharp rise in road accidents on the highway at night.
Courses of action: I. Night patrolling and speed checks should be increased on the highway. II. The highway should be closed at night.

(a) Only course I follows
(b) Only course II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

21.5.3 Advanced level

Q17 Problem: Several cases of food poisoning were reported after a wedding feast.
Courses of action: I. The health department should test food samples from the caterer. II. All weddings should be banned.

(a) Only course I follows
(b) Only course II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q18 Problem: The water level in the city lake has fallen sharply.
Courses of action: I. Residents should be urged to save water. II. The authorities should explore alternative sources of water.

(a) Only course I follows
(b) Only course II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q19 Problem: A large number of fake currency notes have been found in circulation.
Courses of action: I. Banks should be equipped with note-checking machines. II. The government should stop printing currency notes.

(a) Only course I follows
(b) Only course II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q20 Problem: Many pedestrians are injured while crossing a busy road near a school.
Courses of action: I. A zebra crossing with traffic signals should be provided. II. Traffic police should be posted near the school at opening and closing times.

(a) Only course I follows
(b) Only course II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q21 Problem: Some employees of an office reach work late.
Courses of action: I. All the employees should be dismissed. II. The office should be shut down.

(a) Only course I follows
(b) Only course II follows
(c) Either I or II follows
(d) Neither I nor II follows
(e) Both I and II follow

Q22 Based on the passage in 21.4.1, how true is this inference? "Electric two-wheeler sales in the city have increased over the last five years."

(a) Definitely true
(b) Probably true
(c) Data inadequate
(d) Probably false
(e) Definitely false

Q23 Based on the passage in 21.4.1, how true is this inference? "If petrol prices fall sharply, the growth in electric two-wheeler sales may slow down."

(a) Definitely true
(b) Probably true
(c) Data inadequate
(d) Probably false
(e) Definitely false

Q24 Based on the passage in 21.4.1, how true is this inference? "Electric two-wheelers are cheaper to maintain than petrol scooters."

(a) Definitely true
(b) Probably true
(c) Data inadequate
(d) Probably false
(e) Definitely false

Q25 Based on the passage in 21.4.1, how true is this inference? "Government subsidies played no part in the rise in sales."

(a) Definitely true
(b) Probably true
(c) Data inadequate
(d) Probably false
(e) Definitely false

21.6 Solutions

Q1 **Answer: (a) Only I follows**
I combines the two facts. II adds information the statement does not give. Answer: **Only I follows**.

Q2 **Answer: (b) Only II follows**
The shop may close on other days too, so I does not follow. II restates the first fact. Answer: **Only II follows**.

Q3 **Answer: (a) Only I follows**
A regular yearly pattern supports I. Nothing is said about summer prices (II). Answer: **Only I follows**.

Q4 **Answer: (b) Only II follows**
II follows directly. I is extreme and about non-smokers, who are not discussed. Answer: **Only II follows**.

Q5 **Answer: (c) Either I or II follows**
Neither is certain alone, but one of them must happen: either I or II follows. Answer: **Either I or II follows**.

Q6 **Answer: (c) Either I or II follows**
From "some" nothing definite follows about Ravi, but I and II are complementary, so either I or II follows. Answer: **Either I or II follows**.

Q7 **Answer: (b) Only II follows**
II follows from the purpose of the book. I is extreme ("not useful at all"); advanced learners might still revise from it. Answer: **Only II follows**.

Q8 **Answer: (a) Only I follows**
I follows from the stated cause. Nothing is said about the train's usual timing (II). Answer: **Only I follows**.

Q9 **Answer: (a) Only argument I is strong**
I is a relevant, important reason. II does not support "No"; if anything it supports a ban. Answer: **Only argument I is strong**.

Q10 **Answer: (e) Both I and II are strong**
Both are relevant and serious considerations, one for each side. Answer: **Both I and II are strong**.

Q11 **Answer: (b) Only argument II is strong**

I is a superficial preference. II points to a real, important consequence. Answer: **Only argument II is strong**.

Q12 **Answer: (a) Only argument I is strong**

I is a relevant reason. II is an unproven, sweeping comparison and is weak. Answer: **Only argument I is strong**.

Q13 **Answer: (a) Only argument I is strong**

I is relevant and important. II is trivial. Answer: **Only argument I is strong**.

Q14 **Answer: (b) Only argument II is strong**

Stress can be reduced in other ways, so I is not a strong reason to abolish exams. II points to the essential purpose of exams. Answer: **Only argument II is strong**.

Q15 **Answer: (a) Only course I follows**

I addresses the cause directly. II is extreme and makes the problem worse. Answer: **Only course I follows**.

Q16 **Answer: (a) Only course I follows**

I is practical and targets the problem. II is extreme and impractical. Answer: **Only course I follows**.

Q17 **Answer: (a) Only course I follows**

I finds the cause and prevents repetition. II is absurdly extreme. Answer: **Only course I follows**.

Q18 **Answer: (e) Both I and II follow**

Both are practical and reduce the effect of the shortage. Answer: **Both I and II follow**.

Q19 **Answer: (a) Only course I follows**

I helps detect fake notes. II is impractical. Answer: **Only course I follows**.

Q20 **Answer: (e) Both I and II follow**

Both are practical and directly reduce the danger. Answer: **Both I and II follow**.

Q21 **Answer: (d) Neither I nor II follows**

Both are extreme and out of proportion to the problem, so neither follows. Answer: **Neither I nor II follows**.

Q22 **Answer: (a) Definitely true**

The passage says sales increased four times: definitely true. Answer: **Definitely true**.

Q23 **Answer: (b) Probably true**

Rising petrol prices are given as a main reason for the growth, so this is likely, but the passage does not state it: probably true. Answer: **Probably true**.

Q24 **Answer: (c) Data inadequate**

Maintenance costs are not mentioned at all: data inadequate. Answer: **Data inadequate**.

Q25 **Answer: (e) Definitely false**

The passage names subsidies as a main reason, so this contradicts it: definitely false. Answer: **Definitely false**.

A conclusion must follow from the statement alone, with no outside facts. A strong argument is relevant, important and really supports its side. A course of action must be practical, address the problem and be proportionate. For passage inferences, grade each statement only by what the passage states or makes likely.

MODULE 22 · ADVANCED

Cause and effect, decision making

Minimum duration **2 days** | Level **Advanced**

Cause-and-effect questions give two events and ask how they are related: one causes the other, both come from a common cause, or they are unrelated. Decision-making questions give selection conditions with exceptions and ask what to do with each candidate. This module covers both formats with clear tests, a decision table method, and 25 solved questions; every decision answer was computed by a program from the stated conditions.

IN THIS MODULE

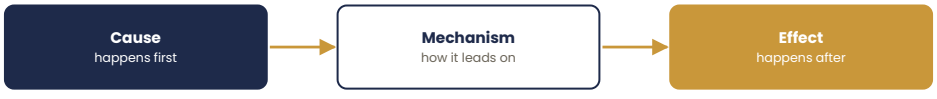
22.1 Cause and effect
22.2 Telling the relationships apart
22.3 Decision making
22.4 Practice questions
22.5 Solutions

22.1 Cause and effect

22.1.1 The five possible relationships

Two statements I and II describe events. Decide how they are linked. The five standard answer choices are:

Choice	Meaning	Example
I → II	I is the cause, II the effect	I: Heavy rain for three days. II: Low-lying areas flooded.
II → I	II is the cause, I the effect	I: Schools declared a holiday. II: A cyclone warning was issued.
Independent causes	both are causes, of different things	rare in exams
Effects of independent causes	unrelated events, each with its own cause	I: Tax rates were cut. II: The cricket team won.
Effects of a common cause	a third event caused both	I: Onion prices rose. II: Tomato prices rose. (poor harvest)



A CAUSE COMES FIRST AND EXPLAINS THE EFFECT

22.2 Telling the relationships apart

22.2.1 Three checks

1. **Time order**: a cause must happen before (or with) its effect.
2. **Mechanism**: you must be able to say in one line how the cause produces the effect.
3. **Third factor**: if both events could come from the same background event (weather, a policy, a season), choose "common cause".

SHORTCUT TRICK
Say "because" between them both ways: "II happened because I happened" or "I happened because II happened". If neither sounds right but both would follow from a third event, it is a common cause; if they have nothing to do with each other, they are effects of independent causes.

WATCH OUT
Correlation is not causation. Two prices rising together usually share a cause (supply, inflation); one does not cause the other.

22.3 Decision making

22.3.1 Conditions, exceptions and the decision table

A company is recruiting management trainees. A candidate must:

1. be a graduate with at least 60% marks;
2. be between 21 and 28 years old (both inclusive) on 1 January 2027;
3. have scored at least 50% in the written test;
4. have at least one year of work experience.

However:

- (A) if a candidate meets all conditions except (1) but has a postgraduate degree with at least 55% marks, **refer to the Director**;
- (B) if a candidate meets all conditions except (4) but scored at least 75% in the written test, **refer to the HR Manager**.

For each candidate the answer is one of: Select, Do not select, Data inadequate, Refer to the Director, Refer to the HR Manager. If information needed for the decision is missing, the answer is "data inadequate"; if the candidate clearly fails a condition that no exception covers, the answer is "do not select" even if other data are missing.

Candidate check	(1) grad ≥ 60	(2) age 21–28	(3) test ≥ 50	(4) exp ≥ 1	Decision
all ✓	✓	✓	✓	✓	Select
only (1) fails, PG ≥ 55%	×	✓	✓	✓	Director
only (4) fails, test ≥ 75%	✓	✓	✓	×	HR Manager

Candidate check	(1) grad ≥ 60	(2) age 21–28	(3) test ≥ 50	(4) exp ≥ 1	Decision
any other failure					Do not select
needed fact missing					Data inadequate

TIP
Make a row for each candidate with ✓, × or ? under each condition. Then read the decision from the pattern: all ✓ → select; one × covered by an exception → refer; any other × → reject; ? that could change the result → data inadequate.

22.4 Practice questions

22.4.1 Basic level

Q1 I. Heavy rain lashed the city for three days.
II. Many low-lying areas of the city were flooded.

(a) I is the cause and II is its effect
(b) II is the cause and I is its effect
(c) Both are independent causes
(d) Both are effects of independent causes
(e) Both are effects of some common cause

Q2 I. The school declared a holiday on Monday.
II. A cyclone warning was issued for Monday.

(a) I is the cause and II is its effect
(b) II is the cause and I is its effect
(c) Both are independent causes
(d) Both are effects of independent causes
(e) Both are effects of some common cause

Q3 I. The price of onions rose sharply this month.
II. The price of tomatoes rose sharply this month.

(a) I is the cause and II is its effect
(b) II is the cause and I is its effect
(c) Both are independent causes
(d) Both are effects of independent causes
(e) Both are effects of some common cause

Q4 I. The company's profits fell sharply this year.
II. The company laid off 200 workers.

(a) I is the cause and II is its effect
(b) II is the cause and I is its effect
(c) Both are independent causes
(d) Both are effects of independent causes
(e) Both are effects of some common cause

Q5 I. The government reduced income tax rates.
II. The national cricket team won the series.

(a) I is the cause and II is its effect
(b) II is the cause and I is its effect
(c) Both are independent causes
(d) Both are effects of independent causes
(e) Both are effects of some common cause

Q6 I. Many students fell ill after eating in the hostel mess.
II. The hostel mess was closed for inspection.

(a) I is the cause and II is its effect
(b) II is the cause and I is its effect
(c) Both are independent causes
(d) Both are effects of independent causes
(e) Both are effects of some common cause

Q7 I. Air pollution in the city reached dangerous levels.
II. Schools in the city were closed for a week.

(a) I is the cause and II is its effect
(b) II is the cause and I is its effect
(c) Both are independent causes
(d) Both are effects of independent causes
(e) Both are effects of some common cause

Q8 I. The price of gold rose sharply.
II. Demand for gold jewellery fell.

(a) I is the cause and II is its effect
(b) II is the cause and I is its effect
(c) Both are independent causes
(d) Both are effects of independent causes
(e) Both are effects of some common cause

22.4.2 Intermediate level

Q9 I. The temperature in the city crossed 45°C.
II. Sales of air conditioners rose sharply.

(a) I is the cause and II is its effect
(b) II is the cause and I is its effect
(c) Both are independent causes
(d) Both are effects of independent causes
(e) Both are effects of some common cause

Q10 I. The bank raised its interest rates on loans.
II. The bank raised its interest rates on deposits.

(a) I is the cause and II is its effect
(b) II is the cause and I is its effect
(c) Both are independent causes
(d) Both are effects of independent causes
(e) Both are effects of some common cause

Q11 I. The teacher was absent for the period.
II. The students played in the ground during the period.

(a) I is the cause and II is its effect
(b) II is the cause and I is its effect
(c) Both are independent causes
(d) Both are effects of independent causes
(e) Both are effects of some common cause

Q12 I. Traffic on the highway was diverted.
II. A major accident blocked the highway.

(a) I is the cause and II is its effect
(b) II is the cause and I is its effect
(c) Both are independent causes
(d) Both are effects of independent causes
(e) Both are effects of some common cause

Q13 Decision set (22.3.1): Anil is 25 years old on 1 January 2027, scored 65% in graduation and 58% in the written test, and has 2 years of work experience. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

Q14 Decision set (22.3.1): Bela scored 58% in graduation and 60% in her postgraduate degree, is 26 years old on 1 January 2027, scored 62% in the written test and has 3 years of experience. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

Q15 Decision set (22.3.1): Chetan scored 70% in graduation and 80% in the written test, is 24 years old on 1 January 2027 and has just completed his degree with no work experience. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

Q16 Decision set (22.3.1): Divya scored 72% in graduation and 66% in the written test, has 4 years of experience and is 30 years old on 1 January 2027. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

22.4.3 Advanced level

Q17 Decision set (22.3.1): Esha is 23 years old on 1 January 2027, scored 68% in graduation and has 2 years of experience. She has taken the written test. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

Q18 Decision set (22.3.1): Farhan scored 61% in graduation and 45% in the written test, is 27 years old on 1 January 2027 and has 2 years of experience. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

Q19 Decision set (22.3.1): Gita scored 55% in graduation and has no postgraduate degree. She is 25 years old on 1 January 2027, scored 70% in the written test and has 2 years of experience. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

Q20 Decision set (22.3.1): Hari scored 64% in graduation and 70% in the written test, is 22 years old on 1 January 2027 and has no work experience. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

Q21 Decision set (22.3.1): Isha is exactly 28 years old on 1 January 2027, scored 75% in graduation and 52% in the written test, and has 1 year of experience. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

Q22 Decision set (22.3.1): Jatin scored 59% in graduation and 54% in his postgraduate degree, is 26 years old on 1 January 2027, scored 60% in the written test and has 2 years of experience. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

Q23 Decision set (22.3.1): Kavya scored 66% in graduation and 76% in the written test and is 21 years old on 1 January 2027. Her work experience is not mentioned. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

Q24 Decision set (22.3.1): Lalit scored 62% in graduation and 80% in the written test, is 29 years old on 1 January 2027 and has no work experience. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

Q25 Decision set (22.3.1): Meena scored 57% in graduation and 58% in her postgraduate degree, scored 77% in the written test, is 24 years old on 1 January 2027 and has no work experience. What should be done?

(a) Select
(b) Do not select
(c) Data inadequate
(d) Refer to the Director
(e) Refer to the HR Manager

22.5 Solutions

Q1 **Answer: (a) I is the cause and II is its effect**

The rain came first and directly caused the flooding. Answer: **I is the cause and II is its effect**.

Q2 **Answer: (b) II is the cause and I is its effect**

The warning (II) led the school to declare the holiday (I). Answer: **II is the cause and I is its effect**.

Q3 **Answer: (e) Both are effects of some common cause**

Neither causes the other; both are likely results of a common cause such as poor rainfall or a transport strike. Answer: **Both are effects of some common cause**.

Q4 **Answer: (a) I is the cause and II is its effect**

Falling profits (I) explain the decision to cut jobs (II). Answer: **I is the cause and II is its effect**.

Q5 **Answer: (d) Both are effects of independent causes**

The two events are unrelated; each has its own cause. Answer: **Both are effects of independent causes**.

Q6 **Answer: (a) I is the cause and II is its effect**

The illness (I) led to the closure for inspection (II). Answer: **I is the cause and II is its effect**.

Q7 **Answer: (a) I is the cause and II is its effect**

The pollution (I) is the reason for closing the schools (II). Answer: **I is the cause and II is its effect**.

Q8 **Answer: (a) I is the cause and II is its effect**

Higher prices (I) reduce demand (II). Answer: **I is the cause and II is its effect**.

Q9 **Answer: (a) I is the cause and II is its effect**

The heat (I) pushed people to buy air conditioners (II). Answer: **I is the cause and II is its effect**.

Q10 **Answer: (e) Both are effects of some common cause**

Both changes typically follow a common cause, such as an increase in the central bank's policy rate. Answer: **Both are effects of some common cause**.

Q11 **Answer: (a) I is the cause and II is its effect**

The teacher's absence (I) allowed the students to play (II). Answer: **I is the cause and II is its effect**.

Q12 **Answer: (b) II is the cause and I is its effect**

The accident (II) caused the diversion (I). Answer: **II is the cause and I is its effect**.

Q13 **Answer: (a) Select**

All four conditions are met (65% ≥ 60, age 25, test 58% ≥ 50, 2 years). Answer: **Select**.

Q14 **Answer: (d) Refer to the Director**

Only condition (1) fails (58%), and her postgraduate score is 60% ≥ 55%: exception (A). Answer: **Refer to the Director**.

Q15 **Answer: (e) Refer to the HR Manager**

Only condition (4) fails (no experience), and his test score is 80% ≥ 75%: exception (B). Answer: **Refer to the HR Manager**.

Q16 **Answer: (b) Do not select**

She fails the age condition (30), which no exception covers. Answer: **Do not select**.

Q17 **Answer: (c) Data inadequate**

Her written-test score is not given, and it decides whether she is selected or rejected. Answer: **Data inadequate**.

Q18 **Answer: (b) Do not select**

His test score (45%) is below 50%, and no exception covers condition (3). Answer: **Do not select**.

Q19 **Answer: (b) Do not select**

She fails (1) with 55% and has no postgraduate degree, so exception (A) does not apply. Answer: **Do not select**.

Q20 **Answer: (b) Do not select**

He fails (4), and his test score (70%) is below the 75% needed for exception (B). Answer: **Do not select**.

Q21 **Answer: (a) Select**

All conditions are met at the boundaries: age 28 is allowed, test 52% ≥ 50, 1 year of experience. Answer: **Select**.

Q22 **Answer: (b) Do not select**

He fails (1), and his postgraduate score (54%) is below the 55% needed for exception (A). Answer: **Do not select**.

Q23 **Answer: (c) Data inadequate**

With at least one year of experience she would be selected; with none she would be referred to the HR Manager (test 76%). The data do not say which. Answer: **Data inadequate**.

Q24 **Answer: (b) Do not select**

He fails the age condition (29); exception (B) needs all conditions except (4) to be met. Answer: **Do not select**.

Q25 **Answer: (b) Do not select**

She fails both (1) and (4); each exception requires all other conditions to be met. Answer: **Do not select**.

For cause and effect, check time order, a clear mechanism, and whether a third event explains both (common cause) or nothing links them (independent causes). For decision making, mark each condition ✓, × or ?; select if all are ✓, refer if exactly the excepted condition fails and the exception is met, reject for any other failure, and say data inadequate only when the missing fact could change the decision.

MODULE 23 · ADVANCED

Data sufficiency

Minimum duration **2 days** | Level **Advanced**

Data-sufficiency questions do not ask for the answer itself; they ask whether the given statements contain enough information to find it. This module explains the five standard options, a fixed checking order (I alone, II alone, then together), how to recognise sufficiency without computing, and the common traps, using the reasoning topics already covered (ranking, directions, blood relations, coding, seating and numbers), with 25 solved questions.

IN THIS MODULE

23.1 Basics
23.2 The checking order
23.3 Topic patterns
23.4 Traps
23.5 Practice questions
23.6 Solutions

23.1 Basics

23.1.1 The five options

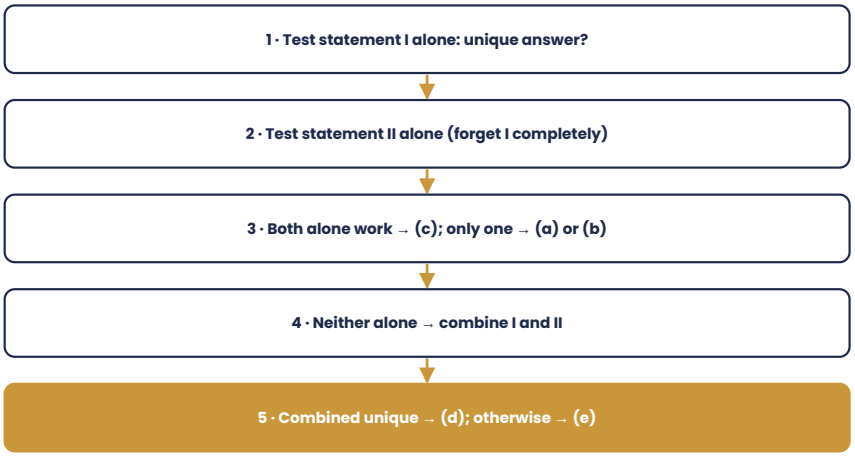
Each question has a question stem and two statements, I and II. You decide whether the statements give a **unique** answer, not what the answer is.

Option	Meaning
(a)	I alone is sufficient, II alone is not
(b)	II alone is sufficient, I alone is not
(c)	Either I alone or II alone is sufficient
(d)	I and II together are needed (neither alone is enough)
(e)	Even I and II together are not sufficient

Sufficient means the information leads to exactly one answer. For a yes/no question, a definite “no” is just as sufficient as a definite “yes”.

23.2 The checking order

23.2.1 I alone, II alone, then together



THE DATA-SUFFICIENCY FLOW

SHORTCUT TRICK

When you test statement II alone, cover statement I with your hand. Carrying information from I into the test of II is the most common way to get these questions wrong.

WORKED EXAMPLE

How many students are in the class? I: Ravi is 12th from the top and 18th from the bottom. II: There are 20 girls in the class.

I alone: total = 12 + 18 − 1 = 29, unique. II alone gives no information about boys. Answer (a).

23.3 Topic patterns

23.3.1 What each topic needs

Topic	Usually sufficient when you have …
Ranking	positions from both ends, or a position and the total
Directions	a connected chain of distances and directions from one point to the other
Blood relations	a chain linking the two people, with genders where the answer depends on them
Coding	two coded sentences sharing exactly the word asked
Seating	enough fixed positions to leave one possibility for the asked seat
Numbers	conditions that leave exactly one value

TIP

You often do not need to finish the calculation. As soon as you can see the answer is unique, stop and move on.

23.4 Traps

23.4.1 Common errors

- Treating a definite "no" as insufficient. "Is A taller than B?" is answered by "B > C > A": the answer is no, and that is sufficient.
- Using outside assumptions (for example assuming people in a family puzzle are of a certain gender).
- Assuming two statements are about the same unknown order, when they could be consistent with several arrangements.
- Forgetting that a combination can still fail: two conditions may still leave two values (option (e)).

23.5 Practice questions

23.5.1 Basic level

Q1 How many students are in the class?
I. Ravi is 12th from the top and 18th from the bottom.
II. There are 20 girls in the class.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q2 In which direction is P from Q?
I. P is 5 km north of R.
II. R is 5 km east of Q.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q3 Who is the tallest among A, B and C?
I. A is taller than B.
II. C is taller than A.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q4 On which day of this week is Ravi's birthday?
I. It is after Tuesday but before Friday.
II. It is on Wednesday.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q5 What is the code for "sky" in a code language?
I. "blue sky" is written "ta na" and "clear sky" is written "na pi".
II. "sky is blue" is written "ta na ro".

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q6 How is M related to N?
I. M is the brother of N's father.
II. N is the daughter of M's sister.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q7 What is Rani's rank from the top in a class of 40 students?
I. She is 15th from the bottom.
II. Exactly 25 students are ranked above her.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q8 How far is the school from the house?
I. The school is 3 km east of the market.
II. The market is 4 km north of the house.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

23.5.2 Intermediate level

Q9 Is today Monday?
I. Yesterday was Sunday.
II. Tomorrow will be Tuesday.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q10 What is the value of the whole number x?
I. x is a prime number between 10 and 20.
II. x is a two-digit number whose digits add up to 8.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q11 P, Q, R, S, T sit in a row facing North. Who sits in the middle?
I. P is at the left end and Q is next to P.
II. R is at the right end and S is next to R.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q12 In which month was Meena born?
I. She was born after March but before June.
II. She was born in a month that has 30 days.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q13 Is A taller than B?
I. C is taller than A.
II. B is taller than C.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q14 What is the code for "red" in a code language?
I. "red ball" is written "ka pa" and "red bat" is written "pa la".
II. "ball game" is written "ka mo".

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q15 In which direction is Seema facing?
I. In the morning, the sun is on her left.
II. In the evening, her shadow falls directly behind her.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q16 A lives in a five-floor building (floors 1 to 5). On which floor does A live?
I. A lives on an odd-numbered floor.
II. A lives above floor 3.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

23.5.3 Advanced level

Q17 Who is the eldest of the five friends P, Q, R, S and T?
I. P is older than Q and R.
II. S is older than P but younger than T.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q18 What is the two-digit number?
I. The sum of its digits is 9.
II. The difference between its digits is 3.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q19 Is the integer n even?
I. 3n is even.
II. n + 2 is even.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q20 How is Q related to P?
I. Q is the wife of P's only brother.
II. P is Q's brother-in-law.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q21 On which date in March was the meeting held?
I. It was held after 15 March and before 20 March.
II. It was held on a Sunday, and 1 March was a Wednesday.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q22 How many people are standing in the queue?
I. Amit is 10th from the front.
II. Amit is 5 places ahead of Ravi, who is 8th from the back.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q23 In which direction is A from B?
I. A is to the north-east of C.
II. C is to the north-east of B.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q24 What is Sunita's position from the left in a row?
I. She is 7th from the right end.
II. Ravi is 4th from the left, and exactly 3 people sit between Ravi and Sunita.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

Q25 What is the value of the whole number X?
I. X is a multiple of 4.
II. X is between 10 and 20.

(a) I alone is sufficient, II alone is not
(b) II alone is sufficient, I alone is not
(c) Either I alone or II alone is sufficient
(d) I and II together are needed (neither alone is enough)
(e) Even I and II together are not sufficient

23.6 Solutions

Q1 **Answer: (a) I alone is sufficient, II alone is not**
I gives 12 + 18 − 1 = 29. II says nothing about the boys. Answer: **(a) I alone is sufficient, II alone is not**.

Q2 **Answer: (d) I and II together are needed (neither alone is enough)**
Each statement alone links P or Q to R only. Together: Q → 5 km east → R → 5 km north → P, so P is north-east of Q. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q3 **Answer: (d) I and II together are needed (neither alone is enough)**
I alone leaves C unknown; II alone leaves B unknown. Together C > A > B, so C is the tallest. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q4 **Answer: (b) II alone is sufficient, I alone is not**
I allows Wednesday or Thursday. II gives the day directly. Answer: **(b) II alone is sufficient, I alone is not**.

Q5 **Answer: (a) I alone is sufficient, II alone is not**
From I, the code common to both phrases is “na” = sky. II alone has three codes for three words, with no way to match them. Answer: **(a) I alone is sufficient, II alone is not**.

Q6 **Answer: (c) Either I alone or II alone is sufficient**
I: M is N's paternal uncle. II: M is the brother of N's mother, N's maternal uncle. Either alone shows M is N's uncle. Answer: **(c) Either I alone or II alone is sufficient**.

Q7 **Answer: (c) Either I alone or II alone is sufficient**

I: 40 − 15 + 1 = 26th. II: 25 above her means she is 26th. Either alone is enough. Answer: **(c) Either I alone or II alone is sufficient**.

Q8 **Answer: (d) I and II together are needed (neither alone is enough)**

Together the house, market and school form a right angle at the market: $\sqrt{(3^2 + 4^2)} = 5$ km. Neither alone links the school to the house. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q9 **Answer: (c) Either I alone or II alone is sufficient**

Each statement alone fixes today as Monday. Answer: **(c) Either I alone or II alone is sufficient**.

Q10 **Answer: (d) I and II together are needed (neither alone is enough)**

I allows 11, 13, 17, 19. II allows 17, 26, 35, 44, 53, 62, 71, 80. Together only 17. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q11 **Answer: (d) I and II together are needed (neither alone is enough)**

Together: P, Q, ?, S, R, so T is in the middle. Either statement alone leaves three people for three seats. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q12 **Answer: (d) I and II together are needed (neither alone is enough)**

I allows April or May. II allows April, June, September or November. Together only April. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q13 **Answer: (d) I and II together are needed (neither alone is enough)**

Together B > C > A, so the answer is a definite "no", which is sufficient. Neither statement alone compares A and B. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q14 **Answer: (a) I alone is sufficient, II alone is not**

From I, the common code "pa" = red. II does not mention red. Answer: **(a) I alone is sufficient, II alone is not**.

Q15 **Answer: (c) Either I alone or II alone is sufficient**

I: morning sun is east, on her left, so she faces North. II: evening sun is west; a shadow behind her means the sun is in front, so she faces West. Each alone gives a unique answer. Answer: **(c) Either I alone or II alone is sufficient**.

Q16 **Answer: (d) I and II together are needed (neither alone is enough)**

I allows 1, 3, 5; II allows 4, 5. Together only floor 5. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q17 **Answer: (d) I and II together are needed (neither alone is enough)**

Together T > S > P > Q, R, so T is the eldest. I alone says nothing about S and T; II alone says nothing about Q and R. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q18 **Answer: (e) Even I and II together are not sufficient**

Together the digits are 6 and 3, but the number could be 63 or 36: not sufficient. Answer: **(e) Even I and II together are not sufficient**.

Q19 **Answer: (c) Either I alone or II alone is sufficient**

I: 3n even means n is even (3 is odd). II: n + 2 even means n is even. Either alone is sufficient. Answer: **(c) Either I alone or II alone is sufficient**.

Q20 **Answer: (a) I alone is sufficient, II alone is not**

I: Q is P's sister-in-law. II: Q could be the wife of P's brother, or the sister of P's wife, or P could be her sister's husband: not unique. Answer: **(a) I alone is sufficient, II alone is not**.

Q21 **Answer: (d) I and II together are needed (neither alone is enough)**

II: Sundays are 5, 12, 19 and 26 March. I: 16 to 19 March. Together only 19 March. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q22 **Answer: (d) I and II together are needed (neither alone is enough)**

II puts Amit 13th from the back. With I, total = 10 + 13 − 1 = 22. Neither alone gives the total. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q23 **Answer: (d) I and II together are needed (neither alone is enough)**

Together, moving north-east twice from B reaches A, so A is north-east of B. Neither alone links A and B. Answer: **(d) I and II together are needed (neither alone is enough)**.

Q24 **Answer: (b) II alone is sufficient, I alone is not**

II: Sunita is 4 + 4 = 8th from the left (to the left of Ravi there is no room for 3 people). I alone needs the total, which is not given. Answer: **(b) II alone is sufficient, I alone is not**.

Q25 **Answer: (e) Even I and II together are not sufficient**

Together X could be 12 or 16: not sufficient. Answer: **(e) Even I and II together are not sufficient**.

Test I alone, then II alone (forgetting I), then both together. A unique answer, including a definite “no”, is sufficient. Stop calculating as soon as uniqueness is clear, avoid outside assumptions, and remember that even combined statements can leave more than one possibility.

MODULE 24 · ADVANCED

Critical reasoning

Minimum duration **2 days** | Level **Advanced**

Critical-reasoning questions give a short argument and ask you to evaluate it: find its assumption, strengthen or weaken it, draw a valid inference, resolve a paradox, or name its flaw. This module breaks an argument into premises and conclusion, explains each question type with a method, lists the common logical fallacies, and gives 25 solved questions.

IN THIS MODULE

24.1 Argument structure
24.2 Question types
24.3 Logical fallacies
24.4 Practice questions
24.5 Solutions

24.1 Argument structure

24.1.1 Premise, conclusion, assumption

An argument has **premises** (the reasons or evidence given), a **conclusion** (what the author wants you to accept), and usually an unstated **assumption** that links them. Find the conclusion first: look for words such as *therefore, so, thus, hence, clearly*.



THE ASSUMPTION BRIDGES THE GAP BETWEEN EVIDENCE AND CLAIM

WORKED EXAMPLE

"Sales of umbrellas rose last month. Therefore it must have rained a lot last month."
Premise: umbrella sales rose. Conclusion: it rained a lot. Assumption: rain is the reason sales rose (and not, for example, a discount sale).

24.2 Question types

24.2.1 What each type asks and how to answer

Type	What you look for	Method
Assumption	the unstated link the argument needs	negate the option: if the argument collapses, it is the assumption
Strengthen	new information that makes the conclusion more likely	support the assumption or rule out another explanation

Type	What you look for	Method
Weaken	new information that makes the conclusion less likely	attack the assumption; offer another cause
Inference	what must be true from the statements	reject anything that needs extra information
Paradox	an explanation for two facts that seem to conflict	the answer makes both facts true at once
Flaw	the error in the reasoning	name the gap: wrong cause, bad sample, etc.

SHORTCUT TRICK
In strengthen and weaken questions, accept the options as true and ask only how they change the **link** between evidence and conclusion. Options that just repeat the premise or discuss something nearby are traps.

WATCH OUT
Correlation is not causation. Many arguments say "A happened with B, so A caused B". The usual weakener is a third factor or a reason why B came first.

24.3 Logical fallacies

24.3.1 The fallacies that appear in exams

Fallacy	Pattern	Example
Hasty generalisation	a few cases → all cases	Two dogs bit me, so all dogs bite.
False cause	A before B → A caused B	I washed my car, then it rained.
Ad hominem	attack the person, not the argument	He failed science, so ignore his view.
Straw man	distort the opponent's view, then attack it	"Fewer cars" becomes "ban all cars".
False dilemma	only two options offered	Either support the law or you do not care.
Circular reasoning	the conclusion is used as the reason	It is best because it is better than all others.
Appeal to popularity	many people believe it → true	Everyone buys it, so it is good.
Slippery slope	one step → chain of disasters	One retest → grades become meaningless.
Appeal to ignorance	not proven false → true	Never proven unsafe, so it is safe.

24.4 Practice questions

24.4.1 Basic level

Q1 Which fallacy does this argument commit? "My neighbour's two dogs bite, so all dogs bite."

(a) Circular reasoning
(b) False cause
(c) Hasty generalisation
(d) Straw man

Q2 Which fallacy does this argument commit? "Every time I wash my car, it rains. So washing my car causes rain."

(a) Ad hominem
(b) Appeal to popularity
(c) Hasty generalisation
(d) False cause

Q3 Which fallacy does this argument commit? "You can't trust Raj's argument about climate change; he failed his science exam."

(a) Ad hominem
(b) Slippery slope
(c) Straw man
(d) False dilemma

Q4 Which fallacy does this argument commit? "Either you support the new law or you don't care about public safety."

(a) Circular reasoning
(b) Ad hominem
(c) Appeal to ignorance
(d) False dilemma

Q5 Which fallacy does this argument commit? "This medicine is the best because it is better than all the others."

(a) Appeal to popularity
(b) Hasty generalisation
(c) False cause
(d) Circular reasoning

Q6 Which fallacy does this argument commit? "Everyone is buying this phone, so it must be good."

(a) Appeal to popularity
(b) Slippery slope
(c) Straw man
(d) Ad hominem

Q7 Which fallacy does this argument commit? "If we let students retake one test, soon they will want to retake every test and grades will mean nothing."

(a) False cause
(b) False dilemma
(c) Slippery slope
(d) Circular reasoning

Q8 Which fallacy does this argument commit? "Priya wants fewer cars in the city centre. So she wants to ban all cars and ruin local businesses."

(a) Ad hominem
(b) Straw man
(c) Appeal to ignorance
(d) Hasty generalisation

24.4.2 Intermediate level

Q9 "Sales of umbrellas rose last month. Therefore it must have rained a lot last month." Which statement most weakens the argument?

(a) A large discount sale on umbrellas was held last month.
(b) Umbrellas are sold in many shops.
(c) Rain had been forecast for last month.
(d) Umbrella sales are usually higher in the monsoon.

Q10 "Students who attend coaching classes score higher in the exam. So coaching classes improve scores." Which statement most weakens the argument?

(a) Coaching classes are expensive.
(b) Some students study on their own.
(c) Students who join coaching classes were already scoring higher before they joined.
(d) The exam has become harder this year.

Q11 "The new traffic signal at the junction has reduced accidents." Which statement most strengthens this claim?

(a) The signal was expensive to install.
(b) Accidents at the junction fell after the signal was installed, although traffic increased and nothing else changed.
(c) Accidents fell in other parts of the city too.
(d) Drivers dislike waiting at signals.

Q12 "Ravi will pass the driving test because he practised every day for a month." Which is the assumption?

(a) A month of daily practice is enough to pass the driving test.
(b) Ravi owns a car.
(c) The driving test is difficult.
(d) Ravi failed the test before.

Q13 "All the shops in the market close at 9 p.m. Shop X is in the market." What must be true?

(a) Shop X opens at 9 a.m.
(b) Shop X is the biggest shop in the market.
(c) All shops in the city close at 9 p.m.
(d) Shop X closes at 9 p.m.

Q14 "The town's population fell over ten years, yet the number of new houses built each year increased." Which statement best explains this?

(a) House prices in the town rose.
(b) Many large families split into smaller households, each needing a separate house.
(c) The town has a new railway station.
(d) Most people in the town own cars.

Q15 "Most successful people wake up early. So waking up early will make you successful." What is the flaw?

(a) It relies on an expert's opinion.

(b) It uses a word with two meanings.

(c) It treats a correlation as proof of cause and effect.

(d) It attacks the people who disagree.

Q16 "The company's new advertisement caused the rise in its sales this quarter." Which statement most weakens this?

(a) A competitor's factory shut down this quarter, leaving the company as the main supplier.

(b) The advertisement won an award.

(c) The company plans another advertisement.

(d) Sales were also high in the same quarter last year.

24.4.3 Advanced level

Q17 "The city should build more bus lanes, because that will reduce traffic jams." Which is the assumption?

(a) Bus lanes will lead some people to use buses instead of cars.

(b) Buses are cheaper than cars.

(c) The city has too many buses.

(d) Traffic jams happen only in the morning.

Q18 "Eating breakfast improves concentration in school." Which evidence most strengthens this?

(a) In a study, students randomly assigned to eat breakfast scored higher on concentration tests than those assigned to skip it.

(b) Many students say they like breakfast.

(c) Breakfast cereals are advertised on television.

(d) Students who eat breakfast usually also sleep longer.

Q19 "No member of the chess club is under 12. Some members of the chess club play cricket." Which must be true?

(a) All cricket players are over 12.

(b) No one under 12 plays cricket.

(c) Some chess club members are under 12.

(d) Some cricket players are at least 12 years old.

Q20 "A survey of 50 people at a gym found that 90% exercise daily, so 90% of the city's people exercise daily." What is the flaw?

(a) The sample is too large.

(b) The sample is not representative of the city.

(c) The survey asked the wrong question.

(d) It attacks people who do not exercise.

Q21 "A hospital with the best doctors in the region has a higher death rate than other hospitals." Which statement best resolves this?

(a) It receives the most serious cases, referred from other hospitals.

(b) Its doctors are paid more.

(c) It has more beds than other hospitals.

(d) It was built recently.

Q22 "Banning plastic bags will reduce litter, since plastic bags are a major part of the litter." Which statement most weakens this?

(a) Plastic bags are cheap.

(b) Some shops already charge for bags.

(c) Shoppers will switch to thin paper bags, which are littered just as often.

(d) Litter is collected every morning.

Q23 "Online classes are as effective as classroom classes, since online students got similar marks." Which information would be most useful to evaluate this?

(a) How many teachers prefer online classes.

(b) The cost of laptops.

(c) Whether online classes start earlier.

(d) Whether both groups took the same exam under similar conditions.

Q24 Which fallacy does this argument commit? "Our product has never been proven unsafe, so it must be safe."

(a) Circular reasoning

(b) False cause

(c) Appeal to ignorance

(d) Appeal to popularity

Q25 "Reading to children every day improves their vocabulary." Which evidence most strengthens this?

(a) Many parents enjoy reading.

(b) Children of similar backgrounds who were read to daily learned more new words in a year than those who were not.

(c) Children's books are colourful.

(d) Some children learn to read early.

24.5 Solutions

Q1 **Answer: (c) Hasty generalisation**

Two cases are used to make a claim about all dogs: **hasty generalisation**.

Q2 **Answer: (d) False cause**

One event following another is taken as proof that it caused it: **false cause**.

Q3 **Answer: (a) Ad hominem**

The argument attacks Raj instead of his reasoning: **ad hominem**.

Q4 **Answer: (d) False dilemma**

Only two options are offered when others exist (one may care about safety and still oppose this law): **false dilemma**.

Q5 **Answer: (d) Circular reasoning**

"Best" and "better than all others" say the same thing; the conclusion is its own reason: **circular reasoning**.

Q6 **Answer: (a) Appeal to popularity**

Popularity is taken as proof of quality: **appeal to popularity**.

Q7 **Answer: (c) Slippery slope**

One small step is claimed to lead inevitably to an extreme outcome: **slippery slope**.

Q8 **Answer: (b) Straw man**

Priya's moderate view is distorted into an extreme one that is easier to attack: **straw man**.

Q9 **Answer: (a) A large discount sale on umbrellas was held last month.**

A discount sale gives another reason for higher sales, breaking the link between sales and rain. Answer: **A large discount sale on umbrellas was held last month.**.

Q10 **Answer: (c) Students who join coaching classes were already scoring higher before they joined.**

If the better students choose coaching, the higher scores may not be caused by coaching at all. Answer: **Students who join coaching classes were already scoring higher before they joined.**.

Q11 **Answer: (b) Accidents at the junction fell after the signal was installed, although traffic increased and nothing else changed.**

It rules out other explanations (less traffic, other changes), so the signal is the likely cause. Answer: **Accidents at the junction fell after the signal was installed, although traffic increased and nothing else changed.**.

Q12 **Answer: (a) A month of daily practice is enough to pass the driving test.**

Negate it: if a month of practice is not enough, the conclusion no longer follows. So it is the assumption. Answer: **A month of daily practice is enough to pass the driving test.**.

Q13 **Answer: (d) Shop X closes at 9 p.m.**

Combining the two statements gives exactly this; the others need extra information. Answer: **Shop X closes at 9 p.m.**.

Q14 **Answer: (b) Many large families split into smaller households, each needing a separate house.**

Fewer people can still need more houses if each household is smaller. This makes both facts true. Answer: **Many large families split into smaller households, each needing a separate house.**.

Q15 **Answer: (c) It treats a correlation as proof of cause and effect.**

Successful people waking early does not show that waking early causes success. Answer: **It treats a correlation as proof of cause and effect.**.

Q16 **Answer: (a) A competitor's factory shut down this quarter, leaving the company as the main supplier.**

It gives another strong reason for the sales rise, weakening the claim that the advertisement caused it. Answer: **A competitor's factory shut down this quarter, leaving the company as the main supplier.**.

Q17 **Answer: (a) Bus lanes will lead some people to use buses instead of cars.**

If nobody switches from cars to buses, bus lanes would not reduce jams. Negating this option breaks the argument. Answer: **Bus lanes will lead some people to use buses instead of cars.**.

Q18 **Answer: (a) In a study, students randomly assigned to eat breakfast scored higher on concentration tests than those assigned to skip it.**

Random assignment rules out other differences between the groups, so the effect is due to breakfast. The last option actually weakens the claim. Answer: **In a study, students randomly assigned to eat breakfast scored higher on concentration tests than those assigned to skip it.**.

Q19 **Answer: (d) Some cricket players are at least 12 years old.**

The chess members who play cricket are at least 12, so some cricket players are at least 12. The other options go beyond the statements or contradict them. Answer: **Some cricket players are at least 12 years old.**.

Q20 **Answer: (b) The sample is not representative of the city.**

People at a gym are far more likely to exercise than the general population: a biased sample. Answer: **The sample is not representative of the city.**.

Q21 **Answer: (a) It receives the most serious cases, referred from other hospitals.**

Treating the sickest patients explains a higher death rate even with the best doctors. Answer: **It receives the most serious cases, referred from other hospitals.**.

Q22 **Answer: (c) Shoppers will switch to thin paper bags, which are littered just as often.**

If bags are simply replaced by other litter, the total litter will not fall. Answer: **Shoppers will switch to thin paper bags, which are littered just as often.**.

Q23 **Answer: (d) Whether both groups took the same exam under similar conditions.**

If the exams or conditions differed, similar marks would not show similar effectiveness. Answer: **Whether both groups took the same exam under similar conditions.**.

Q24 **Answer: (c) Appeal to ignorance**

A lack of proof that it is unsafe is taken as proof that it is safe: **appeal to ignorance**.

Q25 **Answer: (b) Children of similar backgrounds who were read to daily learned more new words in a year than those who were not.**

Comparing similar children isolates the effect of daily reading. Answer: **Children of similar backgrounds who were read to daily learned more new words in a year than those who were not.**.

Find the conclusion, list the premises, and name the unstated assumption that links them. Strengtheners support that link or rule out other causes; weakeners attack it or offer another cause. Inferences must follow with no extra information. A paradox answer makes both facts true. Learn the common fallacies by their pattern.

MODULE 25 · ADVANCED

Formal logic

Minimum duration **2 days** | Level **Advanced**

Formal logic studies statements that are true or false and the rules for combining them. It underlies syllogisms, data sufficiency and critical reasoning, and appears directly in campus tests as connective, conditional, truth-teller and assertion-reason questions. This module covers connectives and truth tables, conditionals (converse, inverse, contrapositive, "only if", "unless"), valid and invalid inferences, knights-and-knaves and alternating truth-teller puzzles, binary logic and assertion-reason, with 25 solved questions checked by a program.

IN THIS MODULE

25.1 Connectives and truth tables
25.2 Conditionals
25.3 Valid and invalid inferences
25.4 Truth-teller puzzles
25.5 Binary logic and assertion-reason
25.6 Practice questions
25.7 Solutions

25.1 Connectives and truth tables

25.1.1 The five connectives

p	q	not p	p and q	p or q	if p then q	p if and only if q
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Read the table row by row. "p and q" is true only when both are true. "p or q" (inclusive) is false only when both are false. "If p then q" is false **only** when p is true and q is false. "p if and only if q" is true when p and q have the same value.

SHORTCUT TRICK

Remember the one false row of "if p then q": a promise is broken only when the condition happens (p true) and the result does not (q false). If the condition never happens, the promise is not broken.

25.1.2 Negations

Statement	Negation
p and q	not p, or not q

Statement	Negation
p or q	not p, and not q
All A are B	Some A are not B
Some A are B	No A are B
If p then q	p and not q

25.2 Conditionals

25.2.1 Converse, inverse and contrapositive

Form	Statement	Same meaning as the original?
Original	If p then q	—
Converse	If q then p	No
Inverse	If not p then not q	No
Contrapositive	If not q then not p	**Yes**

WORKED EXAMPLE

"If it rains, the ground gets wet."
Contrapositive (equivalent): if the ground is not wet, it did not rain. Converse (not equivalent): if the ground is wet, it rained; a sprinkler could have wet it.

25.2.2 "Only if" and "unless"

Sentence	Means
p only if q	if p then q (q is necessary for p)
p if q	if q then p
p unless q	if not q then p
q is necessary for p	if p then q
q is sufficient for p	if q then p

WATCH OUT

"You will pass only if you study" does **not** mean that studying guarantees a pass. It means that if you passed, you must have studied.

25.3 Valid and invalid inferences

25.3.1 Four argument forms

Form	Pattern	Valid?
Modus ponens	If p then q; p; so q	Valid
Modus tollens	If p then q; not q; so not p	Valid
Affirming the consequent	If p then q; q; so p	Invalid
Denying the antecedent	If p then q; not p; so not q	Invalid

TIP
An argument is valid if the conclusion is true in **every** row of the truth table where all premises are true. That is exactly how these forms were checked.

25.4 Truth-teller puzzles

25.4.1 Knights and knaves

On an island, **knights** always tell the truth and **knaves** always lie. Test each possible assignment: a statement made by a knight must be true, and a statement made by a knave must be false.

WORKED EXAMPLE
A says: "B is a knave." B says: "A and I are of the same type."
Try A knight: then B is a knave, and B's claim (same type) is false, consistent. Try A knave: then B is a knight, but B's claim (same type) would be false, which a knight cannot say. So **A is a knight and B is a knave**.

A	B	A's claim (B knave)	B's claim (same type)	Consistent?
knight	knight	F ×		no
knight	knave	T ✓	F ✓ (knave lies)	**yes**
knave	knight	T × (knave must lie)		no
knave	knave	F ✓	T ×	no

25.4.2 Alternating truth-tellers

Some puzzles add an **alternator** whose statements switch between true and false (T, F or F, T). Check each person's pair of statements against the pattern for their type: truth-teller T T, liar F F, alternator T F or F T.

25.5 Binary logic and assertion-reason

25.5.1 Binary logic

Write 1 for true and 0 for false. AND gives 1 only if both are 1; OR gives 0 only if both are 0; NOT flips the value; XOR gives 1 when the two differ.

p	q	p AND q	p OR q	p XOR q
1	1	1	1	0
1	0	0	1	1
0	1	0	1	1
0	0	0	0	0

25.5.2 Assertion and reason

An assertion (A) and a reason (R) are given. Decide whether each is true, and whether R correctly explains A.

Option	Meaning
(a)	Both A and R are true, and R is the correct explanation of A
(b)	Both A and R are true, but R is not the correct explanation of A
(c)	A is true, but R is false
(d)	A is false, but R is true
(e)	Both A and R are false

25.6 Practice questions

25.6.1 Basic level

Q1 If p is true and q is false, what is the truth value of “p and q”?

(a) True **(b)** False
(c) Cannot be determined **(d)** Both

Q2 If p is false and q is true, what is the truth value of “p or q”?

(a) True **(b)** False
(c) Cannot be determined **(d)** Neither

Q3 What is the contrapositive of “If it rains, the ground gets wet”?

(a) If the ground gets wet, it rains.
(b) If it does not rain, the ground does not get wet.
(c) If the ground does not get wet, it did not rain.
(d) It rains and the ground does not get wet.

Q4 What is the converse of "If a number is divisible by 4, it is even"?

(a) If a number is even, it is divisible by 4.
(b) If a number is not even, it is not divisible by 4.
(c) If a number is not divisible by 4, it is not even.
(d) A number is divisible by 4 and not even.

Q5 "You will pass only if you study." Which statement must be true?

(a) If you study, you will pass.
(b) If you pass, you studied.
(c) If you do not pass, you did not study.
(d) Studying guarantees passing.

Q6 Which sentence means the same as "Unless you hurry, you will miss the bus"?

(a) If you hurry, you will catch the bus.
(b) If you miss the bus, you did not hurry.
(c) If you do not hurry, you will miss the bus.
(d) You will miss the bus even if you hurry.

Q7 What is the negation of "All students are present"?

(a) No students are present.
(b) Some students are not present.
(c) All students are absent.
(d) Some students are present.

Q8 "If p then q" is true and q is false. What is p?

(a) True
(b) False
(c) Cannot be determined
(d) Both true and false

25.6.2 Intermediate level

Q9 "If it is a weekday, the shop is open. The shop is open. So it is a weekday." Is this argument valid?

(a) Valid (modus ponens)
(b) Valid (modus tollens)
(c) Invalid (affirming the consequent)
(d) Invalid (denying the antecedent)

Q10 In the truth table of "if p then q", in how many of the four rows is the statement true?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

Q11 When is "p if and only if q" true?

(a) When p and q are both true only
(b) When p and q have the same truth value
(c) When exactly one of them is true
(d) Always

Q12 What is the negation of "If it rains, I stay at home"?

(a) If it does not rain, I go out.
(b) It rains and I do not stay at home.
(c) It does not rain and I stay at home.
(d) If I stay at home, it rains.

Q13 A says: "B is a knave." B says: "A and I are of the same type." (Knights always tell the truth, knaves always lie.) What are A and B?

(a) Both knights
(b) A knight, B knave
(c) A knave, B knight
(d) Both knaves

Q14 A says: "I am a knave or B is a knight." What are A and B?

(a) Both knights
(b) A knight, B knave
(c) A knave, B knight
(d) Both knaves

Q15 With 1 = true and 0 = false, find (1 AND 0) OR (NOT 0).

(a) 0
(b) 1
(c) 10
(d) Cannot be determined

Q16 Assertion (A): Iron rusts in moist air. Reason (R): Rusting of iron needs both oxygen and water.

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true
(e) Both A and R are false

25.6.3 Advanced level

Q17 A says: "All of us are knaves." B says: "Exactly one of us is a knight." C says nothing. What are A, B and C?

(a) A knave, B knight, C knave
(b) A knave, B knave, C knight
(c) A knight, B knave, C knave
(d) All three knaves

Q18 P, Q and R are a truth-teller (always true), a liar (always false) and an alternator (statements alternate true and false), in some order. P: "Q is the alternator. R is the liar." Q: "P is the liar. R is the liar." R: "I am the truth-teller. P is the alternator." Who is the alternator?

(a) P
(b) Q
(c) R
(d) Cannot be determined

Q19 Assertion (A): The sun rises in the east. Reason (R): The earth rotates from west to east.

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true
(e) Both A and R are false

Q20 Assertion (A): Water boils at 100°C at the top of Mount Everest. Reason (R): The boiling point of water falls as air pressure falls.

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true
(e) Both A and R are false

Q21 Which statement is equivalent to "If not p, then q"?

(a) If p, then not q.
(b) If not q, then p.
(c) If q, then not p.
(d) If not q, then not p.

Q22 "All doctors are graduates. Ravi is not a graduate. So Ravi is not a doctor." Is the argument valid?

(a) Valid (modus ponens) **(b)** Valid (modus tollens)
(c) Invalid (affirming the consequent) **(d)** Invalid (denying the antecedent)

Q23 "A meeting is held only if the manager is present. The manager is present only if it is a weekday. Today is Sunday." What follows?

(a) A meeting is held today. **(b)** No meeting is held today.
(c) The manager is present today. **(d)** Nothing can be concluded.

Q24 In the truth table of "(p and q) → r", in how many of the eight rows is the statement false?

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

Q25 In how many of the four rows of a truth table is "p XOR q" true?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

25.7 Solutions

Q1 **Answer: (b) False**

"And" needs both parts true; q is false, so "p and q" is **false**.

Q2 **Answer: (a) True**

"Or" is true when at least one part is true: **true**.

Q3 **Answer: (c) If the ground does not get wet, it did not rain.**

Contrapositive: if not q then not p: **if the ground does not get wet, it did not rain**. (a) is the converse, (b) the inverse, (d) the negation.

Q4 **Answer: (a) If a number is even, it is divisible by 4.**

The converse swaps the parts: **if a number is even, it is divisible by 4**, which is false (6 is even but not divisible by 4).

Q5 **Answer: (b) If you pass, you studied.**

"p only if q" means if p then q: **if you pass, you studied**.

Q6 **Answer: (c) If you do not hurry, you will miss the bus.**

"p unless q" means if not q then p: **if you do not hurry, you will miss the bus**.

Q7 **Answer: (b) Some students are not present.**

The negation of "all A are B" is "some A are not B": **some students are not present**.

Q8 **Answer: (b) False**

Modus tollens: if q is false, p must be false, otherwise "if p then q" would be false. p is **false**.

Q9 **Answer: (c) Invalid (affirming the consequent)**

The shop could also be open at weekends. Inferring p from q is **affirming the consequent**, which is invalid.

Q10 **Answer: (c) 3**

It is false only when p is true and q is false, so it is true in **3** rows.

Q11 **Answer: (b) When p and q have the same truth value**

The biconditional is true when both are true or both are false: **when they have the same truth value**.

Q12 **Answer: (b) It rains and I do not stay at home.**

The negation of "if p then q" is "p and not q": **it rains and I do not stay at home**.

Q13 **Answer: (b) A knight, B knave**

Testing all four cases (25.4.1), only **A knight, B knave** is consistent.

Q14 **Answer: (a) Both knights**

If A were a knave, "I am a knave or ..." would be true, which a knave cannot say. So A is a knight, and his statement is true; "I am a knave" is false, so "B is a knight" must be true: **both knights**.

Q15 **Answer: (b) 1**

1 AND 0 = 0; NOT 0 = 1; 0 OR 1 = **1**.

Q16 **Answer: (a) Both A and R are true, and R is the correct explanation of A**

Both are true, and the need for water and oxygen explains why moist air causes rusting: option **(a)**.

Q17 **Answer: (a) A knave, B knight, C knave**

A cannot be a knight (he would be claiming to be a knave), so A is a knave and "all knaves" is false: at least one is a knight. If B were a knave, only C could be a knight: if C is a knight, "exactly one knight" would be true (a knave cannot say it); if C is a knave, all three would be knaves and A's statement would be true. Both fail. So B is a knight, exactly one knight is true, and C is a knave: **A knave, B knight, C knave**.

Q18 **Answer: (b) Q**

Try P truth-teller: then Q is the alternator and R the liar. Check Q: "P is the liar" (false), "R is the liar" (true): alternates ✓. Check R: both statements false ✓. The other five assignments fail. The alternator is **Q**.

Q19 **Answer: (a) Both A and R are true, and R is the correct explanation of A**

Both are true, and the earth's west-to-east rotation is why the sun appears to rise in the east: option **(a)**.

Q20 **Answer: (d) A is false, but R is true**

At high altitude the pressure is low, so water boils below 100°C: A is false, R is true: option **(d)**.

Q21 **Answer: (b) If not q, then p.**

The contrapositive of "if not p then q" is "if not q then not (not p)", that is, **if not q, then p**.

Q22 **Answer: (b) Valid (modus tollens)**

"If doctor then graduate; not graduate; so not doctor" is **modus tollens**, which is valid.

Q23 **Answer: (b) No meeting is held today.**

Meeting → manager present → weekday. Today is not a weekday, so the manager is not present, so **no meeting is held** (modus tollens twice).

Q24 **Answer: (b) 1**

It is false only when p and q are both true and r is false: **1** row.

Q25 **Answer: (b) 2**

XOR is true when p and q differ: (1, 0) and (0, 1), so **2** rows.

"If p then q" is false only when p is true and q is false; it is equivalent to its contrapositive, not to its converse or inverse. "p only if q" means if p then q; "p unless q" means if not q then p. Modus ponens and modus tollens are valid; affirming the consequent and denying the antecedent are not. For truth-teller puzzles, test every assignment against every statement.

MODULE 26 · ADVANCED

Advanced puzzles

Minimum duration **1 day** | Level **Advanced**

This module collects the puzzle types that appear in harder campus tests and eLitmus-style papers: tournaments (knockout and round robin) and points tables, team selection with conditions, shortest paths in a network, cryptarithmetic, weighing and measuring puzzles, transport and crossing puzzles, and grid counting. Each type gets its key formula or method, and the 25 solved questions were all checked by a program (search, enumeration or brute force).

IN THIS MODULE

26.1 Tournaments and points tables
26.2 Selection, paths and grids
26.3 Cryptarithmetic
26.4 Weighing, measuring and crossing puzzles
26.5 Practice questions
26.6 Solutions

26.1 Tournaments and points tables

26.1.1 Knockout and round robin

Format	Matches	Other facts
Knockout, n teams	n - 1 (every match eliminates one team)	rounds = $\lceil \log_2 n \rceil$; byes = next power of 2 - n
Round robin, each pair once	n(n - 1)/2	each team plays n - 1 matches
Round robin, each pair twice	n(n - 1)	home and away

SHORTCUT TRICK

In a knockout, count losers, not matches: every team except the champion loses exactly once, so there are n - 1 matches whatever the byes.

26.1.2 Points tables

In a round robin where a win gives w points and a draw gives 1 to each side, every match hands out either w points (a result) or 2 points (a draw). With w = 2, the total is always 2 × the number of matches, a useful check on any points table.

26.2 Selection, paths and grids

26.2.1 Team selection with conditions

List the conditions as rules ("P and Q not together", "if R then S", "if T then P"), then enumerate the possible teams systematically, or count with combinations and subtract the forbidden ones.

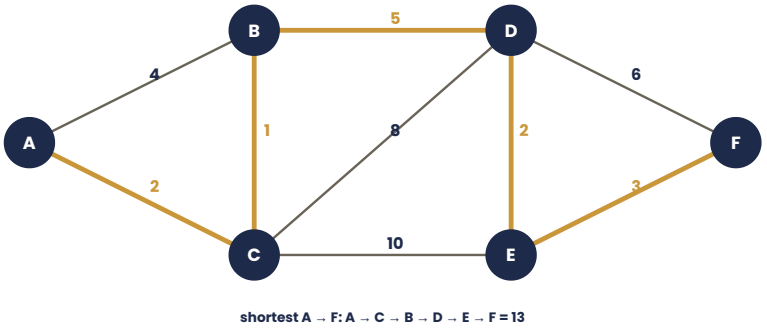
WORKED EXAMPLE

Choose 3 of P, Q, R, S, T. P and Q cannot both be chosen; if R is chosen, S must be chosen; if T is chosen, P must be chosen. How many teams are possible?

Enumerating the 10 possible trios and removing the forbidden ones leaves PRS, PST and QRS: **3** teams.

26.2.2 Shortest paths

To find the shortest route in a small network, keep the best known distance to each point and always extend the closest unfinished point (Dijkstra's method). The direct-looking route is often not the shortest.



A WEIGHTED NETWORK (USED IN QUESTIONS 13 AND 22)

26.2.3 Grid paths

On a grid of m × n squares, the number of shortest paths from one corner to the opposite corner, moving only right or up, is C(m + n, m).

26.3 Cryptarithmetic

26.3.1 Letters standing for digits

Each letter is a different digit and the leading letters are not zero. Start from the column with the most information, usually the leftmost carry.

WORKED EXAMPLE

SEND + MORE = MONEY

MONEY has one more digit than SEND and MORE, so M = 1 (the carry). Then S + 1 (+ carry) gives a two-digit number starting with 1, forcing S = 9 and O = 0. Working through the remaining columns gives **9567 + 1085 = 10652**.

Letter	S	E	N	D	M	O	R	Y
Digit	9	5	6	7	1	0	8	2

TIP
A sum of two n-digit numbers that has n + 1 digits always starts with 1. That single fact opens most addition cryptarithms.

26.4 Weighing, measuring and crossing puzzles

26.4.1 Balance puzzles

A balance has three outcomes (left heavy, right heavy, equal), so each weighing can split the coins into three groups. With one heavier coin among N, the minimum number of weighings is the smallest k with $3^k \geq N$. If the odd coin may be heavier **or** lighter, 3 weighings handle up to 12 coins.

Coins (one heavier)	3	9	27	81
Weighings	1	2	3	4

26.4.2 Measuring with jugs

With jugs of a and b litres you can fill, empty or pour one into the other. A target t can be measured if it is a multiple of the highest common factor of a and b (and fits in a jug). A breadth-first search over (amount in jug 1, amount in jug 2) gives the minimum number of steps.

WORKED EXAMPLE
Measure 4 litres using a 3-litre and a 5-litre jug.
Fill 5 → pour into 3 (5: 2 left) → empty 3 → pour 2 into 3 → fill 5 → pour into 3 until full (1 litre moves): the 5-litre jug now holds **4** litres. That is 6 steps, the minimum.

26.4.3 Crossing puzzles

In the bridge-and-torch puzzle, people cross a bridge at night with one torch, at most two at a time, and a pair walks at the slower person's speed. With times 1, 2, 5, 10 minutes, the best plan sends the two fastest first, so that the two slowest cross together: 2 + 1 + 10 + 2 + 2 = 17 minutes.

26.5 Practice questions

26.5.1 Basic level

Q1 In a knockout tournament with 16 teams, how many matches are played?
(a) 8 **(b)** 15 **(c)** 16 **(d)** 32

Q2 A knockout tournament has 20 teams. How many byes are needed in the first round?
(a) 4 **(b)** 8 **(c)** 12 **(d)** 16

Q3 In a round robin with 8 teams, each pair plays once. How many matches are played?
(a) 16 **(b)** 28 **(c)** 56 **(d)** 64

Q4 In a league of 6 teams, each pair plays twice (home and away). How many matches are played?
(a) 15 **(b)** 30 **(c)** 36 **(d)** 12

Q5 Among 9 identical-looking coins one is heavier. What is the minimum number of weighings on a balance to find it?

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

Q6 Among 27 coins one is heavier. What is the minimum number of weighings needed?

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

Q7 Using a 3-litre and a 5-litre jug, what is the minimum number of steps (fill, empty or pour) to measure exactly 4 litres?

(a) 4 **(b)** 5 **(c)** 6 **(d)** 7

Q8 10 people each shake hands once with every other person. How many handshakes are there?

(a) 45 **(b)** 50 **(c)** 90 **(d)** 100

26.5.2 Intermediate level

Q9 In SEND + MORE = MONEY (each letter a different digit), what is M?

(a) 0 **(b)** 1 **(c)** 2 **(d)** 9

Q10 In SEND + MORE = MONEY, what is the value of SEND?

(a) 9567 **(b)** 9576 **(c)** 8567 **(d)** 9564

Q11 In a round robin of 4 teams (win 2 points, draw 1 point each, loss 0), what is the total number of points awarded?

(a) 8 **(b)** 12 **(c)** 16 **(d)** 6

Q12 Choose 3 of P, Q, R, S, T. P and Q cannot both be chosen; if R is chosen, S must be chosen; if T is chosen, P must be chosen. How many teams are possible?

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

Q13 In the network of 26.2.2, what is the length of the shortest route from A to F?

(a) 12 **(b)** 13 **(c)** 14 **(d)** 15

Q14 A farmer must take a wolf, a goat and a cabbage across a river in a boat that holds him and one item. The wolf cannot be left alone with the goat, nor the goat with the cabbage. What is the minimum number of crossings?

(a) 5 **(b)** 7 **(c)** 9 **(d)** 11

Q15 On a grid of 3 × 3 squares, how many shortest paths go from the bottom-left corner to the top-right corner, moving only right or up?

(a) 9 **(b)** 12 **(c)** 20 **(d)** 27

Q16 Among 12 coins one is counterfeit and may be heavier or lighter. What is the minimum number of weighings that always finds it (and tells whether it is heavier or lighter)?

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

26.5.3 Advanced level

Q17 AB and BA are two-digit numbers with different non-zero digits A and B, A > B, and AB + BA = 121. How many such pairs (A, B) are there?

(a) 3 **(b)** 4 **(c)** 5 **(d)** 9

Q18 In SEND + MORE = MONEY, what is N + D?

(a) 11 **(b)** 12 **(c)** 13 **(d)** 14

Q19 In a round robin of 5 teams (win 3 points, draw 1, loss 0), team A won 3 matches and drew 1. How many points did A get?

(a) 9 **(b)** 10 **(c)** 11 **(d)** 12

Q20 In a round robin of 6 teams with 2 points for a win, what is the maximum number of points a single team can score?

(a) 8 **(b)** 10 **(c)** 12 **(d)** 15

Q21 Four people must cross a bridge at night with one torch; at most two cross at a time, and a pair walks at the slower person's speed. Their times are 1, 2, 5 and 10 minutes. What is the minimum total time?

(a) 17 min **(b)** 18 min **(c)** 19 min **(d)** 20 min

Q22 In the network of 26.2.2, what is the length of the shortest route from A to E?

(a) 10 **(b)** 11 **(c)** 12 **(d)** 13

Q23 A committee of 3 is chosen from 4 men and 3 women. In how many ways can it be chosen if it must include at least one woman?

(a) 28 **(b)** 31 **(c)** 34 **(d)** 35

Q24 A knockout tournament has 64 players. How many rounds are needed to find the champion?

(a) 5 **(b)** 6 **(c)** 7 **(d)** 8

Q25 Using a 4-litre and a 9-litre jug, what is the minimum number of steps (fill, empty or pour) to measure exactly 6 litres?

(a) 6 **(b)** 7 **(c)** 8 **(d)** 10

26.6 Solutions

Q1 **Answer: (b) 15**

Every match eliminates one team, and 15 teams must be eliminated: **15**.

Q2 **Answer: (c) 12**

The next power of 2 is 32, so 32 - 20 = **12** byes.

Q3 **Answer: (b) 28**

8 × 7 / 2 = **28**.

Q4 **Answer: (b) 30**

6 × 5 = **30**.

Q5 **Answer: (b) 2**

Each weighing splits into three groups: 9 → 3 → 1, so **2** weighings.

Q6 **Answer: (b) 3**

$3^3 = 27$, so **3** weighings.

Q7 **Answer: (c) 6**

The sequence in 26.4.2 takes **6** steps, and a breadth-first search confirms no shorter sequence exists.

Q8 **Answer: (a) 45**

10 × 9 / 2 = **45**.

Q9 **Answer: (b) 1**

The sum has one more digit, so the carry M = **1**.

Q10 **Answer: (a) 9567**

The unique solution is 9567 + 1085 = 10652, so SEND = **9567**.

Q11 **Answer: (b) 12**

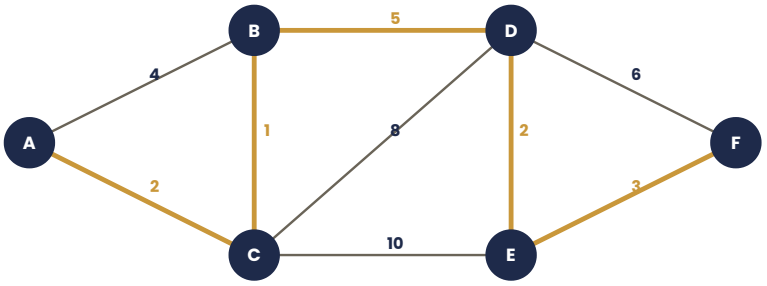
There are 4 × 3 / 2 = 6 matches, each giving away exactly 2 points: **12**.

Q12 **Answer: (b) 3**

The valid teams are PRS, PST and QRS: **3**.

Q13 **Answer: (b) 13**

Route A → C → B → D → E → F: 2 + 1 + 5 + 2 + 3 = **13**. The route through C → E (2 + 10 + 3 = 15) is longer.



Q14 **Answer: (b) 7**

Goat over, back, wolf over, goat back, cabbage over, back, goat over: **7** crossings.

Q15 **Answer: (c) 20**

Each path is 3 rights and 3 ups in some order: C(6, 3) = **20**.

Q16 **Answer: (b) 3**

Two weighings give at most 9 outcomes, fewer than the 24 possibilities; a known three-weighing plan works: **3**.

Q17 **Answer: (b) 4**

AB + BA = 11(A + B) = 121, so A + B = 11: (9, 2), (8, 3), (7, 4), (6, 5): **4** pairs.

Q18 **Answer: (c) 13**

N = 6 and D = 7, so N + D = **13**.

Q19 **Answer: (b) 10**

A plays 4 matches: 3 wins × 3 + 1 draw × 1 = **10** points.

Q20 **Answer: (b) 10**

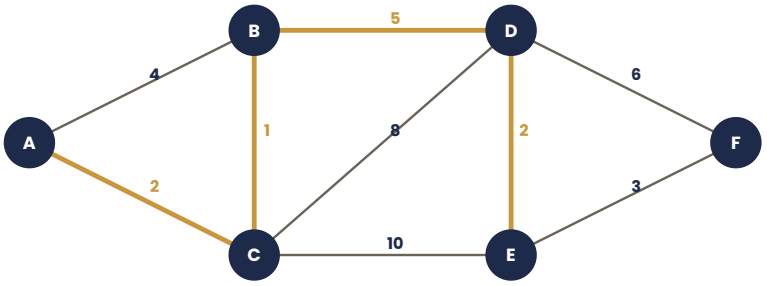
Each team plays 5 matches; winning all gives 5 × 2 = **10**.

Q21 **Answer: (a) 17 min**

1 and 2 cross (2), 1 returns (1), 5 and 10 cross (10), 2 returns (2), 1 and 2 cross (2): **17 minutes**. A search over all plans confirms it.

Q22 **Answer: (a) 10**

A → C → B → D → E = 2 + 1 + 5 + 2 = **10**, shorter than A → C → E = 12.



Q23 **Answer: (b) 31**

All committees C(7, 3) = 35, minus all-men C(4, 3) = 4: **31**.

Q24 **Answer: (b) 6**

64 = 2^6, so **6** rounds.

Q25 **Answer: (c) 8**

A breadth-first search over jug states gives **8** steps, for example: fill 9, pour into 4, empty 4, pour into 4 (9-litre jug has 1), empty 4, pour the 1 into 4, fill 9, pour into 4 until full (3 moves), leaving 6.

Knockout: n - 1 matches; round robin: n(n - 1)/2. Points tables: each match gives a fixed number of points. For selection, write conditions as rules and enumerate; for networks use Dijkstra; grid paths C(m + n, m). In cryptarithms the extra leading digit is 1. Balance puzzles use powers of 3, jug puzzles use search, and crossing puzzles pair the slowest people together.

MODULE 27 · ADVANCED

Exam strategy and mock practice

Minimum duration **1 day** | Level **Advanced**

The last module turns the topics into exam performance: how reasoning sections are usually built in campus, eLitmus-style and bank exams, how to choose and order questions under time pressure, how to keep and use an error log, a revision plan for the final two weeks, and a 25-question mixed mock test drawn from the whole syllabus, fully solved.

IN THIS MODULE

27.1 How reasoning sections are built
27.2 Time management
27.3 Error log and revision plan
27.4 Mixed mock test
27.5 Solutions

27.1 How reasoning sections are built

27.1.1 Typical patterns by exam type

Exact patterns change from year to year, so always check the latest official notification. The broad shapes below have been stable for several years and are a good guide for practice.

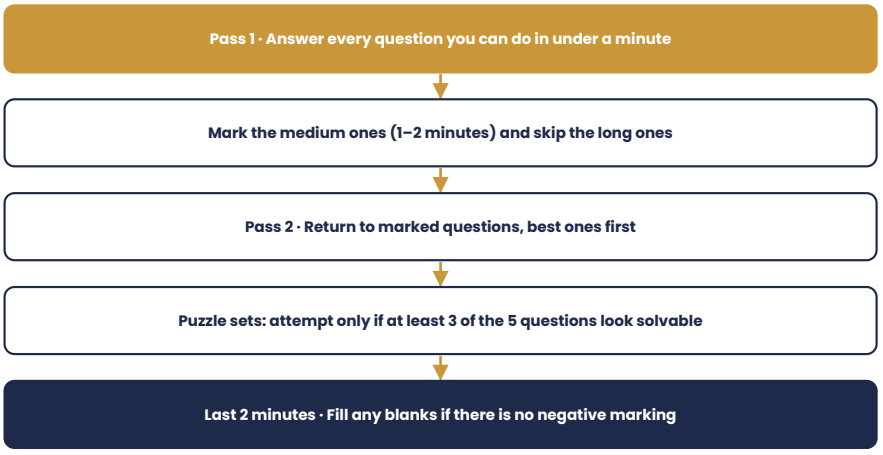
Exam type	What the reasoning section stresses	How to prepare
TCS NQT style	series, coding, blood relations, seating, puzzles, statements; adaptive or sectional timing	speed on Level 1 and 2 topics; practise sectional time limits
Infosys style	harder puzzles, data sufficiency, syllogisms, arrangements, critical reasoning	Level 3 topics; long puzzle sets under time
Wipro, Accenture, Cognizant, Capgemini style	mixed short questions: series, analogy, coding, directions, logical deduction	breadth and accuracy; avoid getting stuck
eLitmus style	few but deep questions: cryptarithmetic, complex arrangements, data sufficiency; negative marking	Module 26 types; attempt only what you can finish
Bank exams	puzzles and seating (circular, linear, floor), inequalities, syllogism, coding, input-output, blood relations	puzzle sets daily; inequalities and syllogism must be fast

WATCH OUT

If the exam has negative marking, a random guess among 4 options with -1/3 per wrong answer has an expected value of zero. Guess only when you can eliminate at least one option.

27.2 Time management

27.2.1 Two-pass method



ORDERING QUESTIONS IN THE EXAM

Topic	Target time per question
Series, analogy, classification, alphabet test	30–40 seconds
Coding, directions, ranking, blood relations, clocks, calendars	45–60 seconds
Inequalities, syllogism, statements	45–60 seconds
Seating and puzzle sets (per set of 5)	6–8 minutes
Data sufficiency, critical reasoning	60–90 seconds

SHORTCUT TRICK
Read the question stem before the data in long puzzle sets. Knowing what is asked tells you which clues to place first and which details you can skip.

27.3 Error log and revision plan

27.3.1 Keeping an error log

For every wrong or slow question, write one line: the topic, the question number, what went wrong (concept, misread, calculation, time), and the rule that would have prevented it. Read the log before every mock. Most marks are lost to repeated mistakes, not new ones.

Date	Topic	Mistake	Rule to remember
—	Seating (south-facing)	used my left instead of theirs	facing South: their left = my right
—	Syllogism	ignored either-or pair	check Some/No and All/Some-not on the same pair
—	Clocks	forgot hour hand moves	angle = |30H - 5.5M|

27.3.2 The final two weeks

Days	Plan
1–4	Re-solve the advanced questions of Modules 12 to 19 (timed)
5–7	Modules 20 to 26 practice sets; review every key-idea box
8–11	One full mock per day + error-log review
12–13	Weak topics only (from the error log)
14	Light revision of formulas and tricks; rest

> **TIP**
> The mixed mock test in 27.4 has 25 questions for 30 minutes. Attempt it in one sitting with a timer, then check the solutions in 27.5 and fill in your error log.

27.4 Mixed mock test

27.4.1 Basic level

Q1 3, 8, 15, 24, 35, ?

(a) 46 (b) 48 (c) 49 (d) 50

Q2 B, D, G, K, P, ?

(a) U (b) V (c) W (d) T

Q3 If BOOK is coded as CPPL, how is DESK coded?

(a) EFTL (b) EFTK (c) CFTL (d) EETL

Q4 A man walks 6 km North and then 8 km East. How far is he from the start?

(a) 14 km (b) 10 km (c) 12 km (d) 2 km

Q5 In a row, Ravi is 9th from the left and 14th from the right. How many people are in the row?

(a) 21 (b) 22 (c) 23 (d) 24

Q6 5 : 26 :: 7 : ?

(a) 49 (b) 50 (c) 51 (d) 56

Q7 Find the odd one out: 2, 3, 5, 9, 11

(a) 2 (b) 3 (c) 9 (d) 11

Q8 Today is Wednesday. What day will it be after 100 days?

(a) Thursday (b) Friday (c) Saturday (d) Sunday

27.4.2 Intermediate level

Q9 What is the angle between the hands of a clock at 4:40?

(a) 90° (b) 100° (c) 110° (d) 120°

Q10 A is B's sister. C is B's father. D is C's mother. How is A related to D?

(a) Daughter **(b)** Granddaughter **(c)** Niece **(d)** Sister

Q11 Statements: P > Q ≥ R = S. Conclusions: I. P > S II. Q > S

(a) Only I follows **(b)** Only II follows
(c) Either I or II follows **(d)** Neither I nor II follows

Q12 Statements: All A are B. No B are C. Conclusions: I. No A are C. II. Some C are not A.

(a) Only I follows **(b)** Only II follows
(c) Neither follows **(d)** Both follow

Q13 P, Q, R, S, T sit in a row facing North. R is in the middle, P is to the immediate left of R, T is at the right end, and Q is not at an end. Who is at the left end?

(a) P **(b)** Q **(c)** S **(d)** T

Q14 If + means × and × means −, what is 6 + 4 × 3?

(a) 21 **(b)** 27 **(c)** 18 **(d)** 30

Q15 What is Ravi's age? I. Ravi is 4 years older than Sita. II. Sita is 20 years old.

(a) I alone is sufficient **(b)** II alone is sufficient
(c) Either alone is sufficient **(d)** Both together are needed

Q16 A clock shows 10:10. What time does its mirror image show?

(a) 1:50 **(b)** 2:50 **(c)** 1:10 **(d)** 2:10

27.4.3 Advanced level

Q17 1, 2, 6, 24, 120, ?

(a) 240 **(b)** 600 **(c)** 720 **(d)** 840

Q18 Find the wrong term: 4, 9, 25, 49, 121, 168

(a) 9 **(b)** 49 **(c)** 121 **(d)** 168

Q19 A machine moves the smallest remaining number to the left in each step (others keep their order). Input: 45 12 78 33 60. What is Step 2?

(a) 12 33 45 78 60 **(b)** 12 45 33 78 60 **(c)** 12 33 45 60 78 **(d)** 12 33 78 45 60

Q20 I. The government cut the import duty on edible oil. II. Edible oil prices fell in the market.

(a) I is the cause and II is its effect **(b)** II is the cause and I is its effect
(c) Both are effects of a common cause **(d)** Both are effects of independent causes

Q21 Notice: "Please use the stairs; the lift is under repair." Assumptions: I. People are able to use the stairs. II. The lift will never work again.

(a) Only I is implicit **(b)** Only II is implicit
(c) Both are implicit **(d)** Neither is implicit

Q22 In a knockout tournament with 32 teams, how many matches are played?

(a) 16 **(b)** 31 **(c)** 32 **(d)** 63

Q23 What is the contrapositive of "If you are a member, you may enter"?

(a) If you may enter, you are a member.

(b) If you are not a member, you may not enter.

(c) If you may not enter, you are not a member.

(d) You are a member and you may not enter.

Q24 1 January 2026 was a Thursday. What day was 1 January 2027?

(a) Thursday **(b)** Friday **(c)** Saturday **(d)** Wednesday

Q25 "A survey of 20 students in the library found that most students read for two hours a day, so most students in the college read for two hours a day." What is the flaw?

(a) The sample is not representative. **(b)** It attacks the students.

(c) It uses circular reasoning. **(d)** It offers only two options.

27.5 Solutions

Q1 **Answer: (b) 48**

Terms are $n^2 - 1$ ($n = 2$ to 6); next $7^2 - 1 =$ **48**.

Q2 **Answer: (b) V**

Gaps +2, +3, +4, +5, then +6: P (16) + 6 = 22 = **V**.

Q3 **Answer: (a) EFTL**

Each letter +1: D→E, E→F, S→T, K→L: **EFTL**.

Q4 **Answer: (b) 10 km**

$\sqrt{(6^2 + 8^2)} =$ **10 km**.

Q5 **Answer: (b) 22**

$9 + 14 - 1 =$ **22**.

Q6 **Answer: (b) 50**

$26 = 5^2 + 1$, so $7^2 + 1 =$ **50**.

Q7 **Answer: (c) 9**

All are prime except **9**.

Q8 **Answer: (b) Friday**

100 mod 7 = 2; Wednesday + 2 = **Friday**.

Q9 **Answer: (b) 100°**

$|30 \times 4 - 5.5 \times 40| = |120 - 220| =$ **100°**.

Q10 **Answer: (b) Granddaughter**

A is C's daughter, and D is C's mother, so A is D's **granddaughter**.

Q11 **Answer: (a) Only I follows**

P > Q ≥ R = S gives P > S (I). Q ≥ S only, so Q > S is not certain (II). Answer **Only I follows**.

Q12 **Answer: (d) Both follow**

All + No gives No A are C (I). Since C has members and none of them is an A, some C are not A (II). **Both follow**.

Q13 **Answer: (c) S**

R 3, P 2, T 5. Q is not at an end, so Q is 4 and **S** is at the left end.

Q14 **Answer: (a) 21**

Rewrite: 6 × 4 − 3 = 24 − 3 = **21**.

Q15 **Answer: (d) Both together are needed**

Neither alone gives Ravi's age; together he is 24. **Both together are needed**.

Q16 **Answer: (a) 1:50**

11:60 − 10:10 = **1:50**.

Q17 **Answer: (c) 720**

Multiply by 2, 3, 4, 5, then 6: 120 × 6 = **720** (factorials).

Q18 **Answer: (d) 168**

The terms are squares of primes 2, 3, 5, 7, 11, 13; the last should be 169, so **168** is wrong.

Q19 **Answer: (a) 12 33 45 78 60**

Step 1: 12 45 78 33 60. Step 2 moves 33: **12 33 45 78 60**.

Q20 **Answer: (a) I is the cause and II is its effect**

Lower import duty reduces the cost of imported oil, so prices fall: **I is the cause and II is its effect**.

Q21 **Answer: (a) Only I is implicit**

The notice assumes the stairs can be used (I). "Under repair" suggests the opposite of II. **Only I is implicit**.

Q22 **Answer: (b) 31**

Every match eliminates one team: 32 − 1 = **31**.

Q23 **Answer: (c) If you may not enter, you are not a member.**

If not q then not p: **if you may not enter, you are not a member**.

Q24 **Answer: (b) Friday**

2026 is an ordinary year: 1 odd day. Thursday + 1 = **Friday**.

Q25 **Answer: (a) The sample is not representative.**

Students found in the library are more likely to read than other students: **the sample is not representative**.

Know the pattern of your target exam, take the easy questions first and mark the rest, attempt puzzle sets only when most of their questions look solvable, and never guess blindly under negative marking. Keep an error log, read it before every mock, and spend the final two weeks on timed mixed practice and weak topics.